

ver 26-1



SH Scientific

Since 1982

Laboratory Equipment

삼흥에너지 공식몰 | 1544-0351
실험실닷컴 www.silhumsil.com

SH Scientific's product line-up

01. Vacuum Drying Oven.....	03
02. Drying Oven.....	16
03. Industrial Drying Oven.....	18
04. Furnace.....	20
05. Mixer / Mill / Dispersion.....	40
06. Ceramic consumable.....	43
07. Plastic Ware & Pipette.....	47
08. Incubator.....	50
09. Incubator Shaker & Shaker.....	52
10. Roller Culture Apparatus.....	56
11. Autoclave.....	58
12. Clean Bench & Colony Counter.....	60
13. Mixer.....	61
14. Stirrer.....	65
15. Rotary Evaporator.....	72
16. Water Bath & Chiller.....	77
17. Measuring Instrument.....	80
18. Climate Chamber & 배터리 시험챔버.....	84
19. Desiccator.....	87
20. Hood.....	88

01. Vacuum Drying Oven

Vacuum Drying Oven



Model	SH-VDO-08NG	SH-VDO-30NG	SH-VDO-70NG
Temp Range	Ambient +5°C to 250°C (Ambient + 9°F to 482°F)		
Dimension Internal (W×D×H)	200×200×200mm 7.87×7.87×7.87"	300×300×300mm 11.81×11.81×11.81"	400×400×400mm 15.75×15.75×15.75"
Dimension External (W×D×H)	385×315×555mm 15.15×12.40×21.85"	605×550×770mm 23.81×21.65×30.31"	705×620×870mm 27.75×24.80×34.25"
Capa	8 L	27 L	64 L



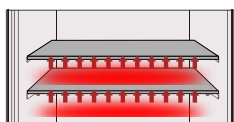
Model	SH-VDO-125NG	SH-VDO-216NG
Temp Range	Ambient +5°C to 250°C (Ambient + 9°F to 482°F)	
Dimension Internal (W×D×H)	500×500×500mm 19.69×19.69×19.69"	600×600×600mm 23.62×23.62×23.62"
Dimension External (W×D×H)	795×770×1025mm 31.30×30.31×40.35"	875×870×1130mm 34.44×34.25×44.49"
Capa	125 L	216 L

Vacuum Drying Oven For Toxic Gas



Model	SH-VDO-30NC	SH-VDO-70NC	SH-VDO-125NC
Temp Range	Ambient +5°C to 150°C (Ambient + 9°F to 302°F)		
Material Internal	STS 316		
Material Packing	Viton Packing		
Dimension Internal (W×D×H)	300×300×300mm 11.81×11.81×11.81"	400×400×400mm 15.75×15.75×15.75"	500×500×500mm 19.69×19.69×19.69"
Dimension External (W×D×H)	605×550×770mm 23.81×21.65×30.31"	705×620×870mm 27.75×24.80×34.25"	795×770×1025mm 31.30×30.31×40.35"
Capa	27 L	64 L	125 L

400°C Vacuum Drying Oven



Direct heating shelves



Model	SH-VDO-30NS	SH-VDO-70NS	SH-VDO-125NS
Temp Range	Ambient +5°C to 400°C (Ambient + 9°F to 752°F)		
Controller	2 EA	3 EA	4 EA
Dimension Internal (W×D×H)	300×300×300mm 11.81×11.81×11.81"	400×400×400 mm 15.75×15.75×15.75"	500×500×500mm 19.68×19.68×19.68"
Dimension External (W×D×H)	800×580×565mm 31.49×22.83×22.24"	900×680×665mm 35.43×26.77×26.18"	1000×790×805mm 39.37×31.10×31.69"

Vacuum Oven For evaporation (Large Capa)



Model	SH-VDO-175NG	SH-VDO-252NG
Temp Range	Ambient +5°C to 150°C (Ambient + 9°F to 302°F)	
Controller	4 EA	5 EA
Dimension External (W×D×H)	865x955x715mm	965x955x825mm
	34.05x37.60x28.15"	38x37.60x32.5"

Cold Bath with Glass Trap 3EA



-20°C

Model	SH-WB-5GDR(-20)
Temp Control Range	Ambient -40°C ~ ambient (-40°F ~ ambient) (Low limit -20°C)
Dimension Internal (W×D×H)	200x200x130mm
	7.87x7.87x5.12"
Dimension External (W×D×H)	283x400x455mm
	11.14x15.75x17.91"



-40°C



-80°C

Model	SH-WB-5GDR(-40)	SH-WB-5GDR(-80)
Temp Control Range	Ambient -60°C ~ -20°C (-108°F ~ -40°F) (Low limit -40°C)	Ambient -100°C ~ -60°C (-180°F ~ -76°F) (Low limit -80°C)
Dimension Internal (Φ×H)	180Φ×210Hmm	180Φ×210Hmm
	7.09Φx8.27H"	7.09Φx8.27H"
Dimension External (W×D×H)	465×560×710mm	505×620×750mm
	18.18x22.04x27.95"	19.88x24.41x29.52"

Cold Bath with Stainless Trap



-20°C

Model	SH-WB-5GDR(-20)
Temp Control Range	Ambient -40°C ~ ambient (-40°F ~ ambient) (Low limit -20°C)
Dimension Internal (W×D×H)	200x200x130mm
	7.87x7.87x5.12"
Dimension External (W×D×H)	283x400x455mm
	11.14x15.75x17.91"



-40°C



-80°C

Model	SH-WB-5GDR(-40)	SH-WB-5GDR(-80)
Temp Control Range	Ambient -60°C ~ -20°C (-108°F ~ -40°F) (Low limit -40°C)	Ambient -100°C ~ -60°C (-180°F ~ -76°F) (Low limit -80°C)
Dimension Internal (Φ×H)	180Φ×210Hmm	180Φ×210Hmm
	7.09Φx8.27H"	7.09Φx8.27H"
Dimension External (W×D×H)	465×560×710mm	505×620×750mm
	18.18x22.04x27.95"	19.88x24.41x29.52"

Large Condensing Trap Bath

NEW



Glass trap 장착



Stainless steel trap 장착(Optional)



Model	SH-WB-68GDR	SH-WB-100GDR
Temp Control Range	-40℃ Ambient (-40°F Ambient)	
Bath capa	68L	100L
Dimension Internal (W×D×H)	1086x675.4x1358(mm)	1438x675x1358(mm)
	42.75x26.59x53.46(inch)	56.61x26.57x53.46(inch)
Material Internal	Stainless Steel	
Material External	Steel Plate with Powder Heating Coated	
Material Trap	Pyrex Glass 4EA	Pyrex Glass 6EA
Power	220V, 1φ	

*Option



CT-GWL



CT-SUSL

Vacuum Oven Package

〈VDO-PK-G Series〉



Package Model	VDO-PK-G1	VDO-PK-G2
Vacuum Dry Oven	SH-VDO-30NG	SH-VDO-70NG
Cold Trap	SH-WB-5GDR(-20)	SH-WB-5GDR(-20)
Vacuum Pump	V10	V10
Oil Mist Trap	VOT-1020	VOT-1020
Dimension(WxDxH)	953x620x1552mm	1053x670x1652mm
	37.52x24.41x61.10"	41.46x26.38x65.04"

Package Model	VDO-PK-G3	VDO-PK-G4
Vacuum Dry Oven	SH-VDO-125NG	SH-VDO-216NG
Cold Trap	SH-WB-5GDR(-20)	SH-WB-5GDR(-20)
Vacuum Pump	V20	V20
Oil Mist Trap	VOT-1020	VOT-1020
Dimension(WxDxH)	1103x768x1738mm	1203x868x1844mm
	43.42x30.24x68.43"	47.36x34.17x72.59"

<VDO-PK-G1,2,3,4CE Series>



Package Model	VDO-PK-G1CE	VDO-PK-G2CE
Vacuum Dry Oven	SH-VDO-30NG	SH-VDO-70NG
Condenser	Vertical Double Hefix (Borosilicate 3.3)	
Cooling Area	0.28 m ²	
Receiving Flask	3L / 0.79gallon	
Chiller	SH-WB-5CDR(-15)	SH-WB-5CDR(-15)
Vacuum Pump	V10	V10
Oil Mist Trap	VOT-1020	VOT-1020
Dimension(WxDxH)	1000x620x1752mm 39.37x24.41x68.97"	1100x670x1852mm 43.31x26.38x72.91"

Package Model	VDO-PK-G3CE	VDO-PK-G4CE
Vacuum Dry Oven	SH-VDO-125NG	SH-VDO-216NG
Condenser	Main + Auxiliary Vertical Double Hefix (Borosilicate 3.3)	
Cooling Area	0.49m ²	
Receiving Flask	5L / 1.32gallon	
Chiller	SH-WB-5CDR(-15)	SH-WB-5CDR(-15)
Vacuum Pump	V20	V20
Oil Mist Trap	VOT-1020	VOT-1020
Dimension(WxDxH)	1266x768x1938mm 49.84x30.24x76.29"	1356x868x2045mm 53.38x34.17x80.51"

*Option : Cold Trap Bath SH-WB-5GDR(-20)

<VDO-PK-C Series>



Package Model	VDO-PK-C1	VDO-PK-C2	VDO-PK-C3
Vacuum Dry Oven	SH-VDO-30NC	SH-VDO-70NC	SH-VDO-125NC
Cold Trap	SH-WB-5GDR(-20)	SH-WB-5GDR(-20)	SH-WB-5GDR(-20)
Vacuum Pump	CV20	CV20	CV20
Oil Mist Trap	VOT-4080	VOT-4080	VOT-4080
Dimension(WxDxH)	953x620x1552mm 37.52x24.41x61.10"	1053x670x1652mm 41.46x26.38x65.04"	1103x768x1738mm 43.42x30.24x68.42"

<VDO-PK-C1,2,3CE Series>



Package Model	VDO-PK-C1CE	VDO-PK-C2CE	VDO-PK-C3CE
Vacuum Dry Oven	SH-VDO-30NC	SH-VDO-70NC	SH-VDO-125NC
Condenser	Vertical Double Hefix (Borosilicate 3.3)		Main+Auxiliary Vertical Double Hefix (Borosilicate 3.3)
Cooling Area	0.28 m ²		0.49m ²
Receiving Flask	3L / 0.79gallon		5L / 1.32gallon
Chiller	SH-WB-5CDR(-15)	SH-WB-5CDR(-15)	SH-WB-12CDR(-25)
Vacuum Pump	CV20	CV20	CV20
Oil Mist Trap	VOT-4080	VOT-4080	VOT-4080
Dimension(WxDxH)	1000x620x1752mm 39.37x24.41x68.97"	1100x670x1852mm 43.31x26.38x72.91"	1266x768x1938mm 49.84x30.24x76.29"

*Option : Cold Trap Bath SH-WB-5GDR(-20)



<VDO-PK-S Series>



Package Model	VDO-PK-S1	VDO-PK-S2	VDO-PK-S3
Vacuum Dry Oven	SH-VDO-30NS	SH-VDO-70NS	SH-VDO-125NS
Cold Trap	SH-WB-5GDR(-40)	SH-WB-5GDR(-40)	SH-WB-5GDR(-40)
Vacuum Pump	V10	V10	V20
Oil Mist Trap	VOT-1020	VOT-1020	VOT-1020
Dimension(WxDxH)	1380x685x1280mm	1465x765x1450mm	1565x895x1535mm
	54.331x26.97x30.39"	57.68x30.11x57.09"	61.61x35.24x60.43"



<VDO-PK-F Series>



Package Model	VDO-PK-F1	VDO-PK-F2
Vacuum Dry Oven	SH-VDO-175NG	SH-VDO-252NG
Condenser	Main + Auxiliary Vertical Double Hefix (CE-10)	
Receiving Flask	5L / 1.32 gallon	
Chiller	SH-WB-12CDR	
Vacuum Pump	PTFEVP130S	DSV330
Dimension(WxDxH)	1445x955x2000mm	1545x955x2000mm
	56.89x37.60x78.74"	60.82x37.60x78.74"

* 다량의 수분 및 유기용매 제거에 최적화

Large Vacuum Oven



Model	SH-VDO-CBD1560
Temp Range	Ambient +5°C to 150°C
Vacuum Resistnace	0 to 76cmHg (0 to 0.1MPa)
Temp controller	Microprocess PID Membrane Touch Digital Controller 4ea
Heating Type	Chamber 내부 4면 Radiant Heating
Heater Capa	3KW x 4Side / 12.0KW
Dimension Internal (WxDxH)	1300x1000x1200mm / 51x39x47"
Dimension External (WxDxH)	1710x1550x1852mm / 67x61x73"

* 사용자 요구사항에 맞게 주문제작 가능합니다.

Glass Condenser for Vacuum Evaporator & Oven



Model	SH-CE5	SH-CE10	SH-CE20
Glass material	Borosilicate 3.3		
Safety Frame	Du-Aluminum & PVC		
Cooling Area	0.28 m ²	0.49 m ²	1.4 m ²
Receiving Flask	3 Liter	5 Liter	10 Liter
Dimension Internal (WxDxH)	400x433x1391mm	500x533x1971mm	600x633x1971mm
	15.75x17.05x54.76"	15.75x17.05x54.76"	23.62x24.92x54.76"
Standard Chiller (Option)	SH-WB-5CDR(-30)	SH-WB-12CDR(-25)	SH-WB-20CDR(-25)
Cooling Capa (Water 기준)	MAX 2L/H, SH-WB-5CDR(-30) 사용시	MAX 3.5 L/H, SH-WB-12CDR(-25) 사용시	MAX 5.0 L/H, SH-WB-20CDR(-25) 칠러 사용시
Recommended for	Rotary evaporator (SH-RE-5L) Vacuum drying oven (SH-VDO-30NG, SH-VDO-70NG, SH-VDO-30NS, SH-VDO-70NS)	Vacuum drying oven (SH-VDO-125NG, SH-VDO-175NG, SH-VDO-252NG, SH-VDO-216NG, SH-VDO-125NS)	Vacuum drying oven (SH-VDO-125NG, SH-VDO-175NG, SH-VDO-252NG, SH-VDO-216NG, SH-VDO-125NS)

Polycarbonate Cubic Vacuum Box

폴리카보네이트 (P.C) Cubic Vacuum Box Cubic 형

NEW

- Cubic 형태의 몰딩제작으로 뛰어난 공간활용성
- Seamless 제작으로 진공유지 우수
- 1mmHg (Torr) 72hr 유지
- Polycarbonate 재질로 투명도 및 부식성이 우수
- 고온유지가 필요없는 진공건조기에 적용
- 일반진공건조기 (히팅형)에 비해 경제적 우수함



Model	SH-PCV-27	SH-PCV-45
Internal Volume	27 L	45 L
Vacuum	Gauge range(Analog)(Mpa)	0 ~ -0.1
	Max.permissible vacuum (Body)(Mpa)	1.33x10 ⁻⁴ (1Torr)
Dimension	Internal(WxDxH)	300x300x300 mm
	Overall(WxDxH)	418x392x338.5 mm
	Nozzle dia	9.5 mm
Shelves	Quantity of shelves (Standard/Max.)	2/4
	Max. load per shelf	5 kg
Net weight		10 kg

폴리카보네이트 (P.C) Cubic Vacuum Box Cubic UV 차단 형



Model	SH-PCV-27U	SH-PCV-45U
Internal Volume	27 L	45 L
Vacuum	Gauge range(Analog)(Mpa)	0 ~ -0.1
	Max.permissible vacuum (Body)(Mpa)	1.33x10 ⁻⁴ (1Torr)
Dimension	Internal(WxDxH)	300x300x300 mm
	Overall(WxDxH)	418x392x338.5 mm
	Nozzle dia	9.5 mm
Shelves	Quantity of shelves (Standard/Max.)	2/4
	Max. load per shelf	5 kg
Net weight		10 kg

Vacuum Accessories



Item	Standard	Material
Hose Barb 11mm	KF(NW)25	Aluminum



Item	Standard	Material
Clamp	KF(NW)16	Aluminum
	KF(NW)25	
	KF(NW)40	
	KF(NW)50	



Item	Standard	Model
Reducer	KF(NW)50x40	SUS 304
	KF(NW)40x25	
	KF(NW)25x16	



Item	Standard	Material
Ceter Ring&Viton Ring	KF(NW)16	SUS 304 Ceter Ring & Viton Ring
	KF(NW)25	
	KF(NW)40	
	KF(NW)50	



Item	Standard	Material
L Connector	KF(NW)25	SUS 304



Item	Standard	Material
Bellows Hose 1M	KF(NW)25	SUS 304
	KF(NW)40	
	KF(NW)50	



Item	Standard	Material
T Connector	KF(NW)25	SUS 304



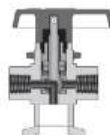
Item	Standard	Material
Blank	KF(NW)25	SUS 304



Item	Standard	Material
Ball Valve 20A	KF(NW)16	SUS 304
Ball Valve 25A	KF(NW)25	



Item	Standard	Material
Vacuum Hose 8x15mm (IDxOD)	Length 5M	SUS 316



Item	Standard	Material
1/4"PT 3-Way Valve	KF(NW)16	SUS 316



Item	Standard	Material
Connector & Tube Lock	KF(NW)25 1/4"Tube	SUS 316
	KF(NW)25 1/8"Tube	

Vacuum Pump

SH-V Series



Option: Oil Mist Trap with Filter
VOT-1020 : V10, V20
VOT-4080 : V40, V60
VOT-100 : V100, V170



Option: Filter for Oil Mist Trap
VOTF-1020 : VOT-1020
VOTF-4080 : VOT-4080
VOTF-100 : VOT-100

Model	SH-V10	SH-V20	SH-V40
Pumping Speed(50/60Hz)	78L/min	156L/min	312L/min
Gas ballast close	5x10 ⁻¹ Pa (3.7x10 ⁻³ torr)	5x10 ⁻¹ Pa (3.7x10 ⁻³ torr)	4x10 ⁻¹ Pa (3.0x10 ⁻³ torr)
Gas ballast open	10 Pa (7.5x10 ⁻² torr)	10 Pa (7.5x10 ⁻² torr)	8x10 ⁻¹ Pa (6.0x10 ⁻³ torr)
Noise level(dB)	≤56	≤56	≤58

Model	SH-V60	SH-V100	SH-V170
Pumping Speed(50/60Hz)	474L/min	960L/min	1,698L/min
Gas ballast close	4x10 ⁻¹ Pa (3.0x10 ⁻³ torr)	4x10 ⁻¹ Pa (3.0x10 ⁻³ torr)	4x10 ⁻¹ Pa (3.0x10 ⁻³ torr)
Gas ballast open	8x10 ⁻¹ Pa (6.0x10 ⁻³ torr)	8x10 ⁻¹ Pa (6.0x10 ⁻³ torr)	8x10 ⁻¹ Pa (6.0x10 ⁻³ torr)
Noise level(dB)	≤58	≤ 62	≤65

SH-VDC (Low noise) Series



Low Noise



Option: Oil Mist Trap with Filter
VOT-1020 : V10, V20
VOT-4080 : V40, V60



Option: Filter for Oil Mist Trap
VOTF-1020 : VOT-1020
VOTF-4080 : VOT-4080

Model	SH-VDC7	SH-VDC10	SH-VDC20
Pumping Speed(50/60Hz)	66L/min	132L/min	183L/min
Gas ballast close	5x10 ⁻¹ Pa (3.7x10 ⁻³ torr)	5x10 ⁻¹ Pa (3.7x10 ⁻³ torr)	5x10 ⁻¹ Pa (3.7x10 ⁻³ torr)
Gas ballast open	10 Pa (7.5x10 ⁻² torr)	10 Pa (7.5x10 ⁻² torr)	10 Pa (7.5x10 ⁻² torr)
Noise level(dB)	≤50	≤50	≤50
Recommended for	Vacuum Drying Ovens (SH-VDO-08NG, SH-VDO-30NG, SH-VDO-30NS) Tube Furnace All Packages	Vacuum Drying Ovens (SH-VDO-08NG, SH-VDO-30NG, SH-VDO-70NG, SH-VDO-30NS, SH-VDO-70NS) Tube Furnace All Packages	Vacuum Drying Ovens (SH-VDO-70NG, SH-VDO-125NG, SH-VDO-216NG, SH-VDO-70NS, SH-VDO-125NS) Tube Furnace All Packages

SH-CV20



For Chemical



Option: Oil Mist Trap with Filter
VOT-4080 : V40, V60



Option: Filter for Oil Mist Trap
VOTF-4080 : VOT-4080

Model	SH-CV20
Pumping Speed(50/60Hz)	167L/min
Ultimate Total Pressure without gas ballast	5x10 ⁻¹ Pa
Ultimate Total Pressure with gas ballast	3 Pa
Noise level(dB)	≤52

PTFE Diaphragm Pump

SH-PTFE Series

NEW

- PTFE 재질사용으로 내화학성이 우수
- 진공흡입량 조절가능
- 두개의 흡입port로 동시에 두개 챔버 사용가능
- 경량 및 저소음으로 사용 편의
- 드라이 진공 (Oil-less)으로 쾌적한 작업환경 유지



For Chemical

Model	SH-PTFEVP30	PTFEVP60
Pumping Speed(50/60Hz)	33L/min	66L/min
Ultimate pressure(mbar)	6 Torr (8 mbar)	1.5 Torr (2 mbar)
Safety function	Variable frequency permanent magnet motor, motor automatic protection	
Inlet / outlet	PTFE	
Dimensions	245x252x185 mm	335x252x185 mm
Power Supply(w)	180 W	360 W
Cooling Column / For solvent recovery	NA	



NEW



For Chemical

- 펌프 전단에 Cooling Column 장착
- 진공증발되는 유기용매 회수 가능

Model	SH-PTFEVP30C	PTFEVP60C
Pumping Speed(50/60Hz)	33L/min	66L/min
Ultimate pressure(mbar)	6 Torr (8 mbar)	1.5 Torr (2 mbar)
Safety function	Variable frequency permanent magnet motor, motor automatic protection	
Inlet / outlet	PTFE	
Dimensions	245x252x185 mm	335x252x185 mm
Power Supply(w)	180 W	360 W
Cooling Column / For solvent recovery	YES	



For Chemical



NEW

Model	SH-PTFEVP130
Pumping Speed(50/60Hz)	133L/min
Ultimate pressure(mbar)	22.5 Torr (30 mbar)
Safety function	Variable frequency permanent magnet motor, motor automatic protection
Inlet / outlet	PTFE
Dimensions	375x200x310 mm
Power Supply(w)	400 W
Cooling Column / For solvent recovery	NA

NEW



Model	SH-PTFEVP130S
Pumping Speed(50/60Hz)	133L/min
Ultimate pressure(mbar)	0.45 Torr (6 mbar)
Safety function	Variable frequency permanent magnet motor, motor automatic protection
Inlet / outlet	PTFE
Dimensions	467x234x319mm
Power Supply(w)	600 W
Cooling Column / For solvent recovery	NA

WELCH



Optimized for Evaporator

Dry Fast Diaphragm Pump

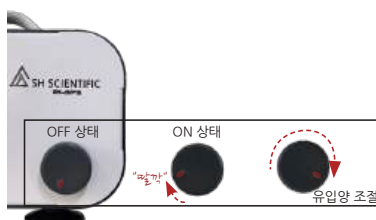
Model	2047
Pumping Speed(50/60Hz)	70L/min
Ultimate Pressure	35 torr
Adjustable Vac. / Gas Ballast	Yes
Recommended for	Rotary evaporator 50~100L

Multistage DryRoots Vacuum Pump



Model	NeoDry15G
Max. Pumping Speed	250 L/min
Ultimate Pressure (Without gas ballast)	0.008 torr (1.0 Pa)
Pumping type	Dry root
Gas ballast N ₂ purge unit (Option)	Gas ballast 밸브 open 시 질소가스 투입가능
Max. allowable moisture	275 g/h
Noise value	≤45 dB(A)

SH-dVP Series



Model	SH-dVP 10	SH-dVP 15	SH-dVP 30
Type	진공(감압)	진공(감압)	컴프레사(가압), 진공(감압)
Pumping Speed(50/60Hz)	10 L/min	0~15 L/min	30 L/min
Vacuum Range	198mmHg (torr) , -0.075MPa	160mmHg (torr) , -0.08MPa	123mmHg(torr) , -0.085MPa
Noise	≤ 40 db	≤ 45 db	≤ 50 db

Anti-Corrosive Vacuum Diaphragm Pump



Model	SH-CDP30
Pump Head	2
Pumping Speed(50/60Hz)	30L/Min
Vacuum	0.02MPa, 150mmHg (torr)
Pressure	30 Psi, 1,551mmHg (torr)

Anti-Corrosive Water Jet Vacuum Pump



Model	SH-WV20
Max. Vacuum Degree	25mmHg(torr), - 0.098Mpa
Pumping Speed(50/60Hz)	20L/Min
Pump Power	180W
Water Bath Volume	15Lit
Flow Rate	80L/Min

Dry Screw Vacuum Pump

NEW

SH-DSV (Oil free) Series



<SH-DSV330>



Model		SH-DSV130	SH-DSV330	SH-DSV530	SH-DSV830
Pumping Speed		130L/min	330L/min	530L/min	830L/min
Extreme vacuum degree		3 Pa (2.25x10 ⁻² Torr / 3x10 ⁻² mbar)			
Gas Purge	Dry nitrogen/Dry compressed air	NA	0~100 slm		
	Pressure		0.05~0.1 Mpa		
	Joint		NA	One touch fitting Φ6mm	
Connection	Inlet port	25 KF			40 KF
	Exhaust port	16 KF	25 KF		25 KF
Max water vapor pumping rate		134 g/h	200 g/h		250 g/h
Oil quantity		80 ml			
Noise		≤58 dB(A)	≤60 dB(A)		≤63 dB(A)
Ambient temperature		5~40 °C			
Operating humidity		Below 90%			
Storage temperature		0~50 °C			
Cooling mode		Air cooling			

SH-DV (Oil free) Series



<SH-DV300, 600>

Model	SH-DV300	SH-DV600
Pumping Speed(50/60Hz)	305L/min	623L/min
Ultimate Pressure	2.5 Pa ($\leq 1.9 \times 10^{-2}$ torr)	0.9 Pa ($\leq 7.5 \times 10^{-3}$ torr)
Noise Level (dB)	≤ 60	≤ 61
Recommended for	Vacuum Drying Ovens (SH-VDO-125NG, SH-VDO-216NG, SH-VDO-175NG, SH-VDO-252NG, SH-VDO-125NS)	Vacuum Drying Ovens (SH-VDO-125NG, SH-VDO-216NG, SH-VDO-175NG, SH-VDO-252NG, SH-VDO-125NS)

High Vacuum Pump

Diffusion & Oil Rotary Pump

NEW

HV40



HV40T



Model	SH-HV40	SH-HV40T
Type	Diffusion & Oil Rotary	6" Turbo & Oil Rotary
Pumping Speed(50/60Hz)	312L/min	
Ultimate Total Pressure	2.6×10^{-4} Pa (2.0×10^{-6} Torr)	4.6×10^{-5} Pa (3.5×10^{-7} Torr)
Inlet Port & Outlet Port	Inlet : KF25, KF40, KF50 / Outlet : KF25	Vacuum: 4 inch (ISO 100) / Exhaust Port KF25
Air In Port	6Ø One Touch Type (압력 5kg 이상 필요)	
Water In/out Port	12Ø One Touch Type	
Diffusion Oil Capacity	0.34 L	-
Rotary Oil Capacity	0.9 L ~ 1.5 L	0.9 L ~ 1.5 L
Ambient Temperature	5~40°C	
Noise	≤ 52 dB	

HV100



Model	SH-HV100
Type	Diffusion & Oil Rotary
Pumping Speed(50/60Hz)	960L/min
Ultimate Total Pressure	2.6×10^{-4} Pa (2.0×10^{-6} Torr)
Inlet Port & Outlet Port	Vacuum: 4 inch (ISO 100) Exhaust Port KF40
Air In Port	6Ø One Touch Type (압력 5kg 이상 필요)
Water In/out Port	12Ø One Touch Type
Diffusion Oil Capacity	0.5 L
Rotary Oil Capacity	3.3 L ~ 4.5 L
Ambient Temperature	5~40°C
Noise	≤ 62 dB

Vacuum Filtration Apparatus



SH-VFAL Series



SH-VFA Series

Model	Receiving Flask	Sand Core Diameter	Filter Membrane	Sand Core Aperture
SH-VFA500	500ml	40mm	47-55mm 가능	10 μ m
SH-VFA1000	1000ml	40mm	47-55mm 가능	10 μ m
SH-VFA2000	2000ml	40mm	47-55mm 가능	10 μ m
SH-VFA5000	5000ml	40mm	47-55mm 가능	30-50 μ m

Model	Receiving Flask	Sand Core Diameter	Filter Membrane	Sand Core Aperture
SH-VFAL2000	2000ml	80mm	90-100mm 가능	10 μ m
SH-VFAL5000	5000ml	80mm	90-100mm 가능	10 μ m
SH-VFAL10000	10000ml	80mm	90-100mm 가능	10 μ m



SH-VFAS250

- Borosilicate 3.3 grade glass
- Tight sealing
- Membrane filter 25mm 사용가능
- 소량샘플 Filtration에 적합
- Silicon rubber joint



SH-dVP Series

유기용매용, Anti-Corrosive
Vacuum Diaphragm Pump

Model	Receiving Flask	Mouth Diameter	Sand Core Diameter	Filter Membrane	Sand Core Aperture
SH-VFAS250	250ml	34mm	25mm	25mm 가능	10 μ m

Vacuum Filtration Apparatus Parts



Dust Cover 포함

Item	Note
Glass Funnel 25ml	For SH-VFAS250
Glass Funnel 300ml (Dust Cover 포함)	For SH-VFA500/1000/2000
Glass Funnel 500ml (Dust Cover 포함)	For SH-VFA5000
Glass Funnel 1000ml (Dust Cover 포함)	For SH-VFAL Series



Item	Note
Aluminum Clamp	For SH-VFAS250
Aluminum Clamp	For SH-VFA Series
Aluminum Clamp	For SH-VFAL Series



Item	Note
조인트 Joint	For SH-VFAS250
조인트 Joint	For SH-VFA Series
조인트 Joint	For VFAL2000
조인트 Joint	For VFAL5000
조인트 Joint	For VFAL10000



Item	Note
Receiving Flask 250ml	For SH-VFAS250
Receiving Flask 500ml	For SH-VFA500
Receiving Flask 1000ml	For SH-VFA1000
Receiving Flask 2000ml	For SH-VFA2000
Receiving Flask 5000ml	For SH-VFA5000
Receiving Flask 2000ml	For SH-VFAL2000
Receiving Flask 5000ml	For SH-VFAL5000
Receiving Flask 10000ml	For SH-VFAL10000



Diameter 25mm

Diameter 25mm



Diameter 100mm

MCE(Mixed Cellulose Ester) Membrane Filter


Diameter 25mm

Diameter 55mm



Diameter 100mm

Nylon Membrane Filter

Item	Type	Diameter	Aperture	Packing
25mm MCE(Mixed Cellulose Ester) Membrane Filter 0.22 μ m	MCE(Mixed Cellulose Ester) Membrane Filter	25mm	0.22 μ m	100pcs/case
25mm MCE(Mixed Cellulose Ester) Membrane Filter 0.45 μ m		25mm	0.45 μ m	
50mm MCE(Mixed Cellulose Ester) Membrane Filter 0.22 μ m		50mm	0.22 μ m	50pcs/case
50mm MCE(Mixed Cellulose Ester) Membrane Filter 0.45 μ m		50mm	0.45 μ m	
100mm MCE(Mixed Cellulose Ester) Membrane Filter 0.22 μ m		100mm	0.22 μ m	
100mm MCE(Mixed Cellulose Ester) Membrane Filter 0.45 μ m		100mm	0.45 μ m	
25mm Nylon Membrane Filter 0.22 μ m	Nylon Membrane Filter	25mm	0.22 μ m	100pcs/case
25mm Nylon Membrane Filter 0.45 μ m		25mm	0.45 μ m	
50mm Nylon Membrane Filter 0.22 μ m		50mm	0.22 μ m	50pcs/case
50mm Nylon Membrane Filter 0.45 μ m		50mm	0.45 μ m	
100mm Nylon Membrane Filter 0.22 μ m		100mm	0.22 μ m	
100mm Nylon Membrane Filter 0.45 μ m		100mm	0.45 μ m	

02. Drying Oven

Forced Convection Drying Oven

Basic Model



Model	SH-DO-54FGB	SH-DO-100FGB	SH-DO-149FGB
Temp Range	Ambient +10℃ to 250℃ (Ambient +18°F to 482°F)		
Dimension Internal (W×D×H)	380×340×420mm	480×400×530mm	555×450×600mm
	14.96×13.39×16.54"	18.90×15.75×20.87"	21.85×17.72×23.62"
Dimension External (W×D×H)	535×510×795mm	635×570×905mm	710×615×975mm
	21.06×20.08×31.30"	25×22.44×35.63"	27.95×24.21×38.39"
Capa	54 L	100 L	150 L

Premium Model



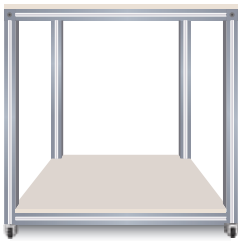
Model	SH-DO-54FG	SH-DO-100FG	SH-DO-149FG	SH-DO-250FG
Temp Range	Ambient +10℃ to 250℃ (Ambient +18°F to 482°F)			
Dimension Internal (W×D×H)	380×340×420mm	480×400×530mm	555×450×600mm	700×450×800mm
	14.96×13.39×16.54"	18.90×15.75×20.87"	21.85×17.72×23.62"	27.56×17.72×31.50"
Dimension External (W×D×H)	500×630×760mm	600×690×870mm	675×740×940mm	820×740×1140mm
	19.68×24.80×29.92"	23.62×27.16×34.25"	26.57×29.13×37.01"	32.28×29.13×44.88"
Capa	54 L	100 L	150 L	250 L

CE



Model	SH-DO-360FG	SH-DO-550FG
Temp Range	Ambient +10°C to 220°C (Ambient +18°F to 428°F)	
Dimension Internal (W×D×H)	600×600×1000mm	700×650×1200mm
	23.62×23.62×39.37"	27.56×25.59×47.24"
Dimension External (W×D×H)	800×900×1420mm	900×950×1610mm
	31.49×35.43×55.90"	35.43×37.40×63.38"
Capa	360 L	546 L

Moving Frame For Oven



Model	SH-DO-MF(Small)	SH-DO-MF(Large)
Dimension External (W×D×H)	730×790×750mm	870×790×750mm
	28.74×31.10×29.52"	34.25×31.10×29.52"
Material	AL-Frame, Foot Master Castor, PVC Plate (Option: SUS Plate)	
Application Model	SH-DO-54FGB, SH-DO-100FGB, SH-DO-149FGB, SH-DO-54FG, SH-DO-100FG, SH-DO-149FG, SH-DO-90FH, SH-DO-150FH	SH-DO-250FG

350°C Drying Oven

CE



Model	SH-DO-90FH	SH-DO-150FH
Temp Range	Ambient +10°C to 350°C (Ambient + 18°F to 662°F)	
Dimension Internal (W×D×H)	450×450×450mm	500×500×600mm
	17.72×17.72×17.72"	19.69×19.69×23.62"
Dimension External (W×D×H)	644×633×935mm	694×683×1110mm
	25.53×24.92×36.81"	27.32×26.89×43.70"
Capa	91 L	150 L

CE



Model	SH-DO-250FH
Temp Range	Ambient +10°C to 350°C (Ambient + 18°F to 662°F)
Dimension Internal (W×D×H)	700×450×800mm
	27.56×17.72×31.50"
Dimension External (W×D×H)	920×820×1230mm
	36.22×32.28×48.42"
Capa	252 L



Model	SH-DO-360FH	SH-DO-550FH
Temp Range	Ambient +10°C to 350°C (Ambient + 18°F to 662°F)	
Dimension Internal (W×D×H)	600x600x1000mm	700x650x1200mm
	23.62x23.62x39.37"	27.56x25.59x47.24"
Dimension External (W×D×H)	820x970x1430mm	920x1020x1630mm
	32.28x38.18x56.29"	36.22x40.15x64.17"
Capa	360 L	546 L

500°C Drying Oven



Model	SH-DO-90FS	SH-DO-150FS	SH-DO-216FS
Temp Range	Ambient +10°C to 500°C (Ambient + 18°F to 932°F)		
Dimension Internal (W×D×H)	450×450×450mm	500×500×600mm	600×600×600mm
	17.72x17.72x17.72"	19.69x19.69x23.62"	23.62 x23.62 x23.62"
Dimension External (W×D×H)	1196×700×925mm	1296×750×1075mm	1440x850x1075mm
	47.09x27.56x36.41"	51.02x29.53x42.32"	56.69x33.46x42.32"
Capa	91 L	150 L	216 L

Drying Oven with HEPA Filter



Model	SH-DO-90FG	SH-DO-150FG	SH-DO-251FG
Temp Range	Ambient +10°C to 250°C (Ambient + 18°F to 482°F)		
Dimension Internal (W×D×H)	450×450×450mm	600×500×500mm	700×600×600mm
	17.72x17.72x17.72"	23.62x19.69x19.69"	27.56x23.62x23.62"
Dimension External (W×D×H)	630×842×1562mm	780×892×1612mm	880×992×1712mm
	24.80x33.15x61.50"	30.71x35.12x63.46"	34.65x39.06x67.40"
Capa	90 L	150 L	250 L

03. Industrial Drying Oven



Model	SH-IDO-864FG
Temp Range	Ambient +10°C to 200°C (Ambient +18°F to 392°F)
Dimension Internal (W×D×H)	900×800×1200mm
	35.43x31.50x47.24"
Dimension External (W×D×H)	1160x1000x1870mm
	45.67x39.37x73.62"
Capa	864 L

CE



Model	SH-IDO-1152FG
Temp Range	Ambient +10°C to 200°C (Ambient +18°F to 392°F)
Dimension Internal (W×D×H)	1200×800×1200mm
	47.24×31.50×47.24"
Dimension External (W×D×H)	1480×1013×1948mm
	58.27×39.88×76.69"
Capa	1152 L

CE



Model	SH-IDO-2250FG	SH-IDO-3612FG
Temp Range	Ambient +10°C to 200°C (Ambient +18°F to 392°F)	
Dimension Internal (W×D×H)	1600×930×1500mm	2150×1120×1500mm
	62.99×36.61×59.06"	84.65×44.09×59.06"
Dimension External (W×D×H)	2300×1145×2090mm	2850×1335×2090mm
	90.55×45.08×82.28"	112.20×52.56×82.28"
Capa	2232 L	3612 L

300°C Industrial Drying Oven

CE



Model	SH-IDO-864FH
Temp Range	Ambient +10°C to 300°C (Ambient +18°F to 572°F)
Dimension Internal (W×D×H)	900×800×1200mm
	35.43×31.50×47.24"
Dimension External (W×D×H)	1160×1000×1870mm
	45.67×39.37×73.62"
Capa	864 L

CE



Model	SH-IDO-1152FH
Temp Range	Ambient +10°C to 300°C (Ambient +18°F to 572°F)
Dimension Internal (W×D×H)	1200×800×1200mm
	47.24×31.50×47.24"
Dimension External (W×D×H)	1480×1013×1948mm
	58.27×39.88×76.69"
Capa	1152 L

CE



Model	SH-IDO-2250FH	SH-IDO-3612FH
Temp Range	Ambient +10°C to 300°C (Ambient +18°F to 572°F)	
Dimension Internal (W×D×H)	1600×930×1500mm	2150×1120×1500mm
	62.99×36.61×59.06"	84.65×44.09×59.06"
Dimension External (W×D×H)	2300×1145×2090mm	2850×1335×2090mm
	90.55×45.07×82.28"	112.20×52.56×82.28"
Capa	2232 L	3612 L

04. Furnace

1050°C Muffle Furnace



*Vent port

*option
석영선반

Model	SH-FU-3MGE	SH-FU-5MGE
Max Temp	1050°C (2192°F)	
Capa	3 L	5.1 L
Dimension Internal (W×D×H)	130x220x110mm	150x260x130mm
	5.12x8.66x4.33"	5.91x10.24x5.12"
Dimension External (W×D×H)	300x380x405mm	320x430x425mm
	11.81x14.96x15.94"	12.60x16.92x16.73"

Model	SH-FU-7MGE	SH-FU-11MGE
Max Temp	1050°C (2192°F)	
Capa	7.4 L	11.2 L
Dimension Internal (W×D×H)	170x290x150mm	200x310x180mm
	6.69x11.42x5.91"	7.87x12.20x7.09"
Dimension External (W×D×H)	340x458x445mm	370x468x475mm
	13.39x18.03x17.52"	14.57x18.43x18.70"

1200°C Muffle Furnace



*Vent port

*option
석영선반

Model	SH-FU-3MG	SH-FU-5MG
Max Temp	1200°C (2192°F)	
Capa	2.9 L	4.5 L
Dimension Internal (W×D×H)	130x250x90mm	150x300x100mm
	5.12x9.87x3.54"	5.91x11.81x3.93"
Dimension External (W×D×H)	370x510x560mm	390x560x570mm
	14.57x20.07x22.05"	15.30x22.05x22.44"

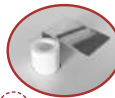
Model	SH-FU-14MG	SH-FU-27MG
Max Temp	1200°C (2192°F)	
Capa	14 L	27 L
Dimension Internal (W×D×H)	250x350x160mm	300x300x300mm
	9.84x13.78x6.30"	11.81x11.81x11.81"
Dimension External (W×D×H)	500x610x660mm	580x590x825mm
	19.69x24.01x25.98"	22.83x23.23x32.48"

Model	SH-FU-64MG	SH-FU-125MG
Max Temp	1200°C (2192°F)	
Capa	64 L	125 L
Dimension Internal (W×D×H)	400x400x400mm	500x500x500mm
	15.74x15.74x15.74"	19.68x19.68x19.68"
Dimension External (W×D×H)	1000x730x910mm	1150x850x1040mm
	39.4x28.74x35.82"	45.27x33.46x40.94"



1200°C Quartz Chamber Muffle Furnace

NEW



*Vent port

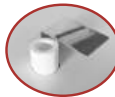


Model	SH-FU-5MGQ	SH-FU-14MGQ
Max Temp	1200°C (2192°F)	
Capa	4.5 L	14 L
Dimension Internal (W×D×H)	Quartz 150x300x100mm	Quartz 250x350x160mm
	5.91x11.81x3.93"	9.84x13.78x6.30"
Dimension External (W×D×H)	390x560x570mm	500x610x660mm
	15.30x22.05x22.44"	19.69x24.01x25.98"

Model	SH-FU-64MGQ
Max Temp	1200°C (2192°F)
Capa	64 L
Dimension Internal (W×D×H)	Quartz 400x400x400mm
	15.74x15.74x15.74"
Dimension External (W×D×H)	1000x730x910mm
	39.4x28.74x35.82"

1500°C Muffle Furnace

CE



*Vent port



Model	SH-FU-4MH	SH-FU-11MH
Max Temp	1500°C (2732°F)	
Capa	4.5 L	11 L
Dimension Internal (W×D×H)	150x200x150mm	200x270x200mm
	5.91x7.87x5.91"	7.87x10.63x7.87"
Dimension External (W×D×H)	820x550x680mm	870x620x740mm
	32.28x21.65x26.77"	34.25x24.41x29.13"

Model	SH-FU-22MH	SH-FU-36MH
Max Temp	1500°C (2732°F)	
Capa	22 L	36 L
Dimension Internal (W×D×H)	250x350x250mm	300x400x300mm
	9.84x13.78x9.84"	11.81x15.75x11.81"
Dimension External (W×D×H)	920x700x870mm	970x750x900mm
	36.22x27.56x34.25"	38.19x29.53x35.43"

1700°C / 1800°C Muffle Furnace

CE



*Vent port



Model	SH-FU-4MS	SH-FU-11MS	SH-FU-22MS
Max Temp	1550°C (Mosi 2 1700°C)		
	1650°C (Mosi 2 1800°C)		
Capa	4.5 L	11 L	22 L
Dimension Internal (W×D×H)	150x200x150mm	200x270x200mm	250x350x250mm
	5.91x7.87x5.91"	7.87x10.63x7.87"	9.84x13.78x9.84"
Dimension External (W×D×H)	920x610x770mm	1000x680x840mm	1080x760x910mm
	36.22x24.01x30.31"	39.37x26.77x33.07"	42.52x29.92x35.83"

1900°C Muffle Furnace



*Vent port



Model	SH-FU-4MS	SH-FU-11MS	SH-FU-22MS
Max Temp	1750°C (Mosi 2 1900°C)		
Capa	4.5 L	11 L	22 L
Dimension Internal (W×D×H)	150x200x150mm	200x270x200mm	250x350x250mm
	5.91x7.87x5.91"	7.87x10.63x7.87"	9.84x13.78x9.84"
Dimension External (W×D×H)	920x610x770mm	1000x680x840mm	1080x760x910mm
	36.22x24.01x30.31"	39.37x26.77x33.07"	42.52x29.92x35.83"

1200°C Vacuum Muffle Furnace



NEW



Option



<MFC>

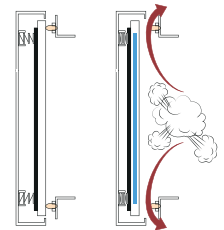


<BPR>



<Vacuum Pump>

• Over pressure safety door



Model	SH-FU-1.5MGV	SH-FU-10MGV	SH-FU-31MGV
Max Temp	1200°C (2192°F)		
Capa	1.5L	10L	31L
Dimension Internal (W×D×H)	115x115x115mm	215x215x215mm	315x315x315mm
	4.53x4.53x4.53"	8.46x8.46x8.46"	12.40x12.40x12.40"
Dimension External (W×D×H)	740x610x590 mm	840x610x700mm	940x710x820mm
	29.13x24.01x23.23"	33.07x24.01x27.56"	37.01x27.95x32.28"

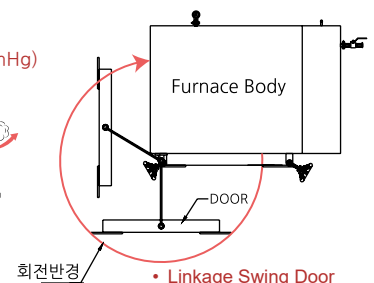
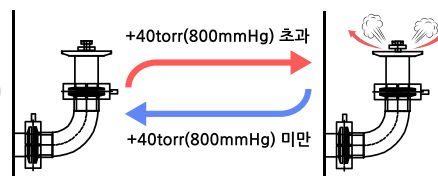
NEW



• 이중 과압 안전장치 : 과압방지용 Relief Port(KF25)에 Relief Valve를 장착하여 +40torr (800mmHg) 자동압력해제 및 유지



Relief valve

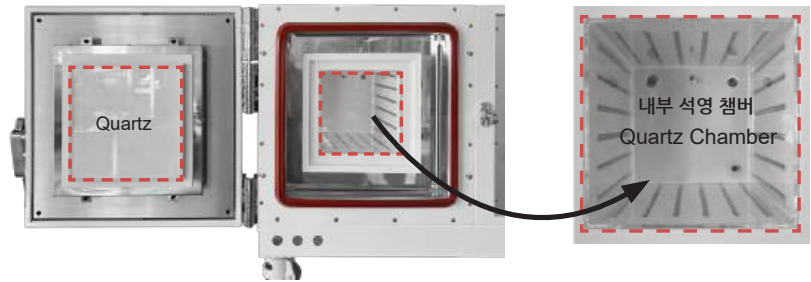


• Linkage Swing Door

Model	SH-FU-64MGV	SH-FU-125MGV
Max Temp	1200°C (2192°F)	
Capa	64 L	125 L
Dimension Internal (W×D×H)	400x400x400mm	500x500x500mm
	15.74x15.74x15.74"	19.68x19.68x19.68"
Dimension External (W×D×H)	1488x1169x1239mm	1588x1269x1339mm
	58.58x46.02x48.78"	62.52x49.96x52.71"

1200°C Vacuum Furnace with Quartz Chamber

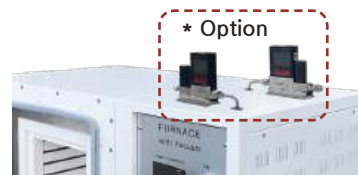
CE



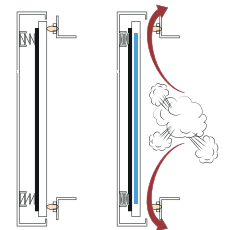
Model	SH-FU-1.5MGVQ	SH-FU-10MGVQ	SH-FU-31MGVQ
Max Temp	1200°C (2192°F)		
Temp Controller	Programmable Controller(SP590)		
Capa	1.5L	10L	31L
Dimension Internal (W×D×H)	(석영챔버 3T) 109x109x109mm	(석영챔버 5T) 205x205x205mm	(석영챔버 5T) 305x305x305mm
	4.29x4.29x4.29"	8.07x8.07x8.07"	12.00x12.00x12.00"
Dimension External (W×D×H)	740x610x590 mm	840x610x700mm	940x710x820mm
	29.13x24.01x23.23"	33.07x24.01x27.56"	37.01x27.95x32.28"

1500°C Vacuum Muffle Furnace

CE

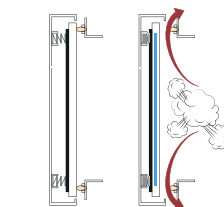


• Over pressure safety door



Model	SH-FU-4.5MHV	SH-FU-18.7MHV
Operating Temperature Rang	With Only Vacuum - 1150°C or below With Only Gas - 1350°C or below	
Capa	4.5 L	18.7 L
Dimension Internal (W×D×H)	150x200x150mm	250x300x250mm
	5.91x7.87x5.91"	9.84x11.81x9.84"
Dimension External (W×D×H)	630x590x830mm	1050x900x940mm
	24.80x23.23x32.68"	41.33x35.43x37.00"

1900°C Vacuum Muffle Furnace

Cooling Water Inlet & Outlet for
Protecting door sealing gasket

• Over pressure safety door

Cooling Water Inlet & Outlet for
Accelerated Body Cooling (Option)

Model	SH-FU-4.5MSV	SH-FU-10MSV	SH-FU-18.7MSV
Operating Temperature Rang	With Only Vacuum - 1400°C or below With Only Gas - 1700°C or below		
Volume	4.5 L	10 L	20 L
Dimension Internal (W×D×H)	150x200x150mm	200x250x200 mm	200x350x250 mm
	5.9x7.9x5.9"	7.9x9.8x7.9"	9.8x13.8x9.8"
Dimension External (W×D×H)	1020x670x770 mm	1070x770x820 mm	1120x870x890 mm
	40.2x26.4x30.3"	42.1x30.3x32.3"	44.1x34.3x35.0"

2400°C / 3000°C Furnace



3000°C Induction Furnace

Model		SH-FU-1.8MS3000	SH-FU-9.4MS3000	SH-FU-35MS3000
Temp Range	Max Temp	3000°C (5432°F)		
	Running Max Temp	With Only Vacuum 2400°C / With Only Gas 2700°C		
Dimension Internal (D×H)		100Φx150H mm	200Φx300H mm	300Φx500H mm
Heater & Insulation		Graphite Heating & Graphite (Hard & Soft)		
Door Open Type		Top Door open by cylinder		
Vacuum System		Diffusion + Roots + Oil Rotary Pump / 6x 10(-5) Torr		

2400°C Graphite Furnace

Model		SH-FU-4MS2400	SH-FU-14MS2400	SH-FU-25MS2400
Temp Range	Max Temp	2400°C (4352°F)		
	Running Max Temp	With Only Vacuum 2000°C / With Only Gas 2200°C		
Dimension Internal (D×H)		160Φx200H mm	260Φx200H mm	320Φx200H mm
Heater & Insulation		Graphite Heating & Graphite (Hard & Soft)		
Door Open Type		Top Door open by cylinder		
Vacuum System		Diffusion + Roots + Oil Rotary Pump		

1200°C Elevator Furnace



Model		SH-FU3.5MGU	SH-FU6.5MGU
Max Temp		1200°C (2192°F)	
Capa		3.5 L	6.5 L
Plate Size (mm)		140x140mm	160x160mm
Dimension Internal (W×D×H)		150x150x180mm	170x170x200mm
		5.90x5.90x7.08"	6.69x6.69x7.87"
Dimension External (W×D×H)		405x490x720mm	425x510x760mm
		15.95x19.29x28.34"	16.73x20.08x29.92"

1500°C Elevator Furnace



Model	SH-FU-2MHU	SH-FU-4MHU	SH-FU-6MHU
Max Temp	1500°C (2732°F)		
Capa	2.2 L	3.8 L	6.2 L
Plate Size (mm)	120X120mm	140X140mm	160X160mm
Dimension Internal (W×D×H)	130X130X150mm	150x150x170mm	170X170X190mm
	5.12x5.12x5.90"	5.90x5.90x6.69"	6.69x6.69x7.48"
Dimension External (W×D×H)	445X600X850mm	465X600X890mm	485X600X930mm
	17.52x23.62x33.46"	18.31x23.62x35.04"	19.09x23.62x36.61"

1700°C / 1800°C Elevator Furnace



Model(1700°C/1800°C)	SH-FU-2MSU	SH-FU-4MSU	SH-FU-6MSU
Max Temp	1700°C (3092°F) / 1800°C (3272°F)		
Capa	2.2 L	3.8 L	6.2 L
Plate Size (mm)	Φ120mm	Φ140mm	Φ160mm
Dimension Internal (W×D×H)	130X130X150mm	150x150x170mm	170X170X190mm
	5.12x5.12x5.90"	5.90x5.90x6.69"	6.69x6.69x7.48"
Dimension External (W×D×H)	380X550X950mm	400X570X1030mm	450X620X1110mm
	14.96x21.65x37.40"	15.74x22.44x40.55"	17.72x24.41x43.70"

1200°C RTCVD Tube Furnace (Gas Flow Package)

300mm Length Hot Zone

NEW



Model	SH-RTCVD-100TG300
Max Temp	1200°C (2192°F)
Applicable Tube Diameter	100 Φ
Dimension Hot Zone	Length 300 mm
Dimension External (W×D×H)	1625x470x1128mm
	63.97x18.50x44.10"

200mm Length x 3 Hot Zone

NEW

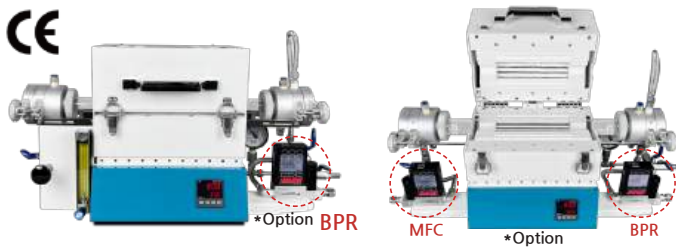


Model	SH-RTCVD-100TG200-3
Max Temp	1200°C (2192°F)
Applicable Tube Diameter	100 Φ
Dimension Hot Zone	200mm x 3 Hot Zone = Length 600 mm
Dimension External (W×D×H)	1950x470x1128mm
	76.77x18.50x44.10"

1200°C CVD Tube Furnace (Gas Flow Package)

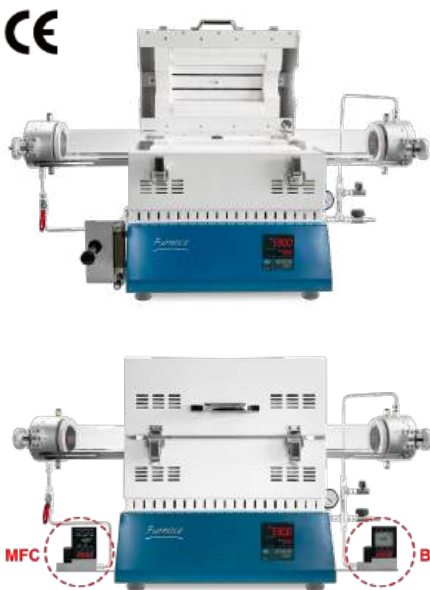
150mm Length Hot Zone

NEW



Model	SH-CVD-30TG150
Furnace	SH-FU-30TG150
Components	Gas Flowmeter 1EA
	Stainless Steel Pipe & Connector
	Gas Sealing Mask
	Quartz Tube (Length 500 mm)

300mm Length Hot Zone



Option



Model	SH-CVD-50TG300	SH-CVD-80TG300	SH-CVD-100TG300
Furnace	SH-FU-50TG300	SH-FU-80TG300	SH-FU-100TG300
Components	Gas Flowmeter 1EA		
	Stainless Steel Pipe & Connector		
	Gas Sealing Mask		
	Quartz Tube (Length 1000 mm)		

Model	SH-CVD-120TG300	SH-CVD-200TG300	SH-CVD-250TG300
Furnace	SH-FU-120TG300	SH-FU-200TG300	SH-FU-250TG300
Components	Gas Flowmeter 1EA		
	Stainless Steel Pipe & Connector		
	Gas Sealing Mask		
	Quartz Tube (Length 1000 mm)	Quartz Tube (Length 1200 mm)	

600mm Length Hot Zone



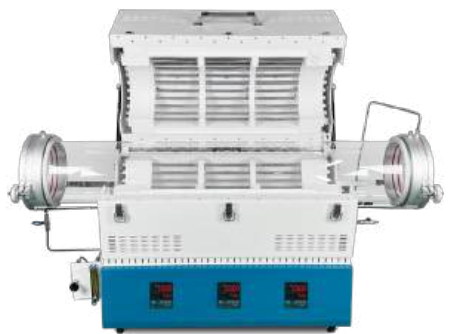
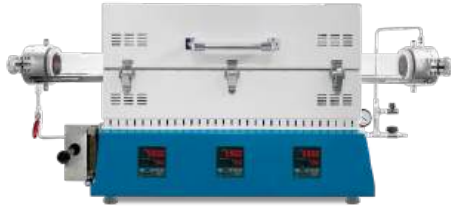
Option



Model	SH-CVD-50TG600	SH-CVD-80TG600	SH-CVD-100TG600
Furnace	SH-FU-50TG600	SH-FU-80TG600	SH-FU-100TG600
Components	Gas Flowmeter 1EA		
	Stainless Steel Pipe & Connector		
	Gas Sealing Mask		
	Quartz Tube (Length 1200 mm)		

Model	SH-CVD-120TG600	SH-CVD-200TG600	SH-CVD-250TG600
Furnace	SH-FU-120TG600	SH-FU-200TG600	SH-FU-250TG600
Components	Gas Flowmeter 1EA		
	Stainless Steel Pipe & Connector		
	Gas Sealing Mask		
	Quartz Tube (Length 1200 mm)	Quartz Tube (Length 1500 mm)	

200mm Length x 3 Hot Zone



Option



Model	SH-CVD-50TG200-3	SH-CVD-80TG200-3
Furnace	SH-FU-50TG200-3	SH-FU-80TG200-3
Components	Gas Flowmeter 1EA	
	Stainless Steel Pipe & Connector	
	Gas Sealing Mask	
	Quartz Tube (Length 1200 mm)	

Model	SH-CVD-100TG200-3	SH-CVD-120TG200-3
Furnace	SH-FU-100TG200-3	SH-FU-120TG200-3
Components	Gas Flowmeter 1EA	
	Stainless Steel Pipe & Connector	
	Gas Sealing Mask	
	Quartz Tube (Length 1200 mm)	

Model	SH-CVD-200TG200-3	SH-CVD-250LTG200-3
Furnace	SH-FU-200TG200-3	SH-FU-250TG200-3
Components	Gas Flowmeter 1EA	
	Stainless Steel Pipe & Connector	
	Gas Sealing Mask	
	Quartz Tube (Length 1500 mm)	

1500°C CVD Tube Furnace (Gas Flow Package)



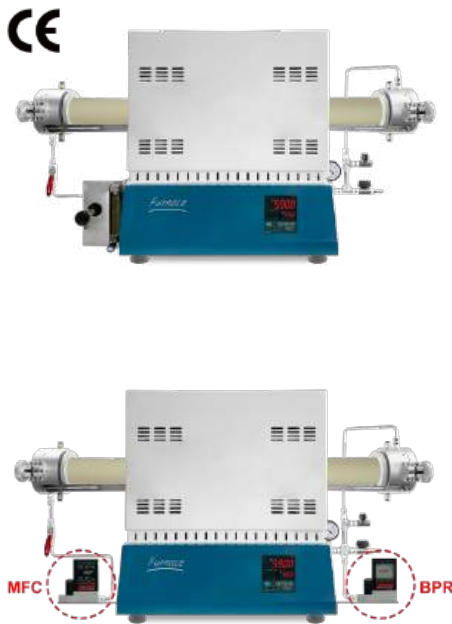
Option



Model	SH-CVD-50TH300	SH-CVD-80TH300
Furnace	SH-FU-50TH300	SH-FU-80TH300
Components	Gas Flowmeter 1EA	
	Stainless Steel Pipe & Connector	
	Gas Sealing Mask	
	Alumina Tube (Length 1000 mm)	

Model	SH-CVD-100TH300	SH-CVD-120TH300
Furnace	SH-FU-100TH300	SH-FU-120TH300
Components	Gas Flowmeter 1EA	
	Stainless Steel Pipe & Connector	
	Gas Sealing Mask	
	Alumina Tube (Length 1000 mm)	

1700°C/1800°C CVD Tube Furnace (Gas Flow Package)



Option



Model	SH-CVD-50TS300	SH-CVD-80TS300
Furnace	SH-FU-50TS300	SH-FU-80TS300
Components	Gas Flowmeter 1EA	
	Stainless Steel Pipe & Connector	
	Gas Sealing Mask	
	Alumina Tube (Length 1000 mm)	

Model	SH-CVD-100TS300	SH-CVD-120TS300
Furnace	SH-FU-100TS300	SH-FU-120TS300
Components	Gas Flowmeter 1EA	
	Stainless Steel Pipe & Connector	
	Gas Sealing Mask	
	Alumina Tube (Length 1000 mm)	

CVD Tube Furnace Package

1200°C Furnace

150mm Length Hot Zone



Option



Package Model	SH-CVD-PK-30TG150
Furnace Model	SH-CVD-30TG150
Chiller	SH-WB-5CDR(-15)
Vacuum Pump	SH-VDC7
Oil Mist Trap	VOT-1020
Dimension	980x734x988mm
	38.58x28.89x38.89"

300mm Length Hot Zone



〈CVD-PK-50/80/100TG300〉



〈CVD-PK-200/250TG300〉

Option



〈MFC〉



〈BPR〉



〈Bullseye〉



〈가스수분제거 Column 설치〉



〈Safety Cover〉



〈Quartz wafer carrier〉



〈Quartz Gas Diffuser〉

Package Model	SH-CVD-PK-50TG300	SH-CVD-PK-80TG300	SH-CVD-PK-100TG300
Furnace Model	SH-CVD-50TG300	SH-CVD-80TG300	SH-CVD-100TG300
Chiller	SH-WB-5CDR(-15)	SH-WB-5CDR(-15)	SH-WB-5CDR(-15)
Vacuum Pump	SH-V10	SH-V10	SH-V10
Oil Mist Trap	VOT-1020	VOT-1020	VOT-1020
Dimension	1265x725x1420mm	1265x725x1440mm	1265x800x1495mm
	49.80x28.54x55.90"	49.80x28.54x56.69"	49.80x31.49x58.85"

Package Model	SH-CVD-PK-120TG300	SH-CVD-PK-200TG300	SH-CVD-PK-250TG300
Furnace Model	SH-CVD-120TG300	SH-CVD-200TG300	SH-CVD-250TG300
Chiller	SH-WB-5CDR(-15)	SH-WB-12CDR(-25)	SH-WB-12CDR(-25)
Vacuum Pump	SH-V10	SH-V20	SH-V20
Oil Mist Trap	VOT-1020	VOT-1020	VOT-1020
Dimension	1265x800x1570mm	1265x800x1600mm	1265x800x1650mm
	49.80x31.49x61.81"	49.80x31.49x61.81"	49.80x31.49x64.96"

600mm Length Hot Zone



〈CVD-PK-50/80/120TG600〉



〈CVD-PK-200/250TG600〉

Option



〈MFC〉



〈BPR〉



〈Bullseye〉



〈가스수분제거 Column 설치〉



〈Safety Cover〉



〈Quartz wafer carrier〉



〈Quartz Gas Diffuser〉

Package Model	SH-CVD-PK-50TG600	SH-CVD-PK-80TG600	SH-CVD-PK-100TG600
Furnace Model	SH-CVD-50TG600	SH-CVD-80TG600	SH-CVD-100TG600
Chiller	SH-WB-5CDR(-15)	SH-WB-5CDR(-15)	SH-WB-5CDR(-15)
Vacuum Pump	SH-V10	SH-V10	SH-V10
Oil Mist Trap	VOT-1020	VOT-1020	VOT-1020
Dimension	1465x725x1420mm	1465x725x1440mm	1465x800x1495mm
	49.80x28.54x55.90"	49.80x28.54x56.69"	49.80x31.49x58.85"

Package Model	SH-CVD-PK-120TG600	SH-CVD-PK-200TG600	SH-CVD-PK-250TG600
Furnace Model	SH-CVD-120TG600	SH-CVD-200TG600	SH-CVD-250TG600
Chiller	SH-WB-5CDR(-15)	SH-WB-12CDR(-25)	SH-WB-12CDR(-25)
Vacuum Pump	SH-V10	SH-V20	SH-V20
Oil Mist Trap	VOT-1020	VOT-1020	VOT-1020
Dimension	1465x800x1570mm	1465x800x1600mm	1465x800x1650mm
	49.80x31.49x61.81"	57.67x31.49x61.81"	57.67x31.49x64.96"

200mm Length x 3 Hot Zone



〈CVD-PK-50/80/120TG200-3〉



〈CVD-PK-200/250TG200-3〉

Option



Package Model	SH-CVD-PK-50TG200-3	SH-CVD-PK-50TG200-3
Furnace Model	SH-CVD-50TG200-3	SH-CVD-80TG200-3
Chiller	SH-WB-5CDR(-15)	SH-WB-5CDR(-15)
Vacuum Pump	SH-V10	SH-V10
Oil Mist Trap	VOT-1020	VOT-1020
Dimension	1515x725x1420mm	1515x725x1440mm
	59.64x28.54x55.90"	59.64x28.54x56.69"

Package Model	SH-CVD-PK-100TG200-3	SH-CVD-PK-120TG200-3
Furnace Model	SH-CVD-100TG200-3	SH-CVD-120TG200-3
Chiller	SH-WB-5CDR(-15)	SH-WB-5CDR(-15)
Vacuum Pump	SH-V10	SH-V10
Oil Mist Trap	VOT-1020	VOT-1020
Dimension	1515x800x1495mm	1515x800x1570mm
	59.64x31.49x58.85"	59.64x31.49x61.81"

Package Model	SH-CVD-PK-200TG200-3	SH-CVD-PK-250TG200-3
Furnace Model	SH-CVD-200TG200-3	SH-CVD-250TG200-3
Chiller	SH-WB-12CDR(-25)	SH-WB-12CDR(-25)
Vacuum Pump	SH-V20	SH-V20
Oil Mist Trap	VOT-1020	VOT-1020
Dimension	1515x800x1600mm	1515x800x1650mm
	59.64x31.49x62.99"	59.64x31.49x64.96"

1500°C Furnace



〈CVD-PK-50/80TH300〉



〈CVD-PK-100/120TH300〉

Option



Package Model	SH-CVD-PK-50TH300	SH-CVD-PK-80TH300
Furnace Model	SH-CVD-50TH300	SH-CVD-80TH300
Chiller	SH-WB-5CDR(-15)	SH-WB-5CDR(-15)
Vacuum Pump	SH-V10	SH-V10
Oil Mist Trap	VOT-1020	VOT-1020
Dimension	1265x725x1485mm	1265x725x1515mm
	49.80x28.54x58.46"	49.80x28.54x59.64"

Package Model	SH-CVD-PK-100TH300	SH-CVD-PK-120TH300
Furnace Model	SH-CVD-100TH300	SH-CVD-120TH300
Chiller	SH-WB-5CDR(-30)	SH-WB-5CDR(-30)
Vacuum Pump	SH-V10	SH-V10
Oil Mist Trap	VOT-1020	VOT-1020
Dimension	1265x800x1580mm	1265x800x1670mm
	49.80x31.49x62.20"	49.80x31.49x65.74"

1700°C / 1800°C Furnace



Option



<MFC>



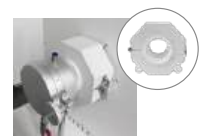
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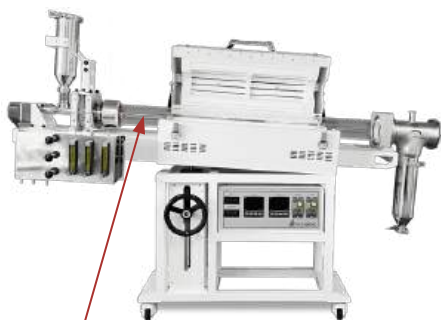


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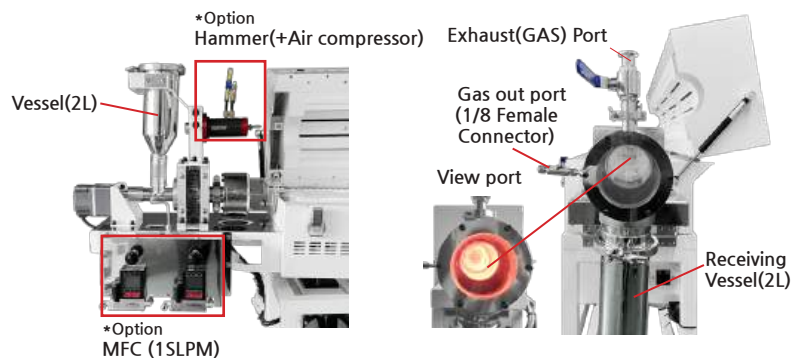
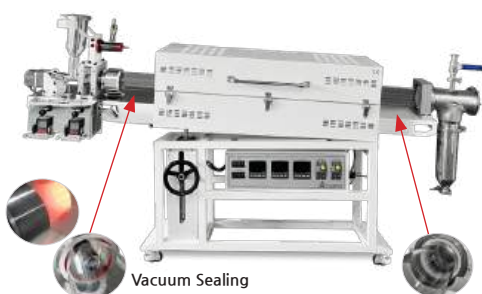
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Furnace Model	SH-CVD-50TS300	SH-CVD-80TS300
Chiller	SH-WB-5CDR(-15)	SH-WB-5CDR(-15)
Vacuum Pump	SH-V10	SH-V10
Oil Mist Trap	VOT-1020	VOT-1020
Dimension (Frame)	1265x725x1885mm	1265x725x1915mm
	49.80x28.54x74.21"	49.80x28.54x75.39"

Package Model	SH-CVD-PK-100TS300	SH-CVD-PK-120TS300
Furnace Model	SH-CVD-100TS300	SH-CVD-120TS300
Chiller	SH-WB-12CDR(-25)	SH-WB-12CDR(-25)
Vacuum Pump	SH-V10	SH-V10
Oil Mist Trap	VOT-1020	VOT-1020
Dimension (Frame)	1265x800x1970mm	1265x800x2050mm
	49.80x31.49x77.56"	49.80x31.49x80.71"

1200°C Rotary Kiln Furnace (Small Scale Rotary Kiln)

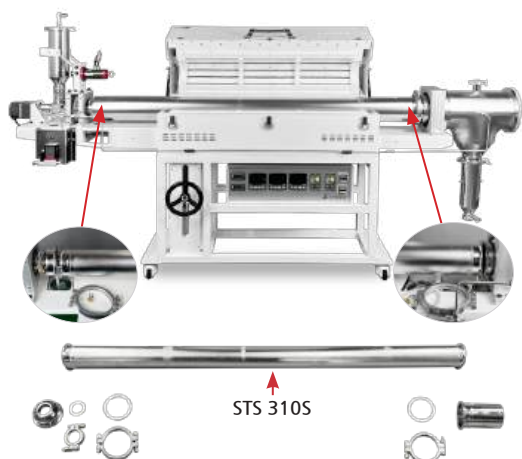


*Option Heating jacket(left, right) with Controller

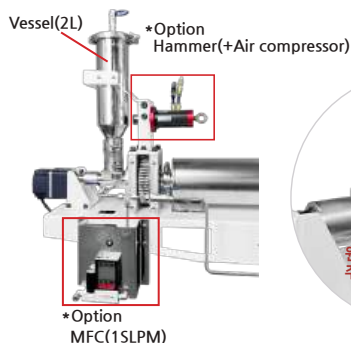


Model	SH-FU-100RKG600	SH-FU-120RKG600
Max Temp	1200°C (2192°F)	
Dimension Hot Zone	300mm x 2 Hot Zone=Length 600 mm	
Applicable Tube Diameter & length	Quartz 100 ϕ x 1480 mm	Quartz 120 ϕ x 1480 mm
Dimension External (WxDxH)	2295x510x1345mm	2295x530x1545mm
	90.35x20.07x52.95"	90.35x20.86x60.82"
Tube Rotation Speed	0 ~ 14 RPM	
Feeder speed	0 ~ 8 RPM	
Tube Angle	0 ~ 8°	

Model	SH-FU-100RKG900	SH-FU-120RKG900
Max Temp	1200°C (2192°F)	
Dimension Hot Zone	300mm x 3 Hot Zone=Length 900 mm	
Applicable Tube Diameter & length	Quartz 100 ϕ x 1800 mm	Quartz 120 ϕ x 1800 mm
Dimension External (WxDxH)	2620x510x1345mm	2620x530x1545mm
	103.15x20.07x52.95"	103.14x20.86x60.82"
Tube Rotation Speed	0 ~ 14 RPM	
Feeder speed	0 ~ 8 RPM	
Tube Angle	0 ~ 8°	



Model	SH-FU-140RKG600	SH-FU-140RKG900
Max Temp	1200°C (2192°F)	
Dimension Hot Zone	300 mm x 2 Hot Zone=Length 600 mm	300mm x 3 Hot Zone=Length 900 mm
Applicable Tube Diameter & length	STS 310S 140 φ x 1400 mm	STS 310S 140 φ x 1720 mm
Dimension External (W×D×H)	2434x550x1385mm	2759x550x1385mm
	95.82x21.65x54.53"	108.62x21.65x54.53"
Heater	6kW	9kW
Tube Rotation Speed	0 ~ 14 RPM	
Feeder speed	0 ~ 8 RPM	
Tube Angle	0 ~ 8 °	



1500°C Rotary Kiln Furnace (Small Scale Rotary Kiln)

NEW



Model	SH-FU-100RKH600	SH-FU-120RKH600
Max Temp	1500°C (2732°F)	
Dimension Hot Zone	300 mm x 2 Hot Zone=Length 600 mm	
Applicable Tube Diameter & length	고순도 Al ₂ O ₃ 100 φ x 1480 mm	고순도 Al ₂ O ₃ 120 φ x 1480 mm
Dimension External (W×D×H)	2335x630x1400mm	2335x650x1400mm
	91.92x24.80x55.11"	91.92x25.59x55.11"
Heater	18.6 kW	26.6 kW
Tube Rotation Speed	0 ~ 14 RPM	
Feeder speed	0 ~ 8 RPM	
Tube Angle	0 ~ 8 °	

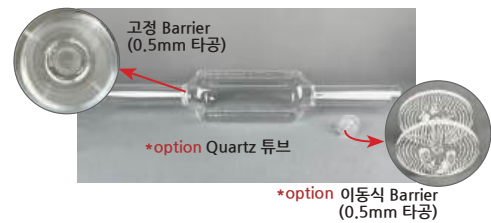


Model	SH-FU-100RKH900	SH-FU-120RKH900
Max Temp	1500°C (2732°F)	
Dimension Hot Zone	300mm x 3 Hot Zone=Length 900 mm	
Applicable Tube Diameter & length	고순도 Al ₂ O ₃ 100 φ x 1800 mm	고순도 Al ₂ O ₃ 120 φ x 1800 mm
Dimension External (W×D×H)	2660x630x1400mm	2660x650x1400mm
	104.72x24.80x55.11"	104.72x25.59x55.11"
Heater	28 kW	40 kW
Tube Rotation Speed	0 ~ 14 RPM	
Feeder speed	0 ~ 8 RPM	
Tube Angle	0 ~ 8 °	

1200°C Batch Rotary Tube Furnace

배치식 회전튜브전기로

NEW

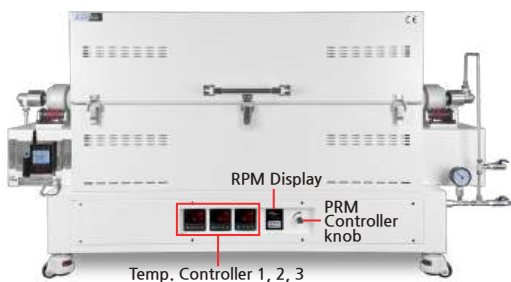


1 Hot Zone



Model		SH-FU-120RTG300	
Max Temp		1200°C (2192°F)	
Temp Controller		Programmable Controller (SP570)	
Tube Diameter & length(Fuse Type) total length(700mm)	Heating Middle	114.3 Φ x 3T x 220mm (STS310S) 120 Φ x 3T x 220mm (Quartz)	
	Heating Reduce	114.3 Φ x 3T > 60 Φ x 2.8T x 40mm (STS310S) 120 Φ x 3T > 60 Φ x 2.5T x 40mm (Quartz)	
	Cooling End	60 Φ x 2.8T x 200mm (STS310S) 60 Φ x 2.5T x 200mm (Quartz)	
Dimension External (WxDxH)		940 x 500 x 640 mm	
		37.0 x 19.6 x 25.2"	
Heater Capa		4.0 kW	
Tube Rotation Speed		0 ~ 10rpm	
Model		SH-FU-200RTG300	SH-FU-250RTG300
Max Temp		1200°C (2192°F)	
Temp Controller		Programmable Controller (SP570)	
Tube Diameter & length(Fuse Type) total length(800mm)	Heating Middle	216.3 Φ x 4T x 220 mm (STS310S) 200 Φ x 5T x 220 mm (Quartz)	267.4 Φ x 4T x 220 mm (STS310S) 250 Φ x 5T x 220 mm (Quartz)
	Heating Reduce	216.3 Φ x 4T > 60 Φ x 2.8T x 40mm (STS310S) 200 Φ x 5T > 60 Φ x 2.5T x 40mm (Quartz)	267.4 Φ x 4T > 60 Φ x 2.8T x 40mm (STS310S) 250 Φ x 5T > 60 Φ x 2.5T x 40mm (Quartz)
	Cooling End	60 Φ x 2.8T x 250mm (STS310S) 60 Φ x 2.5T x 250mm (Quartz)	60 Φ x 2.8T x 250mm (STS310S) 60 Φ x 2.5T x 250mm (Quartz)
Dimension External (WxDxH)		1040 x 580 x 720mm	
		40.9 x 22.8 x 27.9"	
Heater Capa		6.0 kW	8.0 kW
Tube Rotation Speed		0 ~ 10rpm	

3 Hot Zone



Model		SH-FU-120RTG900	
Max Temp		1200°C (2192°F)	
Temp Controller		Programmable Controller (SP570)	
Tube Diameter & length(Fuse Type) total length(1350mm)	Heating Middle	114.3 Φ x 3T x 870mm (STS310S) 120 Φ x 3T x 870mm (Quartz)	
	Heating Reduce	114.3 Φ x 3T > 60 Φ x 2.8T x 40mm (STS310S) 120 Φ x 3T > 60 Φ x 2.5T x 40mm (Quartz)	
	Cooling End	60 Φ x 2.8T x 200mm (STS310S) 60 Φ x 2.5T x 200mm (Quartz)	
Dimension External (WxDxH)		1690 x 500 x 640 mm	
		66.5 x 19.6 x 25.2"	
Heater Capa		8.0 kW	
Tube Rotation Speed		0 ~ 10 rpm	
Model		SH-FU-200RTG900	SH-FU-250RTG900
Max Temp		1200°C (2192°F)	
Temp Controller		Programmable Controller (SP570)	
Tube Diameter & length(Fuse Type) total length(1450mm)	Heating Middle	216.3 Φ x 4T x 870 mm (STS310S) 200 Φ x 5T x 870 mm (Quartz)	267.4 Φ x 4T x 870 mm (STS310S) 250 Φ x 5T x 870 mm (Quartz)
	Heating Reduce	216.3 Φ x 4T > 60 Φ x 2.8T x 40mm (STS310S) 200 Φ x 5T > 60 Φ x 2.5T x 40mm (Quartz)	267.4 Φ x 4T > 60 Φ x 2.8T x 40mm (STS310S) 250 Φ x 5T > 60 Φ x 2.5T x 40mm (Quartz)
	Cooling End	60 Φ x 2.8T x 250mm (STS310S) 60 Φ x 2.5T x 250mm (Quartz)	60 Φ x 2.8T x 250mm (STS310S) 60 Φ x 2.5T x 250mm (Quartz)
Dimension External (WxDxH)		1690 x 580 x 720mm	
		66.5 x 22.8 x 28.3"	
Heater Capa		18 kW	24 kW
Tube Rotation Speed		0 ~ 10rpm	

1200°C Tube Furnace

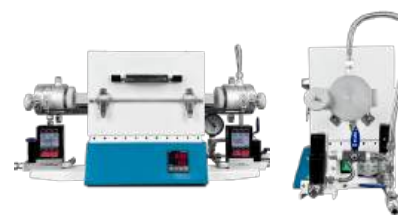
150mm Length Hot Zone



Super Mini

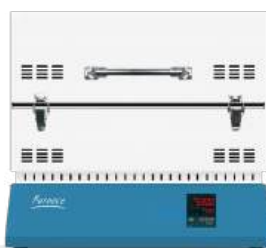
NEW

실제 사이즈 비교



Model	SH-FU-30TG150
Max Temp	1200°C (2192°F)
Temp Controller	Programmable Controller(SP 540)
Applicable Tube Diameter	30 φ
Dimension Hot Zone	Length 150 mm
Dimension External (W×D×H)	280x234x303mm
	11.02x9.21x11.92"

300mm Length Hot Zone



Model	SH-FU-50TG300	SH-FU-80TG300	SH-FU-100TG300
Max Temp	1200°C (2192°F)		
Applicable Tube Diameter	50 φ	80 φ	100 φ
Dimension Hot Zone	Length 300 mm		
Dimension External (W×D×H)	500x370x540mm	500x400x570mm	502x397x557mm
	19.69x14.57x21.26"	19.69x15.75x22.44"	19.76x15.63x21.93"

Model	SH-FU-120TG300	SH-FU-200TG300	SH-FU-250TG300
Max Temp	1200°C (2192°F)		
Applicable Tube Diameter	120 φ	200 φ	250 φ
Dimension Hot Zone	Length 300 mm		
Dimension External (W×D×H)	502x455x630mm	500x540x830mm	500x600x880mm
	19.76x17.91x24.80"	19.69x21.26x32.67"	19.69x23.62x34.64"

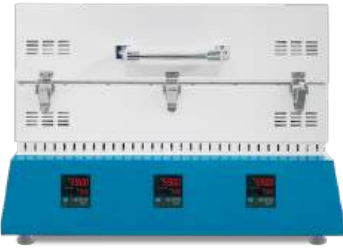
600mm Length Hot Zone



Model	SH-FU-50TG600	SH-FU-80TG600	SH-FU-100TG600
Max Temp	1200°C (2192°F)		
Applicable Tube Diameter	50 φ	80 φ	100 φ
Dimension Hot Zone	Length 600 mm		
Dimension External (W×D×H)	800x370x540mm	800x400x570mm	800x430x600mm
	32.50x14.57x21.26"	31.50x15.75x22.44"	31.50x16.93x23.62"

Model	SH-FU-120TG600	SH-FU-200TG600	SH-FU-250TG600
Max Temp	1200°C (2192°F)		
Applicable Tube Diameter	120 φ	200 φ	250 φ
Dimension Hot Zone	Length 600 mm		
Dimension External (W×D×H)	800x460x630mm	800x540x830mm	800x600x880mm
	31.50x18.11x24.80"	31.49x21.26x32.67"	31.49x23.62x34.64"

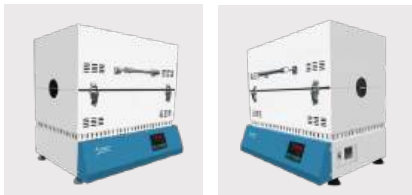
200mm Length x 3 Hot Zone



Model	SH-FU-50TG200-3	SH-FU-80TG200-3	SH-FU-100TG200-3
Max Temp	1200°C (2192°F)		
Applicable Tube Diameter	50 Φ	80 Φ	100 Φ
Dimension Hot Zone	200mm x 3 Hot Zone=Length 600 mm		
Dimension External (W×D×H)	850x370x540mm	850x400x570mm	850x430x600mm
	33.46x14.57x21.26"	33.46x15.75x22.44"	33.46x16.93x23.62"

Model	SH-FU-120TG200-3	SH-FU-200TG200-3	SH-FU-250TG200-3
Max Temp	1200°C (2192°F)		
Applicable Tube Diameter	120 Φ	200 Φ	250 Φ
Dimension Hot Zone	200mm x 3 Hot Zone=Length 600 mm		
Dimension External (W×D×H)	850x460x630mm	850x540x830mm	850x600x880mm
	33.46x18.11x24.80"	33.46x21.26x32.67"	33.46x23.62x34.64"

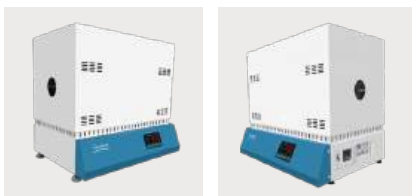
1500°C Tube Furnace



Model	SH-FU-50TH300	SH-FU-80TH300
Max Temp	1500°C (2732°F)	
Applicable Tube Diameter	50 Φ	80 Φ
Dimension Hot Zone	Length 300 mm	
Dimension External (W×D×H)	680x490x660mm	680x540x710mm
	26.77x19.29x25.98"	26.77x21.26x27.95"

Model	SH-FU-100TH300	SH-FU-120TH300
Max Temp	1500°C (2732°F)	
Applicable Tube Diameter	100 Φ	120 Φ
Dimension Hot Zone	Length 300 mm	
Dimension External (W×D×H)	680x540x710mm	680x590x760mm
	26.77x21.26x27.95"	26.77x23.23x29.92"

1700°C / 1800°C Tube Furnace



Model (1700°C/1800°C)	SH-FU-50TS300	SH-FU-80TS300
Max Temp	1550°C (Mosi 2 1700°C)	
	1650°C (Mosi 2 1800°C)	
Applicable Tube Diameter	50 Φ	80 Φ
Dimension Hot Zone	Length 300 mm	
Dimension External (W×D×H)	690x580x890mm	690x580x1010mm
	27.17x22.84x35.04"	27.17x22.83x39.76"

Model (1700°C/1800°C)	SH-FU-100TS300	SH-FU-120TS300
Max Temp	1550°C (Mosi 2 1700°C)	
	1650°C (Mosi 2 1800°C)	
Applicable Tube Diameter	100 Φ	120 Φ
Dimension Hot Zone	Length 300 mm	
Dimension External (W×D×H)	690x600x1030mm	690x640x1110mm
	27.17x23.62x40.55"	27.17x25.20x43.70"

Benchtop Steam Generator

NEW

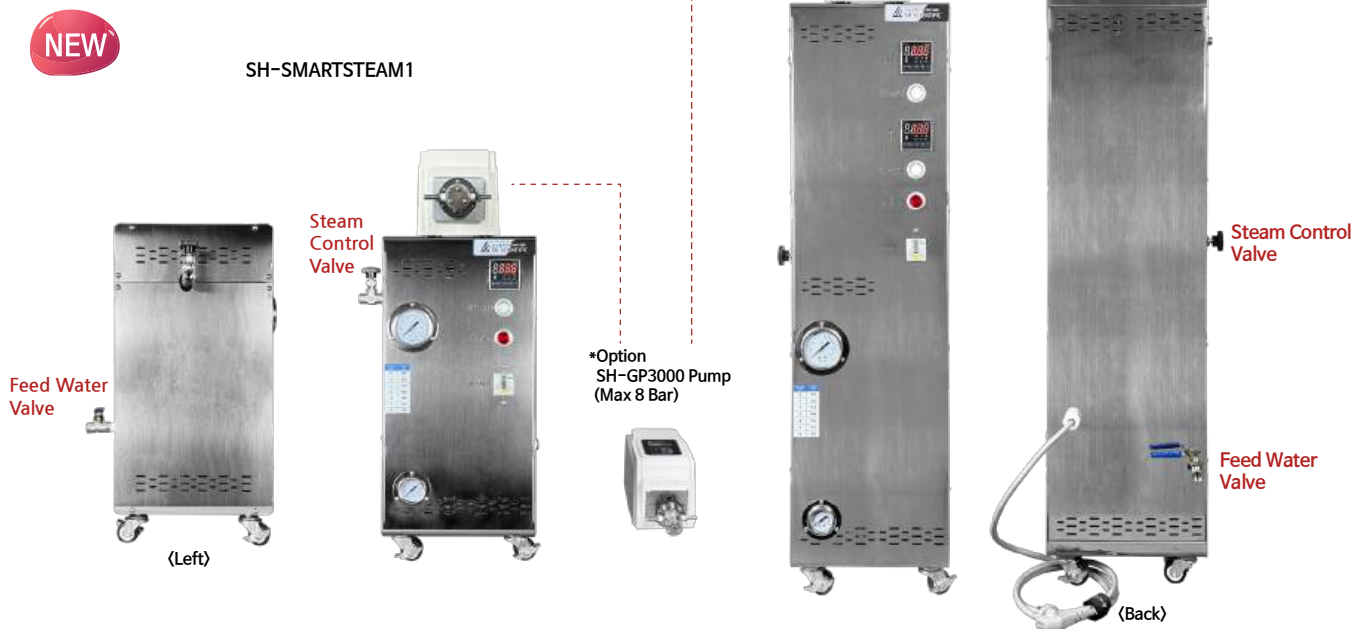
소형 실험실용 스팀 발생장치



Model	SH-LABSTEAM5
Steam Generator	간접 가열식 플라스크 보일링
Max Steam Temp.	100 °C ~ 110°C
Steam Capa	1.5 g/min (200ml/분 , 질소투입시) at 100°C
	0.7 g/min (100ml/분 , 질소투입시) at 100°C
	0.35 g/min (50ml/분 , 질소투입시) at 100°C
Flask Capa	5 L
Steam Supply	캐리어 가스 이송 방식

The Smart & Compact Steam Generator

All-In-One Mobile Steam Generator



Model	SH-SMARTSTEAM1	SH-SMARTSTEAM2
Steam Generation	직접가열식 STS316 밀폐형 보일링	
Max Steam Temp.	165°C (329°F)	500°C(932°F)
Boiling Tank Capacity	0.5 L x 1 tank	0.5 L x 2 tank
Steam Capacity	~ approx. 23g / min (By Steam Control Valve)	
Material	Wet part STS316 / Body STS304 , All Stainless Steel	
Dimension(WxDxH)(mm)	305 x 316 x 546 mm / 12.00 x 12.44 x 21.49"	330 x 316 x 941 mm / 12.99 x 12.44 x 37.04"

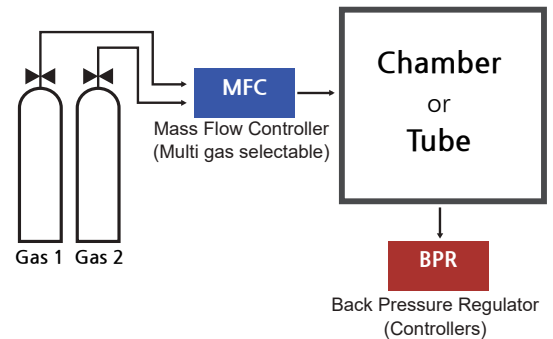
Mass Flow Controller (Multi gas selectable)



하나의 Unit
98 Gas
사용가능



Model	MC Series	MCS Series
Type	General	For Toxi Gas
Flow Range	0 ~ 1 slpm / 0 ~ 5 slpm	
Mass Flow Accuracy at calibration conditions	±0.6% of reading or ±0.1% of full scale	±0.8% of reading or ±0.2% of full scale
Operating Pressure full scale	145 PSIG	
Gas Selectable	98 Gas	128 Gas (Cl ₂ , SO ₂ , HCl 제외)
Communication	RS-232	
Model	MCSS Series	MCV Series
Type	For Toxi Gas(Cl ₂ , SO ₂ , HCl 포함)	For Vacuum
Flow Range	0 ~ 1 slpm / 0 ~ 5 slpm	
Mass Flow Accuracy at calibration conditions	±0.8% of reading or ±0.2% of full scale	
Operating Pressure full scale	145 PSIG	
Gas Selectable	131 Gas (Cl ₂ , SO ₂ , HCl 포함)	98 Gas
Communication	RS-232	



Back Pressure Regulator (Controllers)

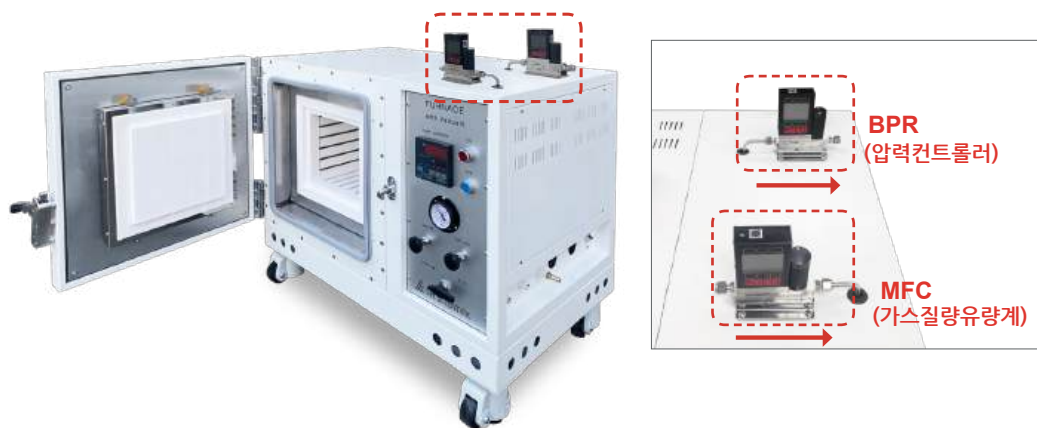


Model	BPR Series	BPRV Series
Type	General	For Vacuum(음압용)
Pressure Range	1PSIG ~ 15PSIG	0PSIG ~ 1PSIG(0 ~ 760torr)
Standard Accuracy	± 0.25% (≥ 1 psi)	
Repeatability	± 0.08% Full Scale	
Typical Response Time	< 100 ms	
Communication	RS-232	
Model	BPRS Series	BPRSS Series
Type	For Toxi GasFor Toxi Gas (Cl ₂ , HCl, SO ₂ 제외)	For Toxi Gas (Cl ₂ , HCl, SO ₂ 포함)
Pressure Range	1PSIG ~ 15PSIG	
Standard Accuracy	± 0.25% (≥ 1 psi)	
Repeatability	± 0.08% Full Scale	
Typical Response Time	< 100 ms	
Communication	RS-232	

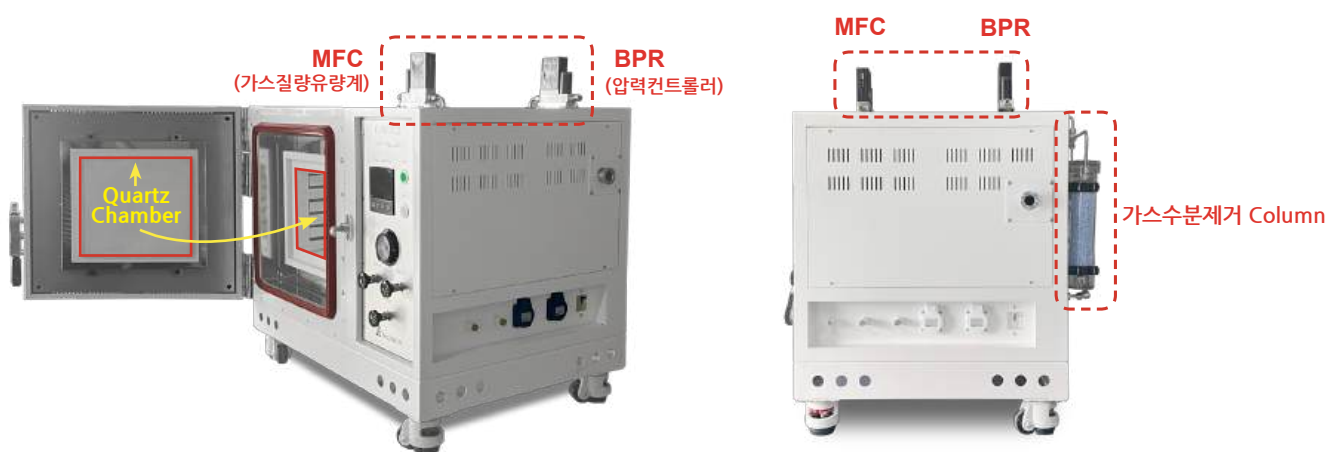
Digital Vacuum Meter(Bullseye)



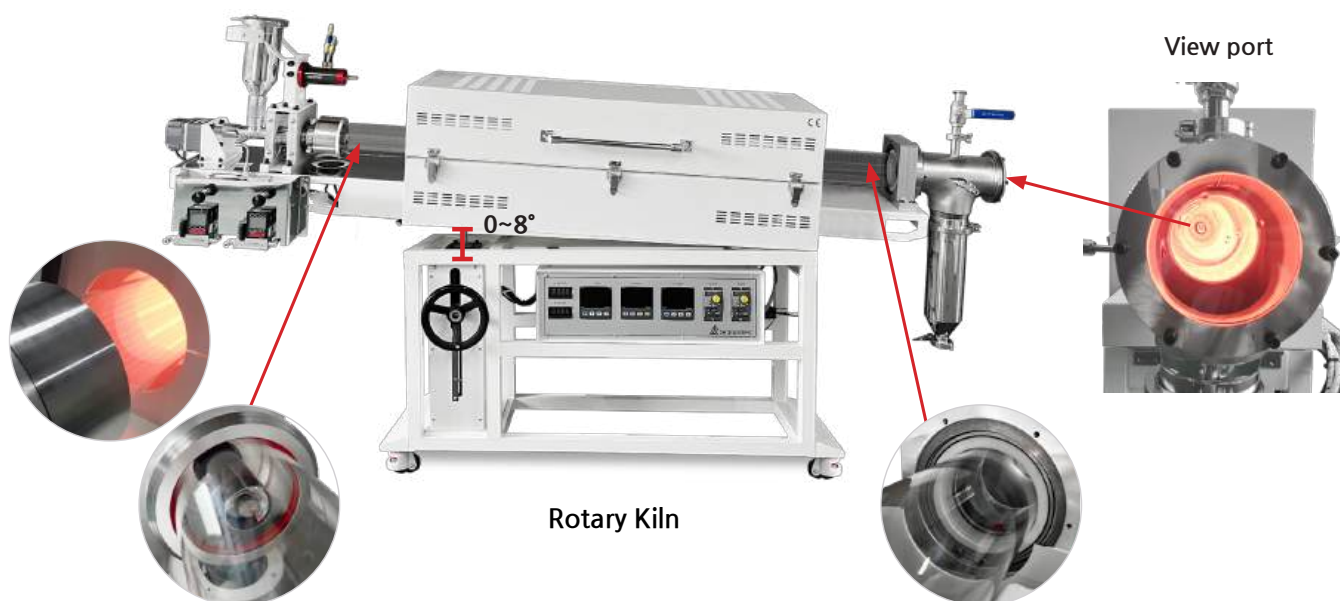
Model	Bullseye
Range	0.001 Torr - 800 Torr
Power	AA x 4 Batteries
Sensor	Agilent 536 Sensor
Accuracy	1~99 mmTorr
	± 2 mmTorr or 20%
	100~2000 mmTorr
	± 10%
	2~6 Torr
	± 25%
Units	Torr, millitorr, microns, mbar, bar, kPa, Pa, mm, mm of Hg, inches of Hg, inches of water, PSIA
Mount	Portable Handheld
Display	LCD
Certifications	CE, RoHS2



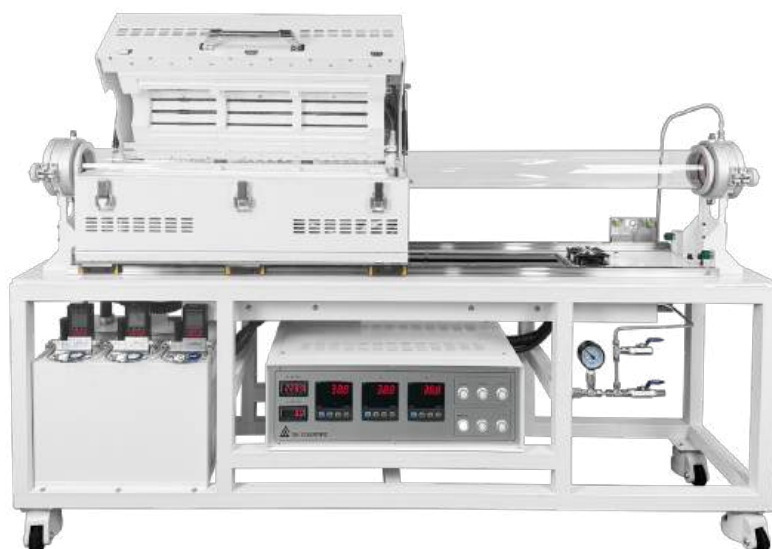
Muffle Furnace with Vacuum



Muffle Furnace with Quartz Chamber



Rotary Kiln



RTCVD Tube Furnace (Gas Flow Package)



Batch Rotary Tube Furnace



CVD Tube Furnace (Gas Flow Package)

05. Mixer / Mill / Dispersion

Jaw Crusher

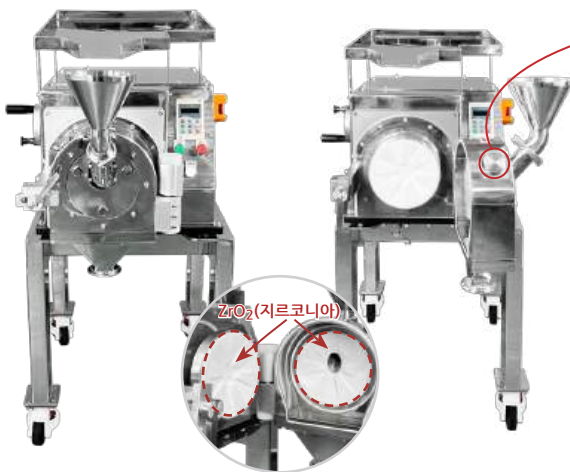
NEW



Model	SH-JC60	SH-JC120
Feed Port Size	60 x 95mm	120 x 200mm
투입 원료 입자 크기	≤ 35mm	≤ 80mm
최종 분쇄 입자 크기	≤ 10mm	≤ 30mm
Capacity (재료에 따라 다름)	Max 30KG/Hr	Max 300KG/Hr
Process	10 Process (Speed, Time 저장)	

Disk Mill

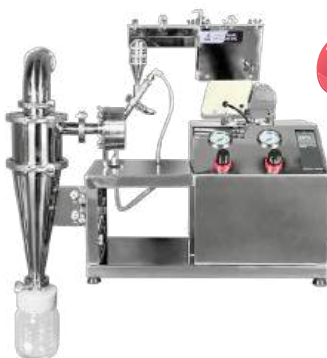
NEW



Model	SH-DISKMILL200
투입 원료 입자 크기	Max 30mm ~ 35mm
최종 분쇄 입자 크기	0.1 ~ 5mm
Material	회전 & 고정 Disk
	ZrO ₂ (지르코니아), 200 dia
	External Case
	Stainless Steel 304
간격 표시용 Gauge	0~30mm, 0.01mm
Disk Speed	100 ~ 1000 R.P.M (권장 500R.P.M) By Inverter
Dimension (WxDxH)	600X960X1250 mm
	23.62X37.80X49.21"
Power Source	380V, 3Φ, 1.5KW

Jet Mill

NEW



- 압축공기 (또는 가스) 연결만으로 분쇄작업
- 불순물 발생이 없어 고순도의 미세입자를 얻을수 있음
- ≤ 5μm 분쇄 가능범위
- 싸이클론 방식적용으로 분쇄효율이 우수
- 탈착이 용이한 투명 회수병 장착으로 분쇄입자 회수가 용이

Model	SH-JETMILL
투입 원료 입자 크기	≤ 5mm
최종 분쇄 입자 크기	≤ 5μm ~ 10μm
분말 회수율	>90%
압력범위	0.4 ~ 1.0 MPa
Dimension (WxDxH)	850x420x760 mm / 33.46x16.53x29.92"
Power	220V, 1Φ

Ball Mill



Model	SH-BALL300-1	SH-BALL700-1
Dimension External (W×D×H)	493x500x350mm	893x500x350mm
	19.41x19.69x13.78"	35.16x19.69x13.78"
Dimension Roller Length, 70Φ	300mm(11.81") x 1stage	700mm(27.56")x1stage



Model	SH-BALL300-2	SH-BALL700-2
Dimension External (W×D×H)	731x450x614mm	1131x450x614mm
	28.78x17.72x24.17"	44.53x17.72x24.17"
Dimension Roller Length, 70Φ	300mm(11.81")x2stage	700mm(27.56")x2stage

Box Ball Mill

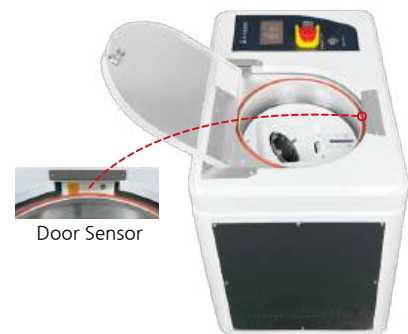


Model	SH-BALL300B	SH-BALL700B
Dimension External (W×D×H)	841x602x1085mm	1241x602x1085mm
	33.11x23.70x42.72"	48.86x23.70x42.72"
Dimension Roller Length, 70Φ	300mm(11.81")x2stage	700mm(27.56")x2stage

Planetary Mixer

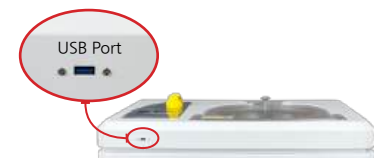
- 공전+자전+진공 (진공펌프 내장) 동시운전
- Program Control, 레시피 설정가능(공전 속도)
- Touch Screen Panel

NEW



350ml Single Cup(P.E) x 1ea 적재

Model	SH-PMC350
Capacity	350ml x 1 cup, 권장용량 (부피 175ml, 무게 200g)
Speed	Revolution : 130 ~ 2000 R.P.M, Rotation : Invariable By Revolution
Vacuum	-99KPa, 120L/min
Program Setting	5 Segment/Pattern, 8 Pattern
Dimension(Wx DxH)	585x405x650 mm
	23.03x15.94x25.59"
Weight	85Kg
Power	220V, 1Φ, 1.5KW, 7A



350ml Single Cup(P.E) x 2ea 적재

Model	SH-PMC700
Capacity	350ml x 2 cup, 권장용량(부피 175ml, 무게 200g)x 2
Speed	Revolution : 130 ~ 3000 R.P.M, Rotation : Invariable By Revolution
Vacuum	-99KPa, 330L/min
Program Setting	3 Segment/Pattern, 5Pattern
Dimension(Wx DxH)	730x510x790 mm
	28.74x20.07x31.10"
Weight	115Kg
Power	220V, 1Φ, 1.75KW, 8A

STS Vessel Planetary Mixer



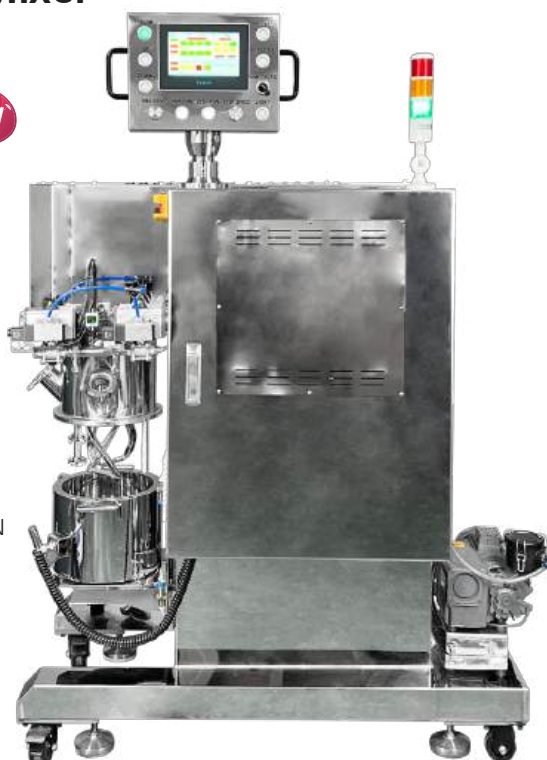
〈SH-DPM02/05/10〉



〈SH-DPM30〉

NEW

UP
↓
DOWN



Model		SH-DPM02	SH-DPM05	SH-DPM10	SH-DPM30
Container Capa		3 Liter	7.5 Liter	14 Liter	40 Liter
Effective Capa		2 Liter	5 Liter	10 Liter	30 Liter
Speed Range	Revolution	0~61 rpm	0~51 rpm	0~49 rpm	0~43 rpm
	Rotation x 2paddle	0~125 rpm	0~112 rpm	0~85 rpm	0~84 rpm
	Dispersion x 1paddle (2 blade) (DPM02~10)	0~7500 rpm	0~9000 rpm	0~6000 rpm	0~4500 rpm
	Dispersion x 2paddle (4 blade) (DPM30)				
Temp Range		-10℃~ 200℃ (Circulation Bath 추가 연결 필요)			
Vacuum Range		-0.098 MPa (25 mmHg , 25 Torr)			
Material	Container	STS304			
	Paddle	Super Duplex Stainless Steel			
	Body	STS304			
Dimension (mm)	Container(Dia x H)	180 Dia x 120 H mm	250 Dia x 160 H mm	300 Dia x 200 H mm	400 Dia x 350 H mm
	External	500 x 800 x 1400mm	700 x 1200 x 1500mm	800 x 1350 x 1600mm	900 x 1650 x 1800mm

Three Roll Mill

NEW



〈SH-ROLL50〉



〈SH-ROLL80〉

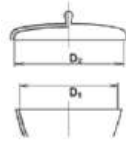


〈SH-ROLL80D〉

Model	SH-ROLL50	SH-ROLL80	SH-ROLL80D
Capa	0.02 ~ 5L/Hour	0.02 ~ 20L / hour	0.02 ~ 20L / hour
Roller Material	ZrO ₂ (지르코니아)	ZrO ₂ (지르코니아)	ZrO ₂ (지르코니아)
Roller	50mm x 150mm (3ea)	80mm x 200mm (3ea)	80mm x 200mm (3ea)
Roller Speed Control	By Knob / digital display	By Knob	By Digital Touch Panel
Roller interval Control	By Handle / digital display	By Handle	By Digital Touch Panel
Dimension (WxDxH)	360 x 190 x 285 mm	600 x 380 x 495 mm	690 x 380 x 495 mm
	14.2 x 7.5 x 11.2"	23.6 x 15.0 x 19.5"	27.2 x 15.0 x 19.5"

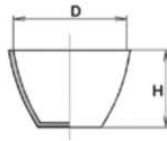
06. Ceramic consumable & furnace accessory

Lid for crucible



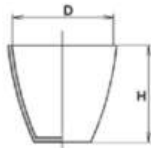
Model	D1xD2(mm)
641 321 000 301 (D/30)	30x34
641 321 000 351 (D/35)	35x39
641 321 000 401 (D/40)	40x44
641 321 000 451 (D/45)	45x49
641 321 000 501 (D/50)	50x54
641 321 000 601 (D/60)	60x64
641 321 000 701 (D/70)	70x74
641 321 000 801 (D/80)	80x84

Crucible (Low form)



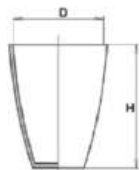
Model	DxH(mm)	V/ml approx.
641 321 100 301 (1/30)	30x19	5
641 321 100 351 (1/35)	35x22	10
641 321 100 401 (1/40)	40x25	17
641 321 100 451 (1/45)	45x28	21
641 321 100 501 (1/50)	50x32	34
641 321 100 601 (1/60)	60x38	62
641 321 100 701 (1/70)	70x44	91

Crucible (Middle form)



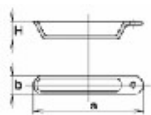
Model	DxH(mm)	V/ml approx.
641 321 200 301 (2/30)	30x25	10
641 321 200 351 (2/35)	35x28	12
641 321 200 401 (2/40)	40x32	20
641 321 200 451 (2/45)	45x36	30
641 321 200 501 (2/50)	50x40	45
641 321 200 601 (2/60)	60x48	80
641 321 200 701 (2/70)	70x56	120
641 321 200 801 (2/80)	80x62	200

Crucible (High form)



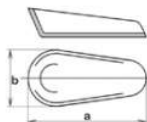
Model	DxH(mm)	V/ml approx.
641 321 300 301 (3/30)	30x38	15
641 321 300 351 (3/35)	35x44	26
641 321 300 401 (3/40)	40x50	35
641 321 300 451 (3/45)	45x56	50
641 321 300 501 (3/50)	50x62	72
641 321 300 601 (3/60)	60x75	120

Combustion boat



Model	axbH(mm)
641 312 214 200 (214/2)	75x13x8
641 312 214 300 (214/3)	105x14x9
641 312 214 400 (214/4)	85x13x8
641 312 214 800 (214/8)	90x13x8

Weighing boat



Model	axb(mm)
641 329 252 001 (252/0)	37x17
641 329 252 101 (252/1)	53x23
641 329 252 201 (252/2)	64x29

Crucible with lid



43(D)x27(H), 19ml

Gooch crucible



30(D)x36(H), 17ml / 35(D)x43(H), 25ml
39(D)x49(H), 35ml / 60(D)x71(H), 120ml

Annealing dish



40(D)x20(H), 21ml / 50(D)x25(H), 35ml
60(D)x30(H), 64ml / 70(D)x35(H), 103ml

Annealing dish



50(a)x35(b)x6(H), 12ml / 62(a)x40(b)x13(H), 12ml
74(a)x49(b)x28(H), 14ml / 95(a)x65(b)x40(H), 14ml

Evaporating dish



50(D)x20(H), 20ml / 63(D)x25(H), 30ml
80(D)x32(H), 60ml / 81(D)x38(H), 107ml
94(D)x42(H), 140ml / 100(D)x40(H), 150ml
125(D)x50(H), 285ml

Evaporating dish



96(D)x30(H), 110ml / 110(D)x37(H), 160ml

Evaporating dish



60(D)x14(H), 20ml / 80(D)x20(H), 55ml
100(D)x25(H), 110ml / 130(D)x30(H), 250ml

Evaporating dish



54(D)x22(H), 22ml / 70(D)x30(H), 62ml
86(D)x33(H), 93ml / 98(D)x40(H), 154ml
112(D)x50(H), 265ml

Unglazed, Mortar rough



56(D)x36(H), 30ml / 63(D)x41(H), 70ml
90(D)x56(H), 160ml / 125(D)x71(H), 400ml

Pestle unglazed



20(D)x88(L)

Pestle unglazed



14(D)x54(L) / 24(D)x115(L)
30(D)x135(L) / 36(D)x150(L)
42(D)x175(L) / 55(D)x210(L)

Pestle glazed



30(D)x135(L) / 36(D)x150(L)

Plate for desiccator DIN



140(D) / 190(D) / 235(D) / 280(D)

Plate for desiccator CSN



189(D) / 241(D) / 290(D)

Plate with cavities



112(a)x81(b)x25(c), 6Qty
115(a)x91(b)x17(c), 12Qty

Funnel Buechner



48(D1)x38(D2)x1(D3)x24(H1)x43(H2), 35ml
 62(D1)x45(D2)x1(D3)x30(H1)x64(H2), 70ml
 77(D1)x58(D2)x1(D3)x35(H1)x64(H2), 120ml
 97(D1)x70(D2)x2(D3)x40(H1)x71(H2), 240ml
 116(D1)x95(D2)x2(D3)x49(H1)x83(H2), 400ml
 130(D1)x110(D2)x2(D3)x52(H1)x85(H2), 600ml

Stirring spatle+scoop



103(L) / 148(L) / 164(L) / 187(L)
 200(L) / 250(L)

Quartz Shelf



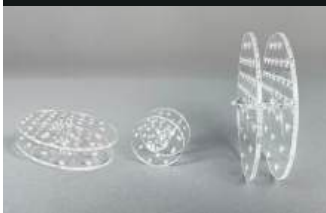
Model No.	Size(WxDxH)	Thickness
SH-QTS-100	100x100x40mm/ 3.93/3.93x1.57"	3t
SH-QTS-190	190x190x40mm/ 7.48/7.48x1.57"	
SH-QTS-290	290x290x40mm/11.41/11.41x1.57"	
SH-QTS-390	390x390x40mm /15.35/15.3x1.57"	4t
SH-QTS-490	490x490x40mm /19.29/19.29x1.57"	

Quartz wafer carrier



Model No.	wafer	Capa	Size(WxDxH)	Temp-Limit
SH-QWC-2	2"	25ea	59x148x20mm / 2.32x5.82x0.78"	Operation Temp1100℃ Max Temp 1250℃
SH-QWC-4	4"	25ea	107x148x40mm / 4.21x5.82x1.57"	
SH-QWC-6	6"	25ea	157x152x55mm / 6.18x5.98x2.16"	
SH-QWC-8	8"	25ea	206x156x70mm / 8.11x6.14x2.75"	

Quartz Gas Diffuser



Model No.	Material
SH-QDS-50	50Φ TUBE용, Quartz gas diffuser, 외경 44Φ
SH-QDS-80	80Φ TUBE용, Quartz gas diffuser, 외경 73Φ
SH-QDS-100	100Φ TUBE용, Quartz gas diffuser, 외경 93Φ
SH-QDS-120	120Φ TUBE용, Quartz gas diffuser, 외경 113Φ
SH-QDS-200	200Φ TUBE용, Quartz gas diffuser, 외경 189Φ
SH-QDS-250	250Φ TUBE용, Quartz gas diffuser, 외경 239Φ

Quartz boat

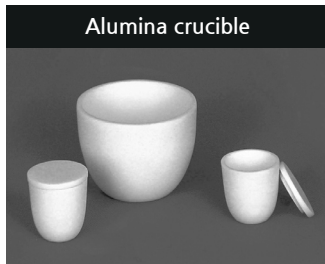


Model No.	Size (ml.LxWxH)	Thickness	Temp-Limit
SH-QB-0430	4ml, 20x50x8mm/0.78x1.96x0.31", for 30TG150	2T	Operation Temp1100℃ Max Temp 1250℃
SH-QB-4550	45ml, 40x100x18mm/1.57x3.93x0.71", for 50TG300	3T	
SH-QB-13080	130ml, 70x100x32mm/2.75x3.93 x1.26", for 80TG300	3T	
SH-QB-250100	250ml, 90x100x40mm/3.54x3.93 x1.57", for 100TG300	3T	
SH-QB-400120	400ml, 110x100x50mm/4.33x3.93 x1.96", for 120TG300	3T	
SH-QB-1300200	1300ml, 186x100x90mm/7.32x3.93 x3.54", for 200TG300	3.5~4T	
SH-QB-2100250	2100ml, 236x100x115mm/9.29x3.93 x4.52", for 250TG300	3.5~4T	

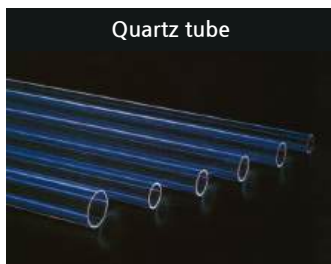
Quartz crucible with lid



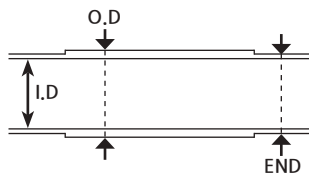
Model No.	Capa(ml)	Size	Temp-Limit
SH-QC-30	30 ml	ø47, ø25xh36mm	Operation Temp1100℃ Max Temp 1250℃
SH-QC-50	50 ml	ø54, ø34xh42mm	
SH-QC-100	100 ml	ø70, ø34xh55mm	
SH-QC-250	250 ml	ø110, ø50xh80mm	
SH-QC-500	500 ml	ø130, ø70xh100mm	



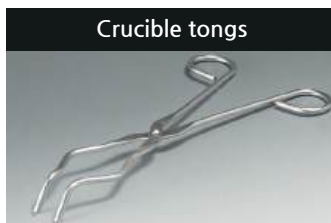
Model No.	Capa(ml)	Size	Max Temp	Lid	
				Model No.	Capa(mm)
SH-ALC-10B	10 ml	ø29xh30mm	1700℃	SH-ALC-10L	10 ml
SH-ALC-20B	20 ml	ø39xh29mm		SH-ALC-20L	20 ml
SH-ALC-30B	30 ml	ø52xh39mm		SH-ALC-30L	30 ml
SH-ALC-50B	50 ml	ø61xh49mm		SH-ALC-50L	50 ml
SH-ALC-100B	100 ml	ø72xh58mm		SH-ALC-100L	100 ml
SH-ALC-150B	150 ml	ø81xh67mm		SH-ALC-150L	150 ml
SH-ALC-200B	200 ml	ø86xh70mm		SH-ALC-200L	200 ml
SH-ALC-250B	250 ml	ø92xh75mm		SH-ALC-250L	250 ml
SH-ALC-300B	300 ml	ø97xh76mm		SH-ALC-300L	300 ml
SH-ALC-500B	500 ml	ø110xh105mm		SH-ALC-500L	500 ml



Model No.	Size(I.D/O.D×L)	Model No.	Size(I.D/O.D×L)
SH-QT-50	46/50×1000mm / 1.81/1.97x39.37"	SH-QT-50(1200)	46/50×1200mm / 1.81/1.97x47.24"
SH-QT-60	55/60×1000mm / 2.17/2.36x39.37"	SH-QT-60(1200)	55/60×1200mm / 2.17/2.36x47.24"
SH-QT-80	74/80×1000mm / 2.91/3.15x39.37"	SH-QT-80(1200)	74/80×1200mm / 2.91/3.15x47.24"
SH-QT-100	94/100×1000mm / 3.70/3.94x39.37"	SH-QT-100(1200)	94/100×1200mm / 3.70/3.94x47.24"
SH-QT-120	114/120×1000mm / 4.49/4.72x39.37"	SH-QT-120(1200)	114/120×1200mm / 4.49/4.72x47.24"
SH-QT-150	114/150×1000mm / 4.49/5.91x39.37"	SH-QT-150(1200)	114/150×1200mm / 4.49/5.91x47.24"
SH-QT-180	174/180×1000mm / 6.85/7.08x39.37"	SH-QT-180(1200)	174/180×1200mm / 6.85/7.08x47.24"
SH-QT-200	193/200×1000mm / 7.60/7.87x39.37"	SH-QT-200(1200)	193/200×1200mm / 7.60/7.87x47.24"
SH-QT-250	240/250×1000mm / 9.45/9.84x39.37"	SH-QT-250(1500)	240/250×1500mm / 9.45/9.84x59.05"
		SH-QT-250(1200)	240/250×1200mm / 9.45/9.84x47.24"



Model No.	Size(I.D/O.D×L)	Alumina Purity : OVER 98%
SH-AT-30	I.D 24/O.D 33/END 30, length 1000mm, max 1600℃	
SH-AT-50	I.D 42/O.D 53/END 50, length 1000mm, max 1600℃	
SH-AT-60	I.D 50/O.D 63/END 60, length 1000mm, max 1600℃	
SH-AT-80	I.D 70/O.D 83/END 80, length 1000mm, max 1600℃	
SH-AT-90	I.D 80/O.D 93/END 90, length 1000mm, max 1600℃	
SH-AT-100	I.D 88/O.D 103/END 100, length 1000mm, max 1600℃	
SH-AT-120	I.D 110/O.D 123/END 120, length 1000mm, max 1600℃	
SH-AT-150	I.D 138/O.D 153/END 150, length 1000mm, max 1600℃	
SH-AT-50(1200)	I.D 42/O.D 53/END 50, length 1200mm, max 1600℃	
SH-AT-60(1200)	I.D 50/O.D 63/END 60, length 1200mm, max 1600℃	
SH-AT-80(1200)	I.D 70/O.D 83/END 80, length 1200mm, max 1600℃	
SH-AT-90(1200)	I.D 80/O.D 93/END 90, length 1200mm, max 1600℃	
SH-AT-100(1200)	I.D 88/O.D 103/END 100, length 1200mm, max 1600℃	
SH-AT-120(1200)	I.D 110/O.D 123/END 120, length 1200mm, max 1600℃	
SH-AT-150(1200)	I.D 138/O.D 153/END 150, length 1200mm, max 1600℃	



Model No.	Opening Size(Φ)	Length
SH-CT-300	2~30 Φ	300mm / 11.81"
SH-CT-400	2~30 Φ	400mm / 15.75"
SH-CT-500	2~30 Φ	500mm / 19.69"

High temp proof gloves



Model No.	Size	Material	Temp-Limit
SH-SG-900	350mm	Para Aramid, Oxy PAN, 면(내부)	Max 900℃

Gas sealing mask



Model No.	Material
SH-SM-50	50Φ TUBE용, Aluminum(Option: SUS), 2ea/pack
SH-SM-60	60Φ TUBE용, Aluminum(Option: SUS), 2ea/pack
SH-SM-80	80Φ TUBE용, Aluminum(Option: SUS), 2ea/pack
SH-SM-100	100Φ TUBE용, Aluminum(Option: SUS), 2ea/pack
SH-SM-120	120Φ TUBE용, Aluminum(Option: SUS), 2ea/pack
SH-SM-200	200Φ TUBE용, Aluminum(Option: SUS), 2ea/pack
SH-SM-274	274Φ TUBE용, Aluminum(Option: SUS), 2ea/pack

07. Plastic Ware

Tips Series

 Low Retention


Pipet Tips			Filter Tips		
Bulk (1,000Tips)	SH200010	10μl	Bulk (1,000Tips)	SH220010	10μl
	SH200200	200μl		SH220011	10μl
	SH201250	1250μl		SH220200	200μl
Rack (96Tips)	SH210010	10μl	Rack (96Tips)	SH220201	200μl
	SH210200	200μl		SH221000	1000μl
	SH211250	1250μl		SH221001	1000μl
				SH230010	10μl
				SH230011	10μl
				SH230200	200μl
				SH230201	200μl
				SH231000	1000μl
				SH231001	1000μl

Pipettes



Single Channel Pipette		Multi Channel Pipette	
01.-2.5μL	0.25 μL, 1.25 μL, 2.5 μL	0.5-10 μL	8 0.50 μL, 5 μL, 10 μL
0.5-10 μL	1 μL, 5 μL, 10 μL	5-50 μL	8 5 μL, 25 μL, 50 μL
2-20 μL	2 μL, 10 μL, 20 μL	30-300 μL	8 30 μL, 150 μL, 300 μL
5-50 μL	5 μL, 25 μL, 50 μL	0.5-10 μL	12 1 μL, 5 μL, 10 μL
10-100 μL	10 μL, 50 μL, 100 μL	5-50 μL	12 5 μL, 25 μL, 50 μL
20-200 μL	20 μL, 100 μL, 200 μL	30-300 μL	12 30 μL, 150 μL, 300 μL
50-200 μL	50 μL, 100 μL, 200 μL		
100-1000 μL	100 μL, 500 μL, 1000 μL		
200-1000 μL	200 μL, 500 μL, 1000 μL		
0.5-5 mL	0.5 mL, 2.5 mL, 5 mL		



Electronic Single Channel Pipette		Electronic Multi Channel Pipette		
0.2 -10 μ L	0.50 μ L, 5 μ L, 10 μ L	0.2 -10 μ L	8	0.5 μ L, 5 μ L, 10 μ L
5-100 μ L	5 μ L, 50 μ L, 100 μ L	2.5-50 μ L	8	5 μ L, 25 μ L, 50 μ L
10-200 μ L	10 μ L, 100 μ L, 200 μ L	15-300 μ L	8	20 μ L, 150 μ L, 300 μ L
50-1000 μ L	100 μ L, 500 μ L, 1000 μ L	0.2-10 μ L	12	0.5 μ L, 5 μ L, 10 μ L
100-5000 μ L	500 μ L, 2500 μ L, 5000 μ L	2.5-50 μ L	12	5 μ L, 25 μ L, 50 μ L
		15-300 μ L	12	20 μ L, 150 μ L, 300 μ L

Pipette Stands



1



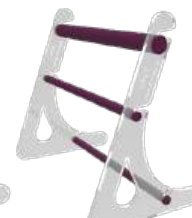
2



3



5



6

Description - Acrylic pipette stands		Color	Sales Unit
1 place pipette stand for manual multi channel pipettes	1	black	Carton/1 pc.
1 place pipette stand for manual multi channel pipettes	2	white	Carton/1 pc.
4 places pipette stand for manual single channel pipettes	3	white	Carton/1 pc.
Description - Acrylic/Aluminium pipette stands		Color	Sales Unit
1 place pipette stand for manual single and multi channel pipettes	5	clear/magenta	Carton/1 pc.
6 places pipette stand for manual single channel pipettes	6	clear/magenta	Carton/1 pc.

Pipette Controller



Description	Sales Unit
pipet4u® Oasis Pipette Controller 0.1-100 mL	Carton/1 pc.
Description - spare parts	
0.2 μ m PTFE membrane filter	Carton/5 pc.
0.45 μ m PTFE membrane filter	Carton/5 pc.
Silicone adapter	Carton/1 pc.
Pipette cone	Carton/1 pc.

Bottle Top Dispenser

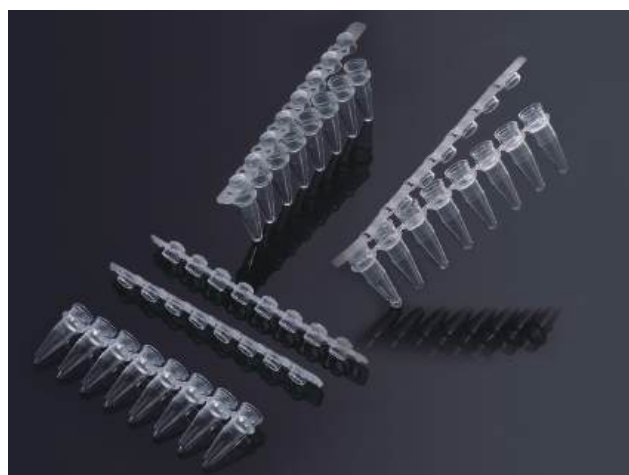


Description - spare parts	Sales Unit
AHN myDpette Bottle Top Dispenser 0.25-2.5 mL	Carton/1 pc.
AHN myDpette Bottle Top Dispenser 0.5-5.0 mL	Carton/1 pc.
AHN myDpette Bottle Top Dispenser 1.0-10.0 mL	Carton/1 pc.
AHN myDpette Bottle Top Dispenser 2.5-25.0 mL	Carton/1 pc.
AHN myDpette Bottle Top Dispenser 5.0-25.0 mL	Carton/1 pc.

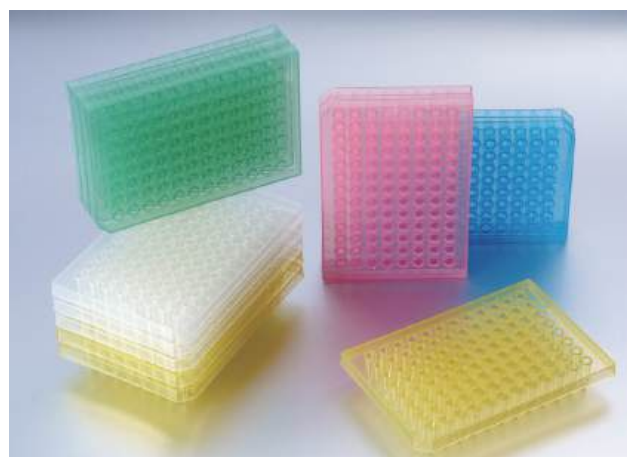
PCR Series



PCR Tube		
SH600082	0.2ml	Flat-top
SH600182	0.2ml	Flat-top
SH600282	0.2ml	Flat-top
SH600382	0.2ml	Flat-top
SH600482	0.2ml	Flat-top
SH600002	0.2ml	Dome-shaped
SH600102	0.2ml	Dome-shaped
SH600202	0.2ml	Dome-shaped
SH600302	0.2ml	Dome-shaped
SH600402	0.2ml	Dome-shaped



8-Strip PCR Tube		
SH600088	0.2ml	Flat
SH600008	0.2ml	Domed



PCR Plate		
SH600056	96-well	Half-Skirted

08. Incubator

Gravity Incubator



Model	SH-CH-54G	SH-CH-88G	SH-CH-150G
Temp Range	Ambient +5℃ to 70℃ (Ambient + 9°F to 158°F)		
Capa	54 L	88 L	150 L
Dimension Internal (W×D×H)	360x360x420mm	420x420x500mm	500x500x600mm
	14.17x16.54x16.54"	16.54x16.54x19.69"	19.69x19.69x23.62"
Dimension External (L×W×H)	490x475x665mm	550x550x755mm	630x630x850mm
	19.29x18.70x26.18"	21.65x21.65x29.72"	24.80x24.80x33.46"

Refrigerated Incubator



Model	SH-CH-149R	SH-CH-250R	SH-CH-480R
Temp Range	-5℃ to 60℃ (23°F to 60°F)		
Capa	150 L	250 L	480 L
IP54 등급 Concent			
Dimension Internal (W×D×H)	555x450x600mm	700x450x800mm	800x600x1000mm
	21.85x17.72x23.62"	27.56x17.72x31.50"	31.50x23.62x39.37"
Dimension External (W×D×H)	715x760x1350mm	860x760x1590mm	975x960x1935mm
	28.15x29.92x53.14"	33.86x29.92x62.59"	38.39x37.80x76.18"

4-Chamber Incubator



Model	SH-CH-500R
Temp Range	-5℃ to 60℃ 23°F to 140°F
Capa	125L x 4 Chambers, Total 500L
IP54 등급 Concent	1EA x 1 Room=4EA
Dimension Internal (W×D×H)	500x500x500mm x 4 Chambers
	19.69x19.69x19.69" x 4 Chambers
Dimension External (W×D×H)	1545x860x1775mm
	60.82x33.85x69.88"

CO₂ Incubators (Dry Heat Sterilization)



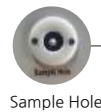
Model	CI-150C	CI-191C
Capa	150 L	191 L
Temp. Range	Ambient +5°C~60°C	
CO ₂ Range	0~20%	
Dimension External (W×D×H)	670x705x860mm	670x705x1010mm
	26.38x27.76x33.86"	26.38x27.76x39.76"
Chamber Dimensions (W×D×H)	540x510x555mm	540x510x705mm
	21.26x20.08x21.85"	21.26x20.08x27.76"
Sterilization	140°C (High Temp)	



- 도어 열선 장착으로 응축 현상 방지



- Alarm 상황 발생시 시각적 & 청각적 경보 제공



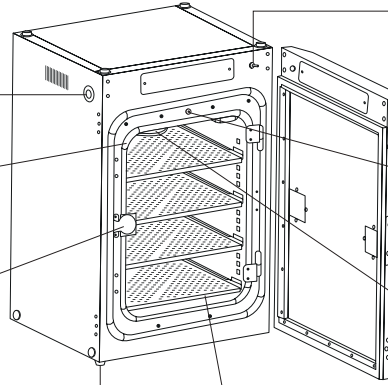
Sample Hole



- 강력한 Inner Glass Door Handle 및 Sealing Gasket 장착으로 누출 방지



- Mirror Stainless Steel과 Round 코너 처리로 세척 용이



Leveling Foot

Tray



- Out Door Glass Door Sensor 장착



- Inner Glass Door Sensor 장착



- HEPA 공기필터로 class 100 이하의 청정공기 제공

CO₂ Incubators (Dry Heat Sterilization)

NEW



Model	CI-150CX	CI-191CX	CI-191CHX
Capa	150 L	191 L	
Temp. Range	Ambient +5°C~60°C		
CO ₂ Range	0~20%		
Humidity Type	Evapor Naturally		RH Control
Max. Humidity	90%(@37℃)		95%(@37℃)
RH Control Range	NA		Ambient Humidity +10%~90%
Dimension External (W×D×H)	670x705x860mm	670x705x1010mm	
	26.38x27.76x33.86"	26.38x27.76x39.76"	
Chamber Dimensions (W×D×H)	540x510x555mm	540x510x705mm	
	21.26x20.08x21.85"	21.26x20.08x27.76"	
Sterilization	140°C (High Temp)		



▼ CI-191CHX



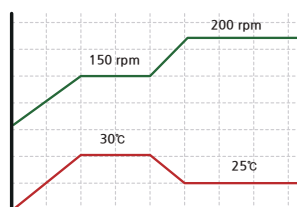
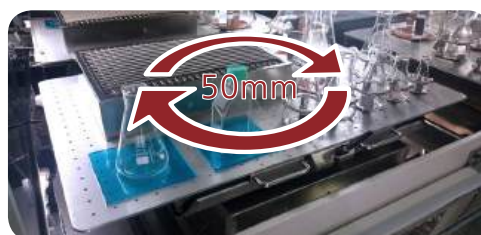
CO₂ Orbital Shakers

NEW

CO₂ Orbital Shaker works in CO₂ Incubators

Model	OSC-26MW
Orbit Diameter	Φ19mm(0.75")
Speed	30rpm~300rpm
Platform Dimensions	305x360mm/12x14.2"
Dimension External (W×D×H)	375x300x94mm/14.8x11.8x3.7"
Dimension Internal (W×D×H)	90x160x26mm/3.5x6.3x1.02"
Ambient	Temp. : 5℃~60℃ / Humidity : < 99%RH

09. Incubator Shaker & Shaker

Stackable Incubator Shakers(Temp / R.P.M / CO₂ / Humi Control)◀12 Program 온도, 속도 설정 가능
Fix / Program mode 운전 가능▼10인치 터치패널을 통한
다양한 운전환경 제공▲광폭 진탕제어 50mm Orbit Diameter
(IS-RDS6C5, 5T/IS-RDS4C5, 5T 모델)▶USB를 이용하여 Data Download 가능
(Excel File 저장)*Technical Specification Comparison Table
Stackable Shaking Incubator

	IS-AX	IS-RDS4 Series					IS-RDS6 Series				
Model	IS-AX	IS-RDS4	IS-RDS4T	IS-RDS4C5	IS-RDS4C5T	IS-RDS4C5H	IS-RDS6	IS-RDS6T	IS-RDS6C5	IS-RDS6C5T	IS-RDS6C5H
Temp Program Control	O	O	O	O	O	O	O	O	O	O	O
R.P.M Program Control	O	O	O	O	O	O	O	O	O	O	O
CO ₂ Program Control	X	X	X	O	O	O	X	X	O	O	O
Humi Program Control	X	X	X	X	X	O	X	X	X	X	O
Humi alarm&display	X	O	O	O	O	O	O	O	O	O	O
Orbital Dia		26mm		50mm			26mm		50mm		
Door type	Leftward	Downward	Upward	Downward	Upward	Downward	Upward	Downward	Upward	Downward	Downward
Room Capacity	190L	250L					401L				
Flask Load Capa (1000mL, Sticky mat)	12ea	15ea					28ea				
Flask Load Capa (250mL, Sticky mat)	30ea	40ea					77ea				

191L

Orbit Diameter



3단 적재시

Model	IS-AX
Capacity	190L
Max. Capacity	Clamp Capacity : 50mlx72 / 100mlx30 / 150mlx30 / 200mlx30 / 250mlx30 / 500mlx16 / 1000mlx12 / 2000mlx6 / 5000ml(Thomson)x4 Sticky Mat : 50mlx80 / 100mlx42 / 150mlx42 / 200mlx30 / 250mlx30 / 500mlx20 / 1000mlx12 / 2000mlx8 / 5000ml(Thomson)x4
Temp Control 12program	Room Temp. - 18℃~60℃ (Lowest Temp. is 4℃)
Speed Control 12program	0 (Stop), 30~300rpm
CO ₂ Control 12 Program	NA
Humidity Alarm	NA
Humidity Controller	NA
Dimension External (WxDxH)	970x780x620mm / 38.19x30.71x24.41"

250L

Orbit Diameter

CO₂ Controller

Humi Controller



2단 적재시

Model	IS-RDS4	IS-RDS4C5	IS-RDS4C5H
Capacity	250L		
Max. Capacity	Clamp Capacity : 50mlx104 / 100mlx45 / 150mlx45 / 200mlx45 / 250mlx40 / 500mlx25 / 1000mlx15 / 2000mlx12 / 5000ml(Thomson)x6 Sticky Mat : 50mlx112 / 100mlx72 / 150mlx60 / 200mlx45 / 250mlx40 / 500mlx28 / 1000mlx15 / 2000mlx12 / 5000ml(Thomson)x6		
Temp Control 12program	Range Room Temp -18℃~60℃ (Lowest Temp. is 4℃), Accuracy ±0.1℃ , Uniformity ±0.5℃ (@37℃)		
Speed Control 12program	Range 0 (Stop), 30~300rpm, Accuracy ±1rpm		
CO ₂ Control 12 Program	NA	Range 0~20%, Accuracy ±0.1%	
Humidity Alarm	Yes		
Humidity Controller	NA	NA	Range 10~95%, Accuracy ±5%
Dimension External (WxDxH)	1240x785x615mm / 48.82x30.91x24.21"		

401L

Orbit Diameter

CO₂ Controller

Humi Controller



3단 적재시

Model	IS-RDS6	IS-RDS6C5	IS-RDS6C5H
Capacity	401L		
Max. Capacity	Clamp Capacity: 50mlx135/100mlx60/150mlx60/200mlx60/250mlx60/500mlx34/1000mlx22/2000mlx15/3000mlx12 Sticky Mat: 50mlx187/100mlx112/150mlx104/200mlx84/250mlx77/500mlx40/1000mlx28/2000mlx15/3000mlx15/5000ml(Thomson)x6		
Temp Control 12program	Range Room Temp -18℃~60℃ (Lowest Temp. is 4℃) , Accuracy ±0.1℃ , Uniformity ±0.5℃ (@37℃)		
Speed Control 12program	Range 0 (Stop), 30~300rpm, Accuracy ±1rpm		
CO ₂ Control 12 Program	NA	Range 0~20%, Accuracy ±0.1%	
Humidity Alarm	Yes		
Humidity Controller	NA	NA	Range 10~95%, Accuracy ±5%
Dimension External (W×D×H)	1400x886x680mm / 55.12x34.88x26.77"		

Chest-Type Incubator Shakers



Model	IS-RDH1	IS-RDH2
Speed	0rpm, 30~300rpm	
Temp. Range	Ambient-15°C~60°C (Low Limit:4°C)	
Dimension External (W×D×H)	1200x700x850mm / 47.24x27.56x33.46"	1200x700x960mm / 47.24x27.56x37.80"
Dimension Internal (W×D×H)	960x560x350mm / 37.80x22.05x13.78"	960x560x420mm / 37.80x22.05x16.54"

Bench-Top Incubator Shakers



Model	IS-RDD3
Speed Range	0rpm, 30~400rpm
Temp. Range	Ambient-15°C~60°C (Low Limit : 4°C)
Dimension External (W×D×H)	700x650x620mm / 29.92x25.59x24.41"
Dimension Internal (W×D×H)	500x460x350mm / 19.69x18.11x13.78"

Mini Incubator Shakers (Double-Tire Extension)



Optimized for PCR

Model	IS-RDD83	IS-RDD84
Temp. Range	Ambient: -10~95°C (Low limit: 5°C)	Ambient: -10~65°C (Low limit: 5°C)
Capacity	4.6L	5L
Speed Range	150~1200rpm	150~1200rpm (Microplate) 150~600rpm (Tube Block)
Platform Dimensions	180x140mm / 7.08x5.51"	
Interior Dimensions (W×D×H)	178x225x115mm / 7x8.85x4.52"	178x225x127mm / 7x8.85x5"
Outside Dimensions (W×D×H)	296x440x316mm / 11.65x17.32x12.44"	

Single Door Double Layers Upright Incubator Shakers



Model	IS-RDV1
Speed Range	0rpm, 30~300rpm
Temp. Range	Ambient-15°C~60°C (Low Limit:4°C)
Dimension External (W×D×H)	700x650x1320mm / 27.56x25.59x51.97"
Dimension Internal (W×D×H)	598x424x676mm / 23.54x16.69x26.61"

Double Doors Double Layers Upright Incubator Shakers



Model	IS-RDV3	IS-RDV35
Speed	0rpm, 30~300rpm	0rpm, 30~250rpm
Temp. Range	Ambient-15°C~60°C (Low Limit:4°C)	
Dimension External (W×D×H)	1102x810x1458mm / 43.39x31.89x57.40"	
Dimension Internal (W×D×H)	1000x590x800mm / 39.37x23.23x31.50"	

Water Bath Shaker



Model	SY-2230
Temp. Range	(Ambient)+5°C~99°C
Speed	20~200rpm
Tank Internal Dimensions (W×D×H)	528x308x226mm / 20.79x12.13x8.90"
Dimension External (W×D×H)	750x380x326mm / 29.53x14.96x12.83"
Max. Power Consumption	1400W

Wrist Shaker



Model	OS-08W
Speed Range	Variable speed: 50~450 rpm
Mixing Angle	1~15° Adjustable
Dimension External (W×D×H)	425x376x413mm / 16.73x14.80x16.26"

Orbit Shaker



Model	SYC-2102
Rotation Method	Circular Orbit, 26mm dia.
Speed	30~300rpm
Platform Dimensions	400x400mm / 15.75x15.75"
Dimension External (W×D×H)	432x535x190mm / 17.01x21.06x7.48"
Ambient	5°C~40°C

Mini Orbital Shaker



Model	OS-03U
Orbit Size	Φ20mm
Speed Range	30~300rpm
Platform Dimensions	307x267mm / 12.09x10.51"
Dimensions (WxDxH)	355x240x123mm / 13.98x9.45x4.84"
Ambient	5°C~40°C

Multi-purpose Shaker



Model	OS-06U
Speed Range	50~450rpm
Platform Dimensions	310x284mm / 12.20x11.18"
Ambient	5°C~40°C
Dimension External (LxWxH)	440x296x129mm / 17.32x11.65x5.08"

Microplate Shaker



Model	OS-07U
Speed Range	150~1000rpm
Platform Dimensions	275x235x12mm / 10.83x9.25x0.47"
Display	LED
Ambient	5°C~40°C
Dimension External (WxDxH)	296x440x151mm / 11.65x17.32x5.94"

10. Roller Culture Apparatus

Roller Culture Apparatus for Monolayer and Suspension Cell Cultures



Model	CG III -45-SF	CG III -40-SF	CG III -30-SF
Bottle Rotation Speed	1~990r/h(Φ110mm Bottle)		
Rotation Mode	Clockwise/Counter Clockwise Alternating Rotation(10 ⁰ ~180 ⁰)		
Bottle Positions per Deck	5	5	5
Bottle Positions	45	40	30
Material	Stainless Steel		
Digital Control Display	Speed/Angle/Time		
Dimension External (WxDxH)	770x594x1850mm 30.31x23.39x72.83"	770x594x1668mm 30.31x23.39x65.67"	770x594x1304mm 30.31x23.39x51.34"



Model	CG III-16-SF
Bottle Rotation Speed	1~990r/h(Φ110mm Bottle)
Rotation Mode	Clockwise/Counter Clockwise Alternating Rotation(10°~180°)
Bottle Positions per Deck	4
Bottle Positions	16
Material	Stainless Steel
Digital Control Display	Speed/Angle/Time
Dimension External (WxDxH)	672x598x936mm 26.46x23.54x36.85"

Roll-in Incubators



Model	PY-16	PY-30	PYC-16	PYC-30
Temp. Range	Environment Temperature +5℃~60℃			
CO ₂ Control Range	N/A		0-20%	
Internal Dimensions (WxDxH)	745x690x1115mm 29.33x27.17x43.90"	870x690x1480mm 34.25x27.17x58.27"	745x690x1115mm 29.33x27.17x43.90"	870x690x1480mm 34.25x27.17x58.27"
Overall Dimensions (WxDxH)	845x885x1349mm 33.27x34.84x53.11"	970x885x1716mm 38.19x34.84x67.56"	845x885x1349mm 33.27x34.84x53.11"	970x885x1716mm 38.19x34.84x67.56"

Bench Top Roller Culture Apparatus



Model	CG I-06-F	CG-06-F
Cell Culture Method	Monolayer Cultures	Suspension Cultures
Bottle Rotation Speed	3~60r/h(Φ110mm Bottle)	60~900r/h(Φ110mm Bottle)
Rotation Mode	Clockwise/Counter Clockwise Alternating Rotation(10°~180°)	
Bottle Positions per Deck	3	
Bottle Positions	6	
Material	Aluminum	
Digital Control Display	Speed/Angle/Time	
Dimension External (WxDxH)	450x360x431.5mm 17.72x14.17x16.99"	

11. Autoclave

Tabletop Autoclave

- Class B AUTOCLAVE



Water tank



Model	SH-AC-18B	SH-AC-23B
Temp Range	Max 138℃ (280.4°F)	
Volume	18 L	23 L
Tray	3 EA	3 EA
Chamber size	Ø247x350mm	Ø247x450mm
Dimension External (WxDxH)	480x700x450mm	480x700x450mm
	18.89x27.56x17.71"	18.89x27.56x17.71"
USB port and USB stick	Standard	
Printer	Option	



Simple opening and closing

터치형 도어 잠금 기능으로 부드러운 터치만으로도 도어의 개폐가 쉬움 (AC-33B 전용)

Model	SH-AC-33B
Temp Range	Max 138℃ (280.4°F)
Volume	33 L
Tray	Stainless Steel Rack
Chamber size	Ø319x417mm
Dimension External (WxDxH)	640x640x560mm
	25.2X25.2X22.0"
USB port and USB stick	Standard
Printer	Standard(Built-in)

- Class N AUTOCLAVE



Model	SH-AC-18N	SH-AC-23N
Temp Range	Max 138℃ (280.4°F)	
Volume	18 L	23 L
Tray	3 EA	3 EA
Chamber size	Ø247x350mm	Ø247x450mm
Dimension External (WxDxH)	480x700x450mm 18.89x27.56x17.71"	480x700x450mm 18.89x27.56x17.71"
USB port and USB stick	Standard	
Printer	Option	

Vertical Autoclave



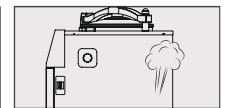
Stainless steel with Plastic Cover



Safe Door Opening Mechanism



Water level sensor



Automatic over-pressure protection function

Model	SH-AC-60M	SH-AC-80M	SH-AC-100M
Temp Range	Max 130℃(266°F)		
Temp&Pressure *주변환경에 따라 변동될 수 있습니다.	121℃(249.8°F) / 20psig (0.14MPa) 125℃(257°F) / 24psig (0.16MPa) 130℃(266°F) / 30psig (0.2MPa)		
Capa	60 L	80 L	100 L
Dimension Internal	344Øx650Hmm 13.54x25.59"	400Øx650Hmm 15.75x25.59"	440Øx650Hmm 17.32x25.59"
Dimension External (WxDxH)	550x680x1090mm 21.65x26.77x42.91"	600x730x1150mm 23.62x28.74x45.28"	600x780x1183mm 23.62x30.71x46.58"



Model	SH-AC-150M
Temp Range	Max 130℃(266°F)
Temp&Pressure *주변환경에 따라 변동될 수 있습니다.	121℃(249.8°F) / 20psig (0.14MPa) 125℃(257°F) / 24psig (0.16MPa) 130℃(266°F) / 30psig (0.2MPa)
Capa	150 L
Dimension Internal	514Øx748.43Hmm 20.34x29.47"
Dimension External (WxDxH)	608x750x1078mm 23.94x29.53x42.44"

12. Clean Bench & Colony Counter

Laminar Clean Bench



Model	SH-HD-900V	SH-HD-1500V	SH-HD-1900V
Type	Laminar Vertical Down Flow		
Dimension Internal (W×D×H)	860x613×690mm	1391x613×690mm	1891x613×690mm
	33.86x24.13x27.17"	54.76x24.13x27.36"	74.45x24.13x27.36"
Dimension External (W×D×H)	966x696×1820mm	1496x696×1820mm	1996x696×1820mm
	35.43x30.71x74.80"	59.06x30.71x74.80"	74.80x30.71x74.80"

Biosafety Cabinet



Model	SH-HD-900B	SH-HD-1500B	SH-HD-1900B
Type	Bio Hazard Type(Class II A2)		
Dimension Internal (W×D×H)	740x600x750mm	1340x600x750mm	1740x600x750mm
	29.13x23.62x29.53"	52.76x23.62x29.53"	68.50x23.62x29.53"
Dimension External (W×D×H)	900x825×1875mm	1500x825×1900mm	1900x825×1900mm
	35.43x32.48x73.81"	59.06x32.48x74.80"	74.80x32.48x74.80"

eCount

Rapid, Accurate Colony Counter



Model	SH-HS120000	SH-HS120228
LxWxH	170x30mm / 7.4x1.2"	CR1225 Battery
Package Includes		
eCount Colony Counter	Hand Tally Adapter	
Black Sharpie® Pen	Weighted Stand	
*Long-life Battery included		

Mini LED Light Box and Magnifier



Model	SH-120661 (Mini Light Box)	SH-HS15092D (Mini Magnifier)
LxWxH	158x150x14mm / 6.2x5.9x0.6"	161x135x207mm / 6.3x5.3x8.1"
*Long-life Battery included		

13. Mixer Tube Revolvers



Model	TR-01
Product Name	Fix Speed Tube Revolver
Rotation Speed	16rpm
Capacity	84x0.5ml tubes, 60x1.5ml/2.0ml tubes, 28x5ml/7ml tubes, 24x10ml tubes, 6x50ml tubes
Dimension External (WxDxH)	225x148x195mm / 8.86x5.83x7.68"

Model	TR-02U
Product Name	Variable Speed Tube Revolver with Digital Display
Rotation Speed	10~40rpm
Capacity	84x0.5ml tubes, 60x1.5ml/2.0ml tubes, 28x5ml/7ml tubes, 24x10ml tubes, 6x50ml tubes
Dimension External (WxDxH)	260x148x195mm / 10.24x5.83x7.68"

Tube Rotator



<TR-03>



<TR-04U>

Model	TR-03	TR-04U
Product Name	Standard Tube Rotator	Digital Tube Rotator
Speed Range	20rpm	2~80rpm
Tilt Angle	90°~180°	
Carousel Capacity	12x50ml/18x15ml/48x2ml/9x50ml & 9x15ml & 12x2ml	
Dimension External (W×D×H)	334x215x405mm / 13.15x8.46x15.94"	

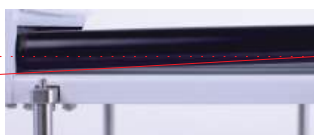


Model	TR-05U
Product Name	High Profile Tube Rotator
Speed Range	5~80rpm
Tilt Angle	95°~180° Six steps : 95°, 112°, 129°, 146°, 163°, 180°
Carousel Capacity	50x9~11mm Tube (1.5/2ml), 30x14~18mm Tube (15ml) 15x27~34mm Tube (50ml), 10x10~15ml, 20x5~7ml Tube 64x18.5mm Drum Carousel, 64x14mm Drum Carousel
Dimension External (W×D×H)	440x296x466mm / 17.32x11.65x18.35"

Mini Roller



Horizontal Line
18mm
(0.7")
Swing Dimensions



(MR-03U Combining both rolling and swinging motion)

Model	MR-01U	MR-03U
Product Name	Stackable Mini Roller	Stackable 3D Mini Roller
Roller Speed	0.5~80rpm	
Max. Load	6.5kg	
Dimension External (W×D×H)	378x360x72mm / 14.88x14.17x2.83"	

Vortex Mixers



Model	VM-01U
Operation Mode	Touch Control, Timed or Continuous Run
Orbital Diameter	Circular Orbit, 4.5mm Dia.
Stirring Speed	200~3000rpm
Speed Display	Panel Marker
Timing Display	LED
Max. Load	0.5 kg
Dimensions(W×D×H)	205x155x108mm / 8.07x6.10x4.25"

CE



Model	VM-02U
Operation Mode	Touch Control, Timed or Continuous Run
Orbital Diameter	Circular Orbit, 4.5mm Dia.
Stirring Speed	200~3000rpm
Speed Display	LED Digital Display
Timing Display	LED
Max. Load	0.5 kg
Dimensions(WxDxH)	205x155x108mm / 8.07x6.10x4.25"

CE



Model	VM-03U	VM-03RU	VM-03GU	VM-03PU
Product Name	Mini Vortex Mixer			
Color	Blue	Red	Green	Purple
Orbital Diameter	Φ4.5mm / Φ0.18"			
Speed	3000rpm			
Dimension External (ΦxH)	96x85mm / 3.78x3.35"			

3D Nutating Shaker

CE



Model	NS-01
Speed	20rpm
Tilt Angle	20°
Max. Load	0.8 kg
Dimension External (L×W×H)	260x260x190mm / 10.24x10.24x7.48"

20° 3D Digital Nutating Mixer

CE



Model	NS-11U
Speed Range	2~60rpm
Tilt Angle	20°
Dimension External (L×W×H)	440x296x255mm / 17.32x11.65x10.04"
Max. Load	0.8kg

20° 3D Nutating Mixer



Model	NS-10
Speed Range	20rpm
Tilt Angle	20°
Dimension External (L×W×H)	440x296x261mm / 17.32x11.65x10.28"
Max. Load	0.8kg

7° 3D Digital Nutating Mixer



Model	NS-12U
Speed Range	2~80rpm
Tilt Angle	7°
Dimension External (L×W×H)	440x296x189mm / 17.32x11.65x7.44"
Max. Load	0.8kg

Multi Modes Nutating Mixer



Model	NS-13U
Speed Range	5~100rpm
Turning Angle(reciprocal mode)	0~360°
Turning Angle(vibration mode)	0~5°
Tilt Angle	7°
Dimension External (L×W×H)	440x296x226mm / 17.32x11.65x8.90"
Max. Load	1 kg

2D Adjustable Rocker



Model	NS-14U
Speed Range	1~50rpm
Tilt Angle	1~15° (adjustable in 1° increments)
Dimension External (W×D×H)	440x296x209mm / 17.32x11.65x8.23"
Max. Load	4.5 kg

2D Rocker



Model	NS-04
Speed	20rpm
Speed Control	N/A
Tilt Angle	±10°
Display	N/A
Dimension External (W×D×H)	355x240x123mm / 13.98X9.45X4.84"



Model	NS-05U
Speed	2~80rpm
Speed Control	1rpm
Tilt Angle	±10°
Display	LED
Dimension External (W×D×H)	355x240x123mm / 13.98X9.45X4.84"

14. Stirrer

Mini Magnetic Stirrer (Motorless Driven Mechanism)



U.S. Patent



Model	SH-120155
Speed Range	200 to 2200 rpm (Infinitely adjustable)
Capacity	Up to 3L H ₂ O
LxWxH	183x132x47mm / 7.2x5.2x1.85"
UOM	1 ea, 8 ea/cs
Noise Level	Full speed, unloaded: 53 dB
Item No. / Color	120155 / ■ Gray, Purple
Package Includes	
Mini Magnetic Stirrer	
2 High quality stir bars (35x9mm)	

Magnetic Induction Stirrer (Motorless Driven Mechanism)



Model	SH-120584
Operating Temp.	Range 2°C to 40°C
Speed Range	10 to 2,000 rpm variable, 1 rpm increments
Capacity	3 liters plus
LxWxH	245x195x75mm / 9.7x7.7x3"
UOM	1 ea, 2 ea/cs
Item No. / Color	120584 / Gray, Purple
Package Includes	
Magnetic Induction Stirrer	
2 Stir bars 35 mm x 12 mm Cylindrical Shape)	
1 Silicone anti-skid plate cover	

Magnetic Stirrer (Motorless Driven Mechanism)



· 점도별 5단계 제어



0001~0005

Model	MS-01U
Stirring Speed	80~2000rpm
Display Screen	LED
Maximum Capacity	2000ml
Platform Dimensions	140x140mm / 5.51x5.51"
Outside Dimensions(WxDxH)	160x200x48mm / 6.30x7.87x1.89"

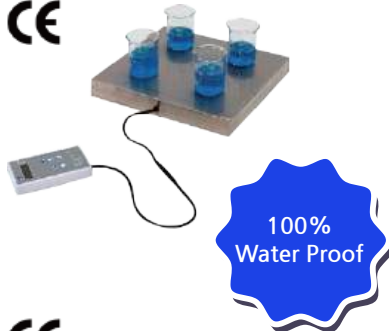
Low profile Magnetic Stirrer (Motorless Driven Mechanism)



Model	MS-01FU
Speed	15~1500rpm
Max.Capacity	500ml
Platform Dimensions	110x110mm / 4.33x4.33"
Max.Load	2 kg
Dimension External (WxDxH)	175x125x20.5mm / 6.89x4.92x0.81"

Position Magnetic Stirrer (Motorless Driven Mechanism)

CE



Model	MS-04BSU
Stirring Speed	80~2000rpm
Number of Stirring Positions	4
Maximum Capacity(Water)	1000ml
Viscosity Level	5
Platform Dimensions	250X270mm / 9.84x10.63"
Dimension External (WxDxH)	280X398X40mm / 11.02x15.67x1.57"

CE



Model	MS-06BSU
Stirring Speed	80~2000rpm
Number of Stirring Positions	6
Maximum Capacity(Water)	1000ml
Viscosity Level	5
Platform Dimensions	385X270mm / 15.16x10.63"
Dimension External (WxDxH)	280X528X40mm / 11.02x20.79x1.57"

CE



Model	MS-15BSU
Stirring Speed	80~2000rpm
Number of Stirring Positions	15
Maximum Capacity(Water)	500mlx8, 200mlx15
Viscosity Level	5
Platform Dimensions	385X270mm / 15.16x10.63"
Dimension External (WxDxH)	280X528X40mm / 11.02x20.79x1.57"

* Accessories



<SH-HSTP>



<SH-HSSR>



<SH-HSHC>

Magnetic Hot Plate Stirrer

-Digital Model



Model		SH-HSD	SH-HSD-Set
Material	Body	Powder Coated Aluminum	
	Plate	Ceramic Coated Aluminum	
Plate Size		180x180mm	
		7.09x7.09"	
Stirrer Speed Range		80 to 1,500rpm	
Maximum Stirring Volume (H ₂ O)		20Lit.	
Heating Temperature		Max. 400°C	
Weight		4 kg	4.6 kg
Accessories		N/A	SH-HSTP, SH-HSSR, SH-HSHC

-Analog Model



Model		SH-HSA
Material	Body	Powder Coated Aluminum
	Plate	Ceramic Coated Aluminum
Plate Size		180x180mm
		7.09x7.09"
Stirrer Speed Range		Max. 1,500rpm
Maximum Stirring Volume (H ₂ O)		20Lit.
Heating Temperature		Max. 400°C
Weight		4 kg
Accessories		N/A

Multi Hotplate Stirrer

NEW



〈SH-HSD-M3〉

Model	SH-HSD-M3	SH-HSD-M3-Set
Stirrer Speed Range	80 to 1,500rpm	
Speed control resolution	5 rpm	
Maximum Stirring Volume(H ₂ O)	20Lit.	
Plate Dimension	140 x 140mm x 3 Place	
heating Temperature	Max. 350°C	
Temp Accuracy	±0.3°C	
Dimensions(WxDxH)	500x240x110mm	
Accessories	-	SH-HSTP, SH-HSSR, SH-HSHC

NEW



〈SH-HSD-M6〉

Model	SH-HSD-M6	
Stirrer Speed Range	80 to 1,500rpm	
Speed control resolution	5 rpm	
Maximum Stirring Volume(H ₂ O)	20Lit.	
Plate Dimension	140 x 140mm x 6(2x3) Place	
heating Temperature	Max. 350°C	
Temp Accuracy	±0.3°C	
Temp Resolution	Display	0.1°C
	Control	0.1°C
Dimensions(WxDxH)	500x430x130mm	

Multi-Position Hot Plate Stirrers

CE



Model	MS-05H
Stirring Speed	0rpm, 100~1200rpm
Temp. Range	Ambient +5°C~120°C
Max. Capacity (water)	500ml
Number of Stirring Positions	5
Dimension External (WxDxH)	130x612x78mm 5.12x24.09x3.07"

CE



Model	MS-10H
Stirring Speed	0rpm, 100~1200rpm
Temp. Range	Ambient +5°C~120°C
Max. Capacity (water)	500ml
Number of Stirring Positions	10
Dimension External (WxDxH)	214x612x78mm 8.43x24.09x3.07"

CE



Model	MS-15H
Stirring Speed	0rpm, 100~1200rpm
Temp. Range	Ambient +5°C~120°C
Max. Capacity (water)	500ml
Number of Stirring Positions	15
Dimension External (WxDxH)	310x612x78mm 12.20x24.09x3.07"

MagFuge®

The Industry's first High-speed

Centrifuge and Magnetic Stirrer in ONE unit



Two stir bars included
Rare-Earth PTFE coated stir bars

Model	SH-120581	
	Centrifuge	Magnetic Stirrer
Operating Temp.	Range 2°C to 40°C	
Speed Range	500 ~ 999 rpm (increments of 1); 1,000 ~ 5,000 rpm (increments of 5); 5,000 ~ 12,500 rpm (increments of 25)	50 ~ 999 rpm (increments of 1); 1,000 ~ 2,500 rpm (increments of 5)
Capacity	12 place 1.5/2.0 mL tube rotor: 24 mL total maximum liquid volume 6 place 5 mL tube rotor: 30 mL total maximum liquid volume	3 liters plus. Diameter of vessel not to exceed 130 mm (6.70")
LxWxH	235x185x120mm / 9.2x7.3x4.7"	
UOM	1/ea, 2ea/cs	
Item No. / Color	120584 /  Gray, Purple	
Package Includes		
Stirrer Rotor	12 Tube Adapters 0.2 mL	5 mL Tube Rotor
2 Stir Bars	12 Tube Adapters 0.5 mL	1.5 mL Tube Rotor

Mini Centrifuges



Model	MLX-104	MLX-106	MLX-110
Rotating Speed	4000rpm	6000rpm	10000rpm
G-force(RCF)	900g	2000g	5000g
Noise Level	≤60dB	≤60dB	≤60dB
Dimension External (WxDxH)	155x140x115mm / 6.10x5.51x4.53"		



Model		MLX-200F		
Rotating Speed		6000rpm		9200rpm
Rotor	G-force(RCF)	5ml	2414g	5677g
		1.5ml	2012g	4731g
		0.2ml	1509g	3549g
Noise Level		≤60dB		
Dimension External (W×D×H)		187x240x139mm / 7.36x9.45x5.47"		

Viscosity Measuring Digital Overhead Stirrer

NEW



Model	SH-OSDV-25	SH-OSDV-25-Set
Stirring Capacity	Max.25 Lit.(H ₂ O)	
Maximum Viscosity	70,000 mPas	
Maximum Torque	130Ncm	
Gear Rate	2.0:1	
Speed Range	30 to 2,000rpm	
Display	Touch Color LCD	
Controller	Digital PID Feedback Control, High Performance MCU	
Viscosity Measurement Range	1/0,000 to 100,000 mPas (15% Deviation)	
Others - Special function	Torque Measuring Function in real time, Viscosity Calculation Target Viscosity Operation, Real Time Monitoring by USB port	
Accessories	-	SH-OSSR030, SH-OSHC010, SH-OSIP010

Model	SH-OSDV-60	SH-OSDV-60-Set
Stirring Capacity	Max.60 Lit.(H ₂ O)	
Maximum Viscosity	150,000 mPas	
Maximum Torque	293Ncm	
Gear Rate	4.5:1	
Speed Range	15 to 800rpm	
Display	Touch Color LCD	
Controller	Digital PID Feedback Control, High Performance MCU	
Viscosity Measurement Range	1/0,000 to 100,000 mPas (15% Deviation)	
Others - Special function	Torque Measuring Function in real time, Viscosity Calculation Target Viscosity Operation, Real Time Monitoring by USB port	
Accessories	-	SH-OSSR030, SH-OSHC010, SH-OSIP030

Overhead Stirrer -Analog Model



-Digital Model



Model	SH-OSA-25	SH-OSA-25-Set	SH-OSA-60	SH-OSA-60-Set
Stirring Capacity	Max.25 Lit. (H ₂ O)		Max. 60Lit. (H ₂ O)	
Maximum Viscosity	70,000mPas		150,000mPas	
Maximum Torque	120 Ncm		270 Ncm	
Speed Range	100 to 2,000rpm		50 to 800rpm	
Speed Control	Stepless			
Safety Device	Over Heat Protector for Motor			
Accessories	N/A	SH-OSSR010, SH-OSIP010, SH-OSHC010	N/A	SH-OSSR010, SH-OSIP030, SH-OSHC010

Model	SH-OSD-25	SH-OSD-25-Set	SH-OSD-60	SH-OSD-60-Set
Stirring Capacity	Max.25Lit.(H ₂ O)		Max.60Lit.(H ₂ O)	
Maximum Viscosity	70,000mPas		150,000mPas	
Maximum Torque	120 Ncm		270 Ncm	
Speed Range	100 to 2,000rpm		50 to 800rpm	
Timer	99hr 59min			
Display	Digital LCD Display with Back-Light Function			
Safety Device	Overload & Heat Protector for Motor			
Accessories	N/A	SH-OSSR030, SH-OSHC010, SH-OSIP010	N/A	SH-OSSR030, SH-OSHC010, SH-OSIP030

* Accessories



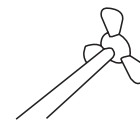
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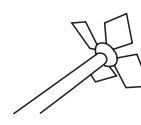
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<SH-OSHC010>



<SH-OSIP010>



<SH-OSIP030>

Model Name	Description
SH-OSSR010	Standard Plate Stand, for Overhead Stirrer and Homogenizer, 25x37cm Plate, with Rod and Silicon Pad
SH-OSSR030	Telescopic Stand for Overhead Stirrer & Homogenizer.
SH-OSHC010	Clamp for Overhead Stirrers and Homogenizers, up to Ø23mm Rod
SH-OSIP010	Propeller, 3 x bladed, Ø8 L500 mm rod, Ø50 mm blade, For SH-OSA-25-Set
SH-OSIP030	Diamond, 4 x bladed, Ø8 L500 mm rod, Ø90 mm blade, For SH-OSA-60-Set

Homogenizer

-Analog Model



Model	SH-HZA	SH-HZA-Set(A, B)
Speed Range	About 27,000rpm(without load)	
Maximum Volume (H ₂ O)	2,500ml	
Display	N/A	
Usable Dispersing Tool	Ø8mm Stator 'SH-HZ1008', Ø10mm Stator 'SH-HZ1010' Ø18mm Stator 'SH-HZ1018', Ø25mm Stator 'SH-HZ1025'	
Accessories	N/A	HZA-Set A : SH-HZ1018, SH-OSSR010, SH-OSHC010 HZA-Set B : SH-HZ1025, SH-OSSR010, SH-OSHC010

Model	SH-HZD	SH-HZD-Set(A, B)
Speed Range	2,000 to 27,000rpm(without load)	
Maximum Volume (H ₂ O)	2,500ml	
Display	3 Digit FND	
Usable Dispersing Tool	Ø8mm Stator 'SH-HZ1008', Ø10mm Stator 'SH-HZ1010' Ø18mm Stator 'SH-HZ1018', Ø25mm Stator 'SH-HZ1025'	
Accessories	N/A	HZD-Set A : SH-HZ1018, SH-OSSR030, SH-OSHC010 HZD-Set B : SH-HZ1025, SH-OSSR030, SH-OSHC010

-Digital Model



* Accessories



<SH-OSSR010>



<SH-OSSR030>



<SH-OSHC010>



<SH-HZ1008>



<SH-HZ1010>



<SH-HZ1018>



<SH-HZ1025>

Model Name	Description	Model Name	Description
SH-OSSR010	Standard Plate Stand, for Overhead Stirrer and Homogenizer, 25x37cm Plate, with Rod and Silicon Pad	SH-HZ1008	Homogenizer Tool, Rotor Ø6.1, Stator Ø8, 1~50 ml
SH-OSSR030	Telescopic Stand for Overhead Stirrer & Homogenizer.	SH-HZ1010	Homogenizer Tool, Rotor Ø7.5, Stator Ø10, 1~100 ml
SH-OSHC010	Clamp for Overhead Stirrers and Homogenizers, up to Ø23mm Rod	SH-HZ1018	Homogenizer Tool, Rotor Ø12.7, Stator Ø18, 10~1000 ml
		SH-HZ1025	Homogenizer Tool, Rotor Ø17, Stator Ø25, 50~2500 ml

Homo Mixer & Disper NEW



<SH-HD>



<SH-HM>

Model	SH-HM-SET	SH-HD-SET
Speed Range	500 to 13,000 rpm(without load)	500 to 8,000 rpm(without load)
Speed Control Resolution	50rpm	10rpm
Maximum Volume (H ₂ O)	Max. 3 Lit.	Max. 5 Lit.
Motor Wattage	300W	
Working Direction	Forward & Reverse (Can change automatically by program operation)	
Motor Type	Brushless DC Motor	
Safety Devices	Overload, Current & Heat Protector for Motor, Electric Error Code Display	
Permissible Temp. & Humidity	5℃~40℃, <80%	
Setting Tool	'SH-HMT' Homo Mixer Tool	'SH-HDT' Homo Disper Tool
Material	Turbine	SUS304
	Shaft	SUS304
Accessories	SH-HMT, SH-OSSR040	SH-HDT, SH-OSSR040

* Accessories



<SH-HMT>



<SH-HDT>



<SH-OSSR040>

Model Name	Description
SH-HMT	Homo Mixer Tool / Tool For Homomixer
SH-HDT	Homo Disper Tool / Tool For Homodisperser
SH-OSSR040	Telescopic Stand for Homo Mixing System, Stand For HM-SET, HD-SET

15. Rotary Evaporator



NEW

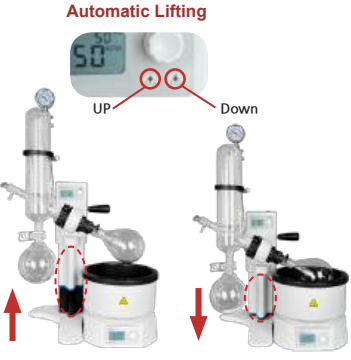
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Manual

SH-RE-1L

<SH-RE-1LA>



Automatic Lifting

UP Down

SH-RE-1LA

Model	SH-RE-1L	SH-RE-1LA
Operating Ambient Temperature	5~35℃ / 41~95°F	
Rotation Speed	20~300rpm	
Evaporating speed	22ml/min	
Bath temp. control range	Ambient~99.9℃/ Ambient~212°F	
Evaporating Flask Capacity	1L / 0.26 gallon	
Head Lift	Manual	Automatic



NEW

Automatic

<SH-RE-10L>



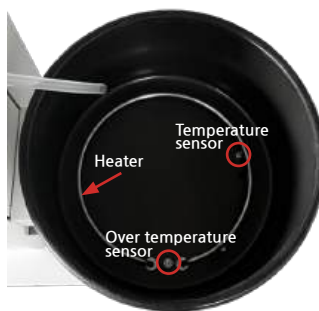
Automatic

<SH-RE-5L>

Model	SH-RE-5L	SH-RE-10L
Operating Ambient Temperature	5~35℃ / 41~95°F	
Rotation Speed	30 ~ 180 rpm	30 ~ 130 rpm
Bath temp. control range	Ambient~99℃/Ambient~210°F	
Evaporating Flask Capacity	5L / 1.32 gallon	10L / 2.64 gallon
Receiving Flask Capacity	3L / 0.79 gallon	5L / 1.32 gallon
Bath Lift	Automatic	



<SH-RE-20L/SH-RE-50L>



Automatic



Model	SH-RE-20L	SH-RE-50L
Operating Ambient Temperature	5~35℃ / 41~95°F	
Rotation Speed	30 ~ 130 rpm	30 ~ 110 rpm
Bath temp. control range	Ambient~99℃/ Ambient~210°F	Ambient ~85℃/ Ambient ~185°F
Evaporating Flask Capacity	20L / 5.28 gallon	50L / 13.20 gallon
Receiving Flask Capacity	10L / 2.64 gallon	20L / 5.28 gallon
Bath Lift	Automatic	

Rotary Evaporator Package



<SH-PK-5L>

Package Model	RE-PK-5L
Rotary Evaporator	SH-RE-5L
Cold Trap(Optional)	SH-WB-5GDR (-40℃ or -80℃)
Vacuum Pump	SH-PTFEVP30C
Chiller	SH-WB-5CDR(-30)



<SH-PK-10L>

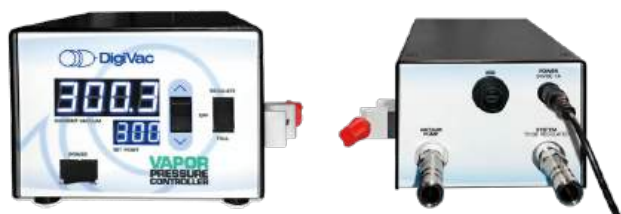
Package Model	RE-PK-10L
Rotary Evaporator	SH-RE-10L
Cold Trap(Optional)	SH-WB-5GDR (-40℃ or -80℃)
Vacuum Pump	SH-PTFEVP30C
Chiller	SH-WB-12CDR



〈SH-PK-50L〉

Package Model	RE-PK-20L	RE-PK-50L
Rotary Evaporator	SH-RE-20L	SH-RE-50L
Cold Trap(Optional)	SH-WB-5GDR (-40℃ or -80℃)	
Vacuum Pump	SH-PTFEVP30C	2047 or SH-PTFEVP60C
Chiller	SH-WB-20CDR	SH-WB-40CDR

Vacuum Controller for SH Rotary Evaporator



Model	SH-VPC
Vacuum Reference	Absolute
Sensor	SEN-775i-NPT-OEM
Resolution	± 0.1 Torr
Units	Torr, mbar
Accuracy	±2.66 mbar; ± 2 Torr
Range	1~1013 mbar; 1~775 Torr
Control band	±2.66 mbar; ±2 Torr
Product Dimensions (WxDxH)	114.3x158.75x85.725mm / 4.5 x 6.25 x 3.375"

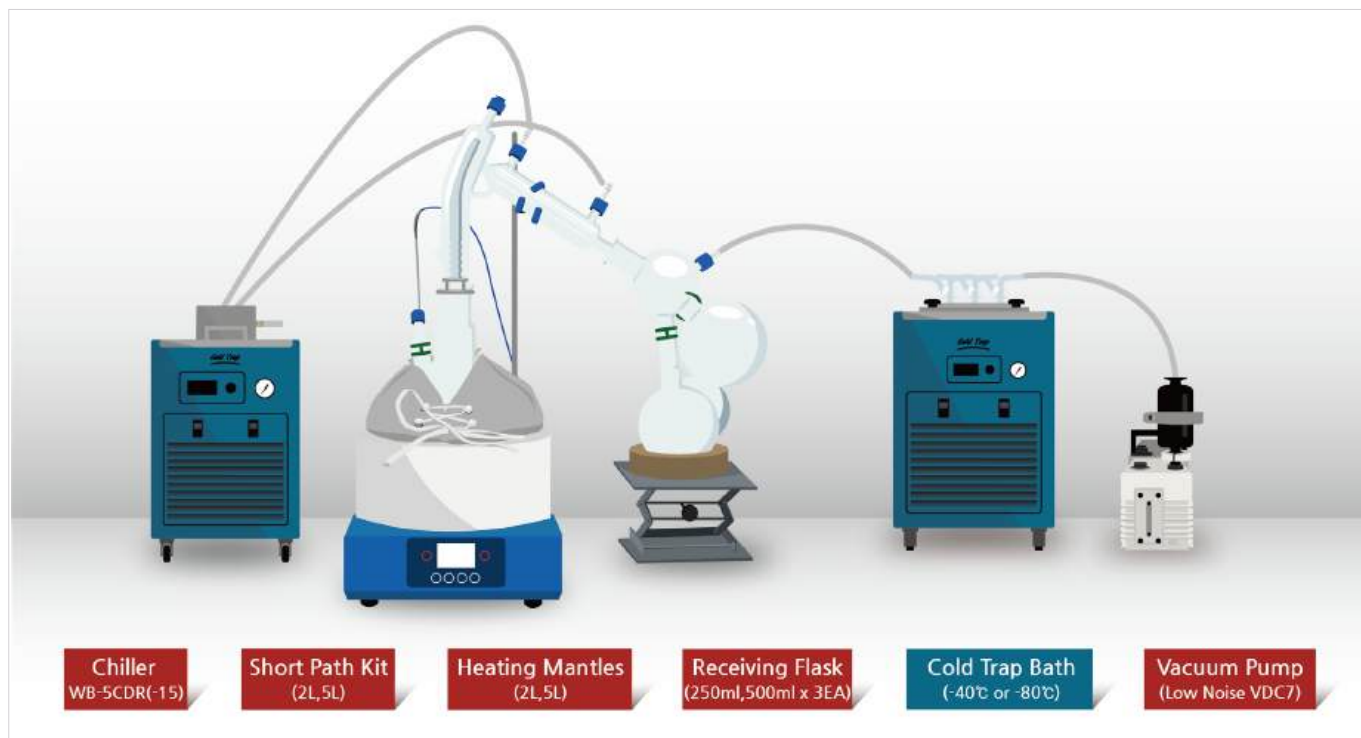
Digital Program Vacuum Controller



〈 세로 설치 시 〉

Model	SH-SVPC
Vacuum Reference	Absolute
Sensor	DPCP, 측정범위 $7.5 \times 10^{(-6)}$ ~ 1000 Torr
Resolution	± 0.1 Torr
Units	Torr, mTorr, KPa
Accuracy	± 5% at Reading
Range	Control 0.01~775Torr / Measure $7.5 \times 10^{(-6)}$ ~1000Torr
Control band	±2.66 mbar; ±2 Torr
Product Dimensions (WxDxH)	가로 설치시 234.95x107.95x120.65mm / 세로 설치시 120.65x107.95x234.95mm

Short Path Full Kit With Cold Trap



Parts for SH Rotary Evaporator



Item	Specifications
1L Evaporating Flask	-
2L Evaporating Flask	-
5L Evaporating Flask	Length 54mm Flange 50mm
10L Evaporating Flask	Flange 125mm
20L Evaporating Flask	Ø370mm Length 97mm Flange 125mm
50L Evaporating Flask	Ø467mm Length 97mm Flange 125mm



Item
PTFE Oil Seal Set



* Only 10L/20L/50L

Item
Evaporating Flask support



Item
Boiling/Evaporating Screw



* Only 10L/20L/50L

Item
PMMA cover for evaporating flask



Item	Specifications
Receiving Flask	-
3L Receiving Flask with Valve	Screw Socket 24mm Flange 38mm
5L Receiving Flask	Flange 60/50mm
10L Receiving Flask	
20L Receiving Flask	



Item	Specifications
Open/Close Valve (SH-RE-5L)	Screw Socket 24mm Flange 38mm
Open/Close Valve (SH-RE-20L/50L)	Screw Socket 24mm Flange 50mm



Item	Specifications
Check Valve	PTFE



Item	Specifications
Condenser Column (SH-RE-1L)	-
Main Condenser (SH-RE-5L)	Outer Diameter 98mm Length 460mm Flange 38mm, 50mm
Main Condenser (SH-RE-10L)	Outer Diameter 120mm Length 290mm Flange 80mm
Main Condenser (SH-RE-20L)	Outer Diameter 130mm Length 680mm Flange 80mm
Main Condenser (SH-RE-50L)	Outer Diameter 160mm Length 735mm Flange 80mm



Item	Specifications
Auxiliary Condenser (SH-RE-10L)	Outer Diameter 130mm Length 505mm Flange 80mm
Auxiliary Condenser (SH-RE-20L)	Outer Diameter 160mm Length 415mm Flange 80mm
Auxiliary Condenser (SH-RE-50L)	Outer Diameter 160mm Length 520mm Flange 80mm



Item	Specifications
Fix Flange 80mm	-
Fix Flange 60mm	-
Fix Flange 40mm	-
Fix Flange 30mm	-



Item
Vacuum Seal Set



Item	Specifications
Three-Way Flask (SH-RE-5L)	Outer Diameter 80mm Length 300mm Flange 50mm
Three-Way Flask (SH-RE-20L/50L)	Outer Diameter 100mm Length 350mm Flange 50mm



Item
Spanner Wrench



Item	Specifications
Drain Valve (SH-RE-5L)	Screw socket 24mm Flange 38mm
Drain Valve (SH-RE-20L/50L)	Screw Socket 37mm Flange 50mm



Item	Specifications
Flange 80mm Gasket	PTFE 52.5/79
Rotation Shaft Gasket	PTFE 41/69
Flange 60mm Gasket	PTFE 33/59
Flange 40mm Gasket	PTFE 20.5/39



Item	Specifications
Rotating Shaft (SH-RE-1L)	-
Rotating Shaft (SH-RE-5L)	Diameter 29mm Length 150mm Flange 40mm
Rotating Shaft (SH-RE-20L/50L)	Diameter 60mm Length 200mm Flange 70mm



Item
Flask Easy Clamp



Item	Specifications
Feeding Valve (SH-RE-1L)	-
Feeding Valve (SH-RE-5L/10L/20L/50L)	Screw Socket 24mm Flange 50mm



Item
Oil Vacuum Gauge with Base



Item	Specifications
Condenser Ring Stand 170mm	Rod Φ 16x160mm
Condenser Ring Stand 170mm	Rod Φ 16x170mm
Condenser Ring Stand 110mm	Rod Φ 16x170mm
Condenser Ring Stand 110mm	Rod Φ 16x120mm



Item	Specifications
Receiving Flask Ring Stand 270mm	Rod Φ 16x160mm
Receiving Flask Ring Stand 220mm	Rod Φ 16x170mm
Receiving Flask Ring Stand 165mm	Rod Φ 12x120mm



Item	Specifications
Condenser Ring Clamp (SH-RE-5L)	Rod Φ 12x120mm Strap design
Condenser Ring Clamp (SH-RE-10/20/50L)	Rod Φ 12x250mm Strap design



분리 시

Item
GL 14 Hose Barb (SH-RE-1L)
GL 18 Hose Barb (SH-RE-5L, 10L, 20L, 50L)

16. Water Bath & Chiller

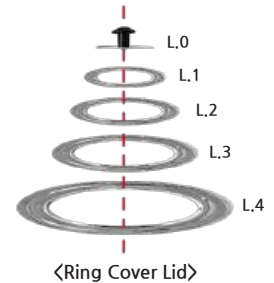
Multi-Purpose Water Bath:



Model	SH-WB-6GDN	SH-WB-11GDN	SH-WB-22GDN
Temp Range	Ambient +5℃ to 99℃ (Ambient + 9°F to 210°F)		
Ring Cover Lid Type	2 holes in 1 row	4 holes in 2 row	6 holes in 2 row
Capa	6.7 L	12.6L	19.3 L
Dimension Internal (W×D×H)	300x160x140mm	300x300x140mm	460x300x140mm
	11.81x6.30x5.51"	11.81x11.81x5.51"	18.11x11.81x5.51"
Dimension External (W×D×H)	345x205x280mm	345x345x280mm	505x345x280mm
	13.58x8.07x11.02"	13.38x13.58x11.02"	19.88x13.38x11.02"



Ring Cover Lid		
NO.	외경(ø,mm)	내경(ø,mm)
L.0	54	-
L.1	73	40
L.2	95	55
L.3	120	75
L.4	140	97



General-Purpose Oil Bath

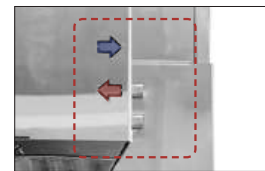


Model	SH-WB-10GDH	SH-WB-20GDH	SH-WB-40GDH
Temp Range	Ambient +10℃ to 300℃ (Ambient +18°F to 572°F)		
Capa	13 L	24 L	48 L
Dimension Internal (W×D×H)	250x300x170mm	350x400x170mm	600x400x200mm
	9.84x11.81x6.69"	13.78x15.75x6.69"	23.62x15.75x7.87"
Dimension External (W×D×H)	635x385x345mm	732x485x345mm	1035x500x390mm
	25.00x15.15x13.58"	28.81x19.09x13.58"	40.74x19.68x13.58"

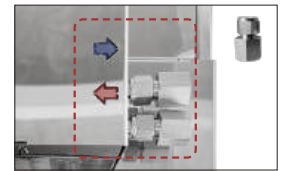
Circulating Water Bath



* Standard(13.0mm(O.D) SUS pipe)

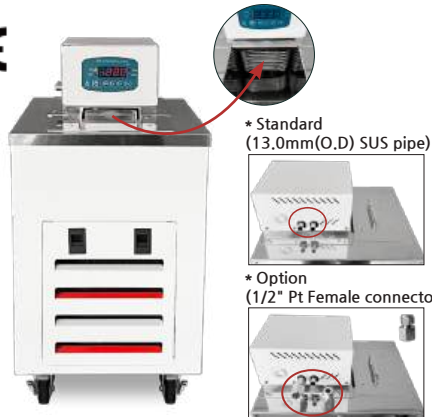


* Option(1/2" Pt Female connector)



Model	SH-WB-11CDN	SH-WB-22CDN	SH-WB-48CDN
Temp Range	Ambient +5℃ to 99℃ (Ambient + 9°F to 210°F)		
Capa	10.8 L	21.8 L	48 L
Dimension Internal (W×D×H)	300x300x140mm	460x300x140mm	600x400x200mm
	11.81x11.81x5.51"	18.11x11.81x5.51"	23.62x15.75x7.87"
Dimension External (W×D×H)	345x345x391mm	505x345x391mm	1005x485x440mm
	13.58x13.58x15.39"	19.88x13.58x15.39"	39.56x19.09x17.32"

Refrigerated Circulating Bath



Model	SH-WB-7CDR	SH-WB-13CDR	SH-WB-22CDR
Temp Range	Ambient -40°C to 150°C (Low Limit -20°C) Ambient -64°F to 302°F		
Capa	6.8 L	13 L	22L
Recirculating Pump	Max flow	2~16 L/min (tested with clean water at 20°C)	
	Max Head	3.5 meter (tested with clean water at 20°C)	
	Max Head	5.1 psig (tested with clean water at 20°C)	
Dimension Internal (W×D×H)	150x300x150mm	250x350x150mm	350x420x150mm
	5.91x11.81x5.90"	9.84x13.78x5.90"	13.78x16.54x5.90"
Dimension External (W×D×H)	340x417x605mm	440x455x645mm	540x520x700mm
	13.38x16.41x23.82"	17.32x17.91x25.39"	21.26x20.47x27.55"

3 in 1 Compact Chiller



Model	SH-WB-5CDR(-15)	
Temp Range	-15°C to ambient (based on ambient 25°C / No load condition)	
Cooling Capacity	Holding temp.	Cooling Power
	10°C	980W
	0°C	650W
	-10°C	330W
Recirculating Pump	Max Flow	16L/min. (tested with distilled water at 20°C)
	Max Head	3.5meter (tested with distilled water at 20°C)
	Max pressure	5.1 psig (tested with distilled water at 20°C)
	Water Proof Level	1px4
Dimension Internal (WxDxH)	200x200x130mm	
	7.87x7.87x5.12"	
Dimension External (WxDxH)	283x400x565mm	
	11.14x15.75x22.24"	
Reservoir Capa	5L	

Recirculating Chiller



Model	SH-WB-5CDR	
Order Number	SH-WB-5CDR(-30)	SH-WB-5CDR(-70)
Temp Control Range	Ambient -50°C to Ambient (Low limit -30°C) Ambient -28°F to Ambient	Ambient -90°C to Ambient (Low limit -70°C) Ambient -162°F to Ambient
Reservoir Capa	5 L	
Recirculating Pump	Max flow	16 L/min (tested with distilled water at 20°C)
	Max Head	3.5 meter (tested with distilled water at 20°C)
	Max pressure	5.1 psig (tested with distilled water at 20°C)
Dimension Internal (Φ×H)	180Φ×210Hmm	
	7.09Φx8.27H"	
Dimension External (W×D×H)	465×560×847mm	505×620×887mm
	18.18x22.04x33.35"	19.88x24.41x34.92"



Model	SH-WB-12CDR	SH-WB-20CDR	SH-WB-40CDR
Order Number	SH-WB-12CDR(-25)	SH-WB-20CDR(-25)	SH-WB-40CDR(-20)
Temp Control Range	Ambient -50°C to Ambient (Low limit -25°C) Ambient -28°F to Ambient		Ambient -45°C to Ambient (Low limit -20°C) Ambient -28°F to Ambient
Reservoir Capa	12 L	20 L	40 L
Recirculating Pump	Max flow	23.0 L/min (tested with distilled water at 20°C)	
	Max Head	9 meter (tested with distilled water at 20°C)	
	Max pressure	12.8psig (tested with distilled water at 20°C)	
Dimension Internal (Φ×H)	220Φ×250Hmm	280Φ×295Hmm	350Φ×390Hmm
	8.66Φx9.84H"	11.02Φx11.61H"	13.78Φx15.35H"
Dimension External (W×D×H)	482x614x811mm	442x644x1135mm	533x764x1179mm
	18.98x24.17x31.93"	17.40x25.35x44.68"	20.98x30.08x46.42"

Digital Ultrasonic Cleaner



Model	SH-WB-2UC
Temperature range	Ambient + 5°C~80°C(41~176°F)
Capacity	2L
Working Frequency	40KHz
Ultrasonic Power	70W
Heating Power	100W
Timing time	10~600s
Inner Tank Dimension (LxWxH)	150x135x100mm / 5.91x5.31x3.93"
Packing Size (LxWxH)	242x227x275mm / 9.52x8.93x10.83"

Power Adjustable Ultrasonic Cleaner



Model	SH-WB-5UC	SH-WB-11UC	SH-WB-20UC	SH-WB-30UC
Temperature range	Ambient + 5°C~80°C(41~176°F)			
Capacity	5L	11L	20L	30L
Working Frequency	40KHz			
Ultrasonic Power	120W	240W	360W	600W
Degas Function	YES			
Heating Power	200W	300W	400W	600W
Timing time	1~99mins			
Power Regulation	0~100%			
Inner Tank Dimension (LxWxH)	240x135x150mm 9.45x5.31x5.91"	300x240x150mm 11.81x9.45x5.91"	330x300x200mm / 12.99x11.81x7.87"	500x300x200mm 19.69x11.81x7.87"
Packing Size (LxWxH)	342x240x355mm 13.46x9.45x13.98"	412x357x380mm 16.22x14.06x14.96"	472x437x470mm / 18.58x17.20x18.50"	657x450x475mm 25.87x17.72x18.70"

Mini Water Bath



Model	WB-02
Temp. Range	Ambient +5°C~95°C
Max.Capacity	240ml
Dimension External (WxDxH)	262x143x96mm / 10.31x5.63x3.78"

Digital Dry Bath



Model	DB-02
Temp. Range	Ambient +5°C~120°C
No.of block	1
Dimension External (WxDxH)	262x143x96mm / 10.31x5.63x3.78"

17. Measuring Instrument

pH 50+ DHS (pH - mV - Temp)



Model	pH 50+ DHS
pH Measuring Range	0.00...14.00 pH
Resolution	0.1/0.01 pH
mV Measuring Range	± 1,000 mV
Resolution	1 mV
Temperature Measuring Range	0...100.0 °C
Resolution / Accuracy	0.1 °C / ± 0.5 °C

pH 8+ DHS

(pH - mV - Temp - GLP - Data Logger USB - Printer RS232)



Model	pH 8+ DHS
pH Measuring Range	-2...16
Resolution / Accuracy	0.1 / 0.01 / ±0.02
mV Measuring Range (redox) / Resolution	± 2000 / 1
Ion Measuring Range	-
Temperature Measuring Range	-10...+110 °C
Resolution / Accuracy	0.1/±0.5 °C

pH 80+ DHS (pH - mV - ION - Temp - GLP - GMP - Data Logger)



Model	pH 80+ DHS
pH Measuring Range	-2...20
Resolution / Accuracy	0.1 / 0.01 / 0.001 / ±0.002
mV Measuring Range (redox) / Resolution	± 2000 / 0.1 / 1
Ion Measuring Range	0.001...19999 ppm
Temperature Measuring Range	-20...+120 °C
Resolution / Accuracy	0.1/±0.5 °C

PC 80+ DHS

(pH - mV - Ion - COND - TDS - Salinity - Resistivity Temperature - GLP GMP - Data Logger)



Model	PC 80+ DHS
pH Measuring Range	-2...20
Resolution / Accuracy	0.1 / 0.01 / 0.001 / ±0.002
mV Measuring Range (redox) / Resolution	± 2000 / 0.1 / 1
Ion Measuring Range	0.001...19999 ppm
COND Measuring Range	0.00...20.00 - 200.0 - 2000 µS 2.00 - 20.00 - 200.0 - 1000 mS
Resolution	Automatic scale
TDS Measuring Range	0.1 mg/l...500g/l
TDS Factor/ Accuracy	0.40 ... 1.00 / ±1 % full scale
Salinity Measuring Range / Accuracy	0.01...100 ppt / ±1 % full scale
Resistivity Measuring Range / Accuracy	1...10 MΩ / ±1 % full scale
Temperature °C Measuring Range / Accuracy	-20.0...120.0 / ± 0.5 °C

pH 7+ DHS (pH - mV/ORP - Temp)



Model	pH 7+ DHS
pH measuring Range	0.00...14.00 pH
Resolution	0.1 / 0.01 pH
mV Measuring Range	± 1,000 mV
Resolution	1 mV
Temperature Measuring Range	0...100.0 °C
Resolution / Accuracy	0.1 °C / ± 0.5 °C

pH 70+ DHS (pH - mV/ORP - Temp - GLP - USB Data Logger)



Model	pH 70+ DHS
pH measuring Range	-2.00...16.00 pH
Resolution	0.1 / 0.01 pH
mV Measuring Range	- 1,000...+1,900 mV
Resolution	0.1 mV (± 200 mV) / 1mV (outside)
Temperature Measuring Range	-10...110 °C
Resolution / Accuracy	0.1 °C / ± 0.5 °C

OXY 7

Polarographic sensor / O₂-Temp / Barometric pressure



Model	OXY 7
Dissolved Oxygen Measuring Range	0.00...19.99 mg/L-ppm / 20.0...50.0 mg/L-ppm
Resolution	0.01 / 0.1
Oxygen Saturation Measuring Range	0.0...199.9 % / 200...400%
Resolution	0.1 / 1 %
Barometric Air Pressure Measuring Range	0.0...1100 mbar
Resolution	1 mbar
Temperature Measuring Range	0.0... 60.0 °C
Resolution	0.1 °C
Salinity Measuring Range	0...50 ppt

OXY 70

Optical LDO sensor / O₂-Temperature - Barometric pressure , GLP - Data logger USB
Mains supply and battery



Model	OXY 70
Dissolved Oxygen Measuring Range	0.00...19.99 mg/L-ppm / 20.0...50.0 mg/L-ppm
Resolution	0.01 / 0.1
Oxygen Saturation Measuring Range	0.0...199.9 % / 200...400%
Resolution	0.1 / 1 %
Barometric Air Pressure Measuring Range	0.0...1100 mbar
Resolution	1 mbar
Temperature Measuring Range	0.0... 60.0 °C
Resolution	0.1 °C
Salinity Measuring Range	0...50 ppt

SH Precision Balances



NEW



Model	SH-BL200	SH-BL220S	SH-BL310	SH-BL610
Capacity	220 g		310 g	610 g
Repeatability	0.0001 g		0.001 g(1 mg)	0.01 g
Liner	± 0.0003 g		± 0.003 g	± 0.03 g
Pan Size	ϕ 80mm		ϕ 90mm	ϕ 130 mm
Dimension	300x200x295mm	340x215x350mm	270x200x270mm	230x190x70mm
Case	ABS Shell	Aluminum Shell	ABS Shell	ABS Shell
Display	LCD	5 Inch touch screen	LCD	LCD

SH Moisture Balance



NEW



Model	SH-MA110E
Capacity	110 g
Readability	0.001 g
Sensor	Common Load Cell
Max temp	160°C
Display	5 Inch touch screen
Heating Space	30mm
Step Heating	NA
Networking	NA
Rapid Reduce Temperature	NA
Case	ABS



Model	SH-MA110C
Capacity	110 g
Readability	0.001g
Sensor	HBM Load Cell
Max temp	199°C
Display	LCD
Heating Space	22mm
Step Heating	NA
Networking	NA
Rapid Reduce Temperature	NA
Case	Aluminum



Model	SH-MA110S
Capacity	110 g
Readability	0.0001 g
Sensor	Magnetic Sensor
Max temp	230°C
Display	7 Inch touch panel
Heating Space	35mm
Step Heating	YES
Networking	YES
Rapid Reduce Temperature	YES
Case	Aluminum

Flue Gas Analyzer

스마트 연소가스 분석기



빠르고 간편한 스마트-터치 기술
5인치의 대형 HD 한글 디스플레이
현장 보고서 작성 및 실시간 이메일 발송
측정 데이터 현장 출력 가능

Model	testo 300
인증	TÜV 인증 (1st German Federal Immission Control Ordinance (BImSchV))
보관 온도	-20 ~ +50°C
작동 온도	-5 ~ +45°C
충전 온도	0 ~ +45°C
습도 적용범위	15 ~ 90% RH(비응축)
전원 공급	충전가능한 배터리, USB 어댑터를 이용한 기기 충전
배터리 수명	10 시간
보호 등급	IP 40
메모리	측정값 1,000,000개 까지 저장 가능
디스플레이	5인치 터치 디스플레이, HD 1280x720 픽셀, IPS (160 k)
무게	약 800 g
크기	길이: 244 mm (프로브 연결) 높이: 59 mm 너비: 98 mm



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<300case>

Flue Gas Analyzer

스마트 연소가스 분석기



Model	testo 340 Nox	testo 340 Sox
메모리	최대 : 100개의 폴더 생성 가능 폴더 당 : 최대 10개의 지점(사이트) 설정 가능 지점(사이트) 당 : 최대 200개의 측정 기록 저장 가능 최대 기록 개수는 폴더 또는 사이트의 수에 의해 결정됩니다.	
사용자 지정 연료	O ₂ , CO, NO, NO ₂	O ₂ , CO, NO, SO ₂
펌프	펌프 유속 : 0.6 l/min (조절 가능) 호스 길이 : 최대 7.8 m (2개의 호스 연장) 최대 양압 : +50 mbar 최대 음압 : -200 mbar	
무게	960 g	
크기	283 x 103 x 65 mm	
보관 온도	-20 ~ +50 °C	
작동 온도	-5 ~ +50 °C	
디스플레이	한글 디스플레이 160 x 240 픽셀	
전원 공급	배터리 3.7 V / 2.4 Ah 충전 케이블 6.3 V / 2 A	
하우징 재질	TPE PC	
보호 등급	IP40	



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<340case>

18. Climate Chamber & 배터리 시험챔버

Temp Control Model

-25℃ ~ 120℃



Model	SH-CH-150U1	SH-CH-288U1	SH-CH-408U1
Temp Range	-25℃ ~ 120℃(-13°F~248°F)		
Dimension Internal (W×D×H)	500x500x600mm	600x600x800mm	850x600x800mm
	19.69x19.69x23.62"	23.62x23.62x31.50"	33.46x23.62x31.50"
Dimension External (W×D×H)	920x995x1555mm	1020x1095x1755mm	1270x1095x1805mm
	36.22x39.17x61.22"	40.15x43.11x69.09"	50.00x43.11x71.06"
Capa	150 L	288 L	408 L

Model	SH-CH-800U1	SH-CH-1200U1
Temp Range	-25℃ ~ 120℃(-13°F~248°F)	
Dimension Internal (W×D×H)	1000x800x1000mm	1200x1000x1000mm
	39.37x31.50x39.37"	47.24x39.37x39.37"
Dimension External (W×D×H)	1520x1325x2100mm	1790x1525x2100mm
	59.84x52.16x82.67"	70.47x60.03x82.67"
Capa	800 L	1200 L

-50℃ ~ 120℃



Model	SH-CH-150U2	SH-CH-288U2	SH-CH-408U2
Temp Range	-50℃ ~ 120℃(-58°F~248°F)		
Dimension Internal (W×D×H)	500x500x600mm	600x600x800mm	850x600x800mm
	19.69x19.69x23.62"	23.62x23.62x31.50"	33.46x23.62x31.50"
Dimension External (W×D×H)	920x995x1555mm	1020x1095x1755mm	1270x1095x1805mm
	36.22x39.17x61.22"	40.15x43.11x69.09"	50.00x43.11x71.06"
Capa	150 L	288 L	408 L

Model	SH-CH-800U2	SH-CH-1200U2
Temp Range	-50℃ ~ 120℃(-58°F~248°F)	
Dimension Internal (W×D×H)	1000x800x1000mm	1200x1000x1000mm
	39.37x31.50x39.37"	47.24x39.37x39.37"
Dimension External (W×D×H)	1520x1325x2100mm	1790x1525x2100mm
	59.84x52.16x82.67"	70.47x60.03x82.67"
Capa	800 L	1200 L

Temp&Humi Control Model

15℃ ~ 120℃



Model	SH-CTH-150SC	SH-CTH-288SC	SH-CTH-408SC
Temp Range	15℃ ~ 120℃(59°F~248°F)		
Humi Range	50% ~ 98% at Temp 20℃ ~ 35℃ (68°F~95°F) 30% ~ 98% at Temp 35℃ ~ 80℃(68°F~176°F) 25% ~ 90% at Temp 80℃ ~ 85℃(176°F~185°F) 20% ~ 80% at Temp 85℃ ~ 90℃(185°F~194°F)		
Dimension Internal (W×D×H)	500x500x600mm	600x600x800mm	850x600x800mm
	19.69x19.69x23.62"	23.62x23.62x31.50"	33.46x23.62x31.50"
Dimension External (W×D×H)	950x995x1595mm	1050x1095x1795mm	1300x1095x1845mm
	37.40x39.17x62.79"	41.22x43.11x70.66"	51.18x43.11x72.63"



Model	SH-CTH-800SC	SH-CTH-1200SC
Temp Range	15℃ ~ 120℃ (59°F~248°F)	
Humi Range	50% ~ 98% at Temp 20℃ ~ 35℃ (68°F~95°F) 30% ~ 98% at Temp 35℃ ~ 80℃(68°F~176°F) 25% ~ 90% at Temp 80℃ ~ 85℃(176°F~185°F) 20% ~ 80% at Temp 85℃ ~ 90℃(185°F~194°F)	
Dimension Internal (W×D×H)	1000x800x1000mm	1200x1000x1000mm
	39.37x31.50x39.37"	47.24x39.37x39.37"
Dimension External (W×D×H)	1615x1325x2100mm	1790x1525x2100mm
	63.58x52.16x82.67"	70.47x60.03x82.67"

-25℃ ~ 120℃



Model	SH-CTH-150SCR1	SH-CTH-288SCR1	SH-CTH-408SCR1
Temp Range	-25℃ ~ 120℃ (-13°F~248°F)		
Humi Range	50% ~ 98% at Temp 20℃ ~ 35℃ (68°F~95°F) 30% ~ 98% at Temp 35℃ ~ 80℃ (95°F~176°F) 25% ~ 90% at Temp 80℃ ~ 85℃ (176°F~185°F) 20% ~ 80% at Temp 85℃ ~ 90℃ (185°F~194°F)		
Dimension Internal (W×D×H)	500x500x600mm	600x600x800mm	850x600x800mm
	19.69x19.69x23.62"	23.62x23.62x31.50"	33.46x23.62x31.50"
Dimension External (W×D×H)	950x995x1595mm	1050x1095x1795mm	1300x1095x1845mm
	37.40x39.17x62.79"	41.22x43.11x70.66"	51.18x43.11x72.63"



Model	SH-CTH-800SCR1	SH-CTH-1200SCR1
Temp Range	-25℃ ~ 120℃ (-13°F~248°F)	
Humi Range	50% ~ 98% at Temp 20℃ ~ 35℃ (68°F~95°F) 30% ~ 98% at Temp 35℃ ~ 80℃ (95°F~176°F) 25% ~ 90% at Temp 80℃ ~ 85℃ (176°F~185°F) 20% ~ 80% at Temp 85℃ ~ 90℃ (185°F~194°F)	
Dimension Internal (W×D×H)	1000x800x1000mm	1200x1000x1000mm
	39.37x31.50x39.37"	47.24x39.37x39.37"
Dimension External (W×D×H)	1620x1445x2100mm	1820x1645x2100mm
	63.78x56.89x82.67"	71.65x64.76x82.67"

-50℃ ~ 120℃



Model	SH-CTH-150SCR2	SH-CTH-288SCR2	SH-CTH-408SCR2
Temp Range	-50℃ ~ 120℃ (-58°F~248°F)		
Humi Range	50% ~ 98% at Temp 20℃ ~ 35℃ (68°F~95°F) 30% ~ 98% at Temp 35℃ ~ 80℃ (95°F~176°F) 25% ~ 90% at Temp 80℃ ~ 85℃ (176°F~185°F) 20% ~ 80% at Temp 85℃ ~ 90℃ (185°F~194°F)		
Dimension Internal (W×D×H)	500x500x600mm	600x600x800mm	850x600x800mm
	19.69x19.69x23.62"	23.62x23.62x31.50"	33.46x23.62x31.50"
Dimension External (W×D×H)	950x995x1595mm	1050x1095x1795mm	1300x1095x1845mm
	37.40x39.17x62.79"	41.22x43.11x70.66"	51.18x43.11x72.63"



Model	SH-CTH-800SCR2	SH-CTH-1200SCR2
Temp Range	-50℃ ~ 120℃ (-58°F~248°F)	
Humi Range	50% ~ 98% at Temp 20℃ ~ 35℃ (68°F~95°F) 30% ~ 98% at Temp 35℃ ~ 80℃ (95°F~176°F) 25% ~ 90% at Temp 80℃ ~ 85℃ (176°F~185°F) 20% ~ 80% at Temp 85℃ ~ 90℃ (185°F~194°F)	
Dimension Internal (W×D×H)	1000x800x1000mm	1200x1000x1000mm
	39.37x31.50x39.37"	47.24x39.37x39.37"
Dimension External (W×D×H)	1620x1445x2100mm	1820x1645x2100mm
	63.78x56.89x82.67"	71.65x64.76x82.67"

Thermal Shock Tester (-50℃ to 180℃)



Model	SH-TST27-2	SH-TST64-2
Type	Air to Air 2-Zone (Elevating type)	
Dimension Internal (WxDxH)	300x300x300(mm)	400x400x400(mm)
	11.8x11.8x11.8"	15.7x15.7x15.7"
Dimension External (WxDxH)	1130x1800x1845(mm)	1230x1900x2045(mm)
	44.4x70.8x72.6"	48.4x74.8x80.5"
Volume of chamber	27 L	62 L
Range	-50℃ ~ 180℃ (Cycle range -40℃ ~ 150℃, Variation : 5 min.)	



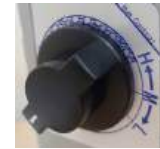
Model	SH-TST27-3	SH-TST64-3
Type	Air to Air 3-Zone (Air substitution type)	
Dimension Internal (WxDxH)	300x300x300(mm)	400x400x400mm
	11.8x11.8x11.8"	15.7 x 15.7 x 15.7"
Dimension External (WxDxH)	1130x1800x1820mm	1230x1900x2020mm
	44.4x 0.8x71.6"	48.4 x 74.8 x 79.5"
Volume of chamber	27 L	62 L
Range	-50℃ ~ 180℃ (Cycle range -40℃ ~ 150℃, Variation : 5 min.)	

19. Desiccator

Desiccator



Temp/Humi gauge(Digital)



Dehumidifier (Inside)



Dehumidifier (Outside)

Model No.	SH-DS-3T		Options	
Dimension Internal	270x300x400mm / 10.63x11.81x15.75"		Purge	Desiccant
Dimension External	320x330x480mm / 12.60x12.99x18.90"		N/A	N/A
Material	Body	PMMA		
	Frame	ABS		
	Shelve	ABS		
	Caster	N/A		
Component	Temp / Humi gauge	Digital built in	N/A	N/A
	Dehumidifier	built in		
	Shelves	3 EA, 270x300mm 10.63x11.81"		



Model No.	SH-DS-10		Options	
Dimension Internal	560x483x731mm / 22.05x19.02x28.78"		Purge	Desiccant
Dimension External	574x535x765mm / 22.60x20.87x30.12"		Essential  valve 2EA Optional  Gas flowmeter	Basic composition  Silca tray Optional  Dehumidifier
Material	Body	PMMA		
	Frame	AL		
	Shelve	SUS		
	Caster	N/A		
Component	Temp / Humi gauge	Digital built in	N/A	N/A
	Dehumidifier	Option		
	Shelves	2 EA, 460x490mm 18.11x19.29"		

*제습기가 주문되지 않은 경우 Silca Tray가 제공됩니다.



Model No.	SH-DS-12		Options	
Dimension Internal	560x483x986mm / 22.05x19.02x38.82"		Purge	Desiccant
Dimension External	574x535x1090mm / 22.60x20.87x42.91"		Essential  Valve 2EA Optional  Gas flowmeter	Basic composition  Silca tray Optional  Dehumidifier
Material	Body	PMMA		
	Frame	AL		
	Shelve	SUS		
	Caster	N/A		
Component	Temp / Humi gauge	Digital built in	N/A	N/A
	Dehumidifier	Option		
	Shelves	4 EA, 460x490(mm) 18.11x19.29"		

*제습기가 주문되지 않은 경우 Silca Tray가 제공됩니다.



Model No.	SH-DS-20		Options	
Dimension Internal	560x483x1486mm / 22.05x19.02x58.50"		Purge	Desiccant
Dimension External	574x535x1770mm / 22.60x20.87x69.69"		Essential  Valve 2EA Optional  Gas flowmeter	Basic composition  Silca tray Optional  Dehumidifier
Material	Body	PMMA		
	Frame	AL		
	Shelve	SUS		
	Caster	PA		
Component	Temp / Humi gauge	Digital built in	N/A	N/A
	Dehumidifier	Option		
	Shelves	4 EA, 460x490mm 18.11x19.29"		



Model No.	SH-DS-40		Options	
Dimension Internal	1138x483x1486mm / 44.80x19.02x58.50"		Purge	Desiccant
Dimension External	1164x535x1770mm / 45.83x20.87x69.69"		Essential	Basic composition
Material	Body	PMMA	 valve 2EA  Gas flow meter  Silica tray  Dehumidifier	Optional
	Frame	AL		
	Shelve	SUS		
	Caster	PA		
Component	Temp / Humi gauge	Digital built in		
	Dehumidifier	Option		
	Shelves	8 EA, 460x490(mm)		

Vacuum Desiccator



Model No.		SH-VDS-12	SH-VDS-19	SH-VDS-36
Dimension Internal (WxDxH)		260x180x260mm	260x200x360mm	260x300x460mm
		10.23x7.09x10.24"	10.24x7.87x14.17"	10.24x11.81x18.11"
Material	Body	PMMA		
Component	Valves	8MM VALVE 2EA		
	Gauge	Vacuum gauge		
	Dehumidifier	Silica tray		
Shelves		2 EA	3 EA	4 EA



Model No.		SH-VDS-29	SH-VDS-43
Dimension Internal (WxDxH)		260x280x410mm	300x280x510mm
		10.24x11.02x16.14"	11.81x11.02x20.18"
Material	Body	PMMA	
Component	Valves	8MM VALVE 2EA	
	Gauge	Vacuum gauge	
	Dehumidifier	Silica tray	
Shelves		4 EA	5 EA

20. Hood

Fume Hood



Option



Model	SH-HD-1200UP	SH-HD-1500UP
Type	Fume Hood Type	
Dimension External (WxDxH)	1200x800x2350mm	1500x800x2350mm
	47.24x31.50x92.52"	59.06x31.50x92.52"
Dimension Internal (WxDxH)	1024x676x1160mm	1324x676x1160mm
	40.31x26.64x45.67"	52.13x26.64x45.67"

Model	SH-HD-1800UP
Type	Fume Hood Type
Dimension External (WxDxH)	1800x800x2350mm
	70.87x31.50x92.52"
Dimension Internal (WxDxH)	1624x676x1160mm
	63.93x26.64x45.67"

Lab Mobile Hood



<풍량 1~10단계 조절가능>



풍량 1단계



풍량 5단계



풍량 10단계

Model	SH-HD-MUP
Power	200W
Air volume	330m ³ /h
Noise	≤63dB(A)
Filtration	0.3μm 99.7%
Size(WxDxH)	265x405x550mm
	10.43x15.94x21.65"
Size of flange & Cover	Ø75mm / Ø260mm
Filter Change Alarm	Yes

Table Top Fume Hood



* Option



Model	SH-HD-700UP
Type	Table Top Ducted Fume Hood Type
Dimension External (W×D×H)	700x620x1150mm
	27.6x24.4x45.3"
Dimension Internal (W×D×H)	640x550x700mm
	25.2x21.7 x27.6"

PP Fume Hood



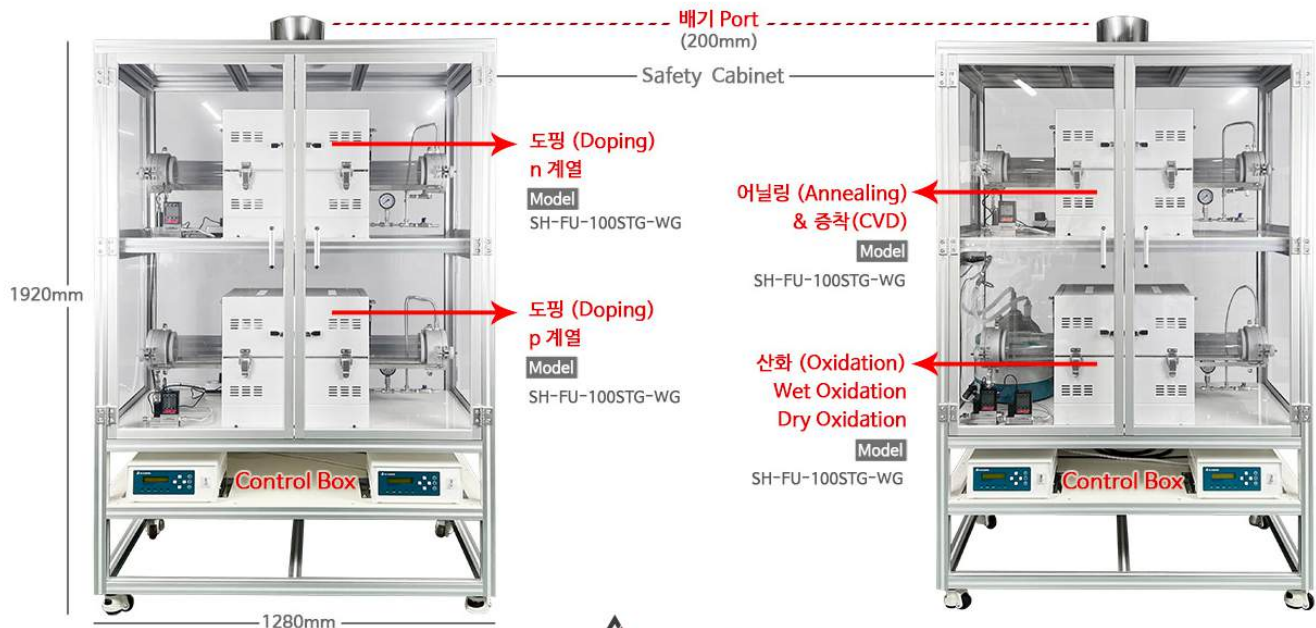
Model	SH-HDPP-1200UP	SH-HDPP-1500UP	SH-HDPP-1800UP
Type	Ducted PP Fume Hood Type		
Dimension External (W×D×H)	1240x800x2200mm	1540x800x2200mm	1840x800x2200mm
	48.8x31.5x86.6"	60.6x31.5x86.6"	72.4x31.5x86.6"
Dimension Internal (W×D×H)	1020x570x872mm	1320x570x872mm	1620x570x872mm
	40.2x22.4x34.3"	52.0x22.4x34.3"	63.8x22.4x34.3"

Custom-Made System

반도체 공정(Doping, Oxidation, Annealing, CVD) 실험용 턴키 시스템

본 설비는 반도체 공정 구현을 위해 통상 4SET의 튜브전기로(튜브, 실링마스크, 가스Flow meter 포함)가 필요한데 협소한 설치공간의 단점 해결을 위해 두 개의 투명 캐비닛에 각각 공정별 2 SET의 튜브전기를 상/하로 설치하여 공간절약 효과와 실험중 튜브파손 및 외부오염 방지 목적으로 제작 되었습니다.

캐비닛 상부에는 200mm 배기 Port 를 설치하여 고온 실험으로 인한 실험실 전체 온도상승을 억제하는 효과도 있습니다.



Doping Turn-Key System 구성품목

Safety Cabinet	1EA
도핑 (Doping) 용 튜브전기로	2SET (n 계열, p 계열)
불활성 가스 주입용 MFC	2EA
BPR (Back Press Regulator)	2EA - 반응기내 가스 양압 유지

Oxidation & CVD Turn-Key 구성품목

Safety Cabinet	1EA
산화 (Oxidation) 용 튜브전기로	1SET
O ₂ 가스 주입용 MFC	1EA - Dry & Wet Oxidation (건/습식 산화)
Carrier 가스 주입용 MFC	1EA
Steam 발생용 Heat unit	1SET - Wet Oxidation (습식산화)
BPR (Back Press Regulator)	1EA - 반응기내 가스 양압 유지
어닐링 (Annealing)& 증착(CVD) 용 튜브전기로	1SET
불활성 가스 주입용 MFC	1EA
BPR (Back Press Regulator)	1EA - 반응기내 가스 양압 유지

Custom-Made System

청정수소 생산실험 수직형 전기로 풀 패키지

연속투입식 수직형 튜브전기로

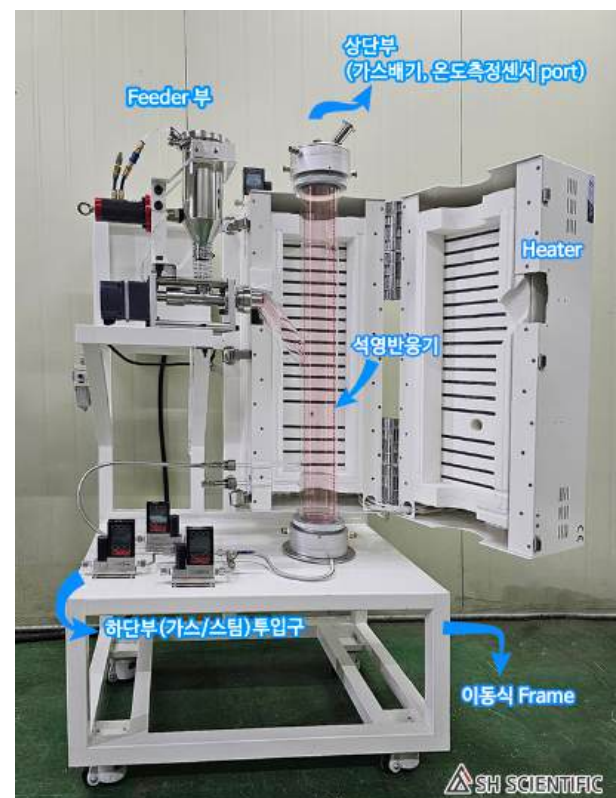
바이오매스 열분해 및 가스화 반응을 이용하여 청정수소가스를 얻는 목적의 실험을 수행할수 있도록 제작된 수직로 장비입니다.

- 구성품목 : 1. 수직형 튜브전기로
 2. 연속투입 Feeder (최소 12 ~ 15g/min 투입가능)
 3. 석영 반응기 100 dia x 1200 L, 실링 마스크 포함 (Gas IN / OUT port)
 4. 가스조절 system (MFC 2ea / 디지털 압력게이지 2ea)
 5. 스팀제너레이터
 6. 냉각수 공급용 chiller

열분해 반응 : 연속투입식 Feeder 를 이용 소량의 바이오매스를 고온(최고 1000도)의 석영반응기로 투입

- 가스화 반응 : 1. SMR (Steam Methane Reforming) -- 스팀 제너레이터 (Steam Generator)이용
 2. 단독 가스(O₂) 투입 (carrer gas 포함) -- 질량유량계 (Mass Flow Controller)이용, N₂ 또는 Ar 캐리어 가스 활용
 3. SMR + 가스 (O₂) 투입 -- 스팀 + 가스 조성을 이용

3가지 공정을 모두 구현가능하도록 제작되었습니다.



Max Temp	: 1200℃
Feeder speed	: 0 ~ 15 rpm
Temp control	: Program controller 1ea
Hot Zone	: 600mm
Mass Flow Controller	: MFC - 2EA (N ₂ , O ₂)
Digital Pressure Gauge	: 2ea
Quartz Tube	: 외경 100 dia (Length 1000 mm)
Feeder 연결구	: 50 dia
220V, 1 Ø, 5kW, 25A	
Gas IN port 2ea, Steam IN port 1ea	
Gas OUT port 1ea, KF 25	

Custom-Made System

Pilot Plant Scale Rotary Kiln

소규모 생산 및 대규모 양산설비 구축전 모델링 로타리 킬른

- Tube Diameter Ø267mm × Length 3140mm

Key Features

- 연속식 공정진행 가능
- Pre Heating zone - 증발 가스 응축 방지
- Gas in / out port & exhaust port
- 램핑기능을 갖춘 투시창 - 공정중 튜브내부 관찰 용이
- STS 310S 튜브 내부에 Baffle 4개 설치 - 균일한 열전달과 균일한 가스접촉
- 손쉬운 튜브교체 - 튜브교체시 측면이 아닌 상,하부 방향으로 튜브교체
튜브교체를 위한 별도의 여유공간 필요없음

* Option

Max Temp 1800℃ - 공정 최고온도 1600℃
 Mass Flow Control & Mass Flow Meter 장착 - 정밀가스제어및 모니터링
 증발증기 응축기 및 배출가스 흡착및 흡수탑 설치

▼ SH-FU-267RKG2400



◀ SH-FU-100RKG900



* Specification

Max Temp	1000 °C
Tube Material & Dimension	STS310S, O.D 267Ø / I.D 259 Ø x 3140 mm (Length)
Tube Tilt & Rotation Speed	Tilt 0 ~ 15°, Speed 0 ~ 5 R.P.M
Pre-Heating Zone	200 mm × 2 zones (left and right end sections)
Main Heating Zone	250 mm × 8 zones, total length 2175 mm (Including 25 mm × 7 Support poles)
Temp Controller & Sensor	Program Controller 10 ea / K-type Sensor 10 ea
Feed & Receiving Vessel	STS304, Capa 7 Liter
Gas In / Out / Exhaust Port	Gas Port 1/4" Tube Lock / Exhaust Port KF100
External Dimension (mm)	4772 Wide x 614 Depth x 1840 High
Electrical Power	380V, 3-phase, 4-wire system, 44 kW

Custom-Made System

UHV 초고진공 10^{-9} Torr 320 Dia 튜브 전기로

UHV (Ultra-High Vacuum) + 800°C Tube Furnace

- UHV Annealing 공정 또는 일부 열분해 공정중 UHV(초고진공) 분위기 필요시 적용되는 장비
- 반도체, 배터리, 연료전지, 디스플레이 전극소재의 어닐링 및 우주항공분야, 방산분야 등의 진공 블레이징 공정에도 사용
- 전극소재의 경우 금속박막 전극, 실리콘 음극, SOFC 전극, ITO 전극의 Annealing 공정에 적용

구성품목 : Openable 3 Heat Zone Tube Furnace
318.5 Ø Stainless Steel 316 Tube
UHV (Ultra High Vacuum) System



Openable 3 Heat Zone Tube Furnace

Max Temp	800°C (1472 °F)
Running Temp	600°C (1112 °F)
Application Tube Diameter	320mm
Temp Controller	45 Steps Programmable Controller x 3zone
Sensor	K Type
Heater Capacity	12kW X 3 Heat Zone / Total 36kW
Heating Length	300mm x 3 Heat Zone / Total 900mm (including 25mm Tube Support, 950mm)
Heat Element & Insulation	Kanthal A-1 & Ceramic
Electrical Power	380V, 3Ø, 54.8 A

318.5 Ø Stainless Steel 316 Tube

Material	Stainless Steel 316
Sealing Flange & Gasket	Stainless Steel 316 With Cooling Water Port & Viton Vacuum Gasket
Vacuum Port	ISO200 / MF flange
Gas In Port	ISO200 / MF flange

UHV (Ultra High Vacuum) System

Cryopump	8"
Oil Rotary Pump	960L/min
Max Vacuum	5x 10 ⁻⁹ Torr
Vacuum Port & Vent Port	ISO200 & KF16

Client List in the Korean Market



Client List in the Global Market



General Laboratory Equipment Manufacturer

Vacuum Drying Oven
Drying Oven
Industrial Drying Oven
Furnace
Ceramic consumable & Furnace accessory
Mixer · Grinding · Mill · Dispersion
Plastic Ware
Incubator
Incubator Shaker & Shaker
Roller Culture Apparatus
Autoclave
Clean Bench & Colony Counter & Hood
Mixer
Stirrer & Mantle
Rotary Evaporator
Water Bath & Chiller
Measuring Instrument
Climate Chamber
Ball Mill
Desiccator



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